MILITARY SPECIFICATION

MANUAL, TECHNICAL, COMBAT SYSTEM TECHNICAL OPERATIONS; REQUIREMENTS FOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification sets forth the Naval Sea Systems Command requirements for the content, general style, and format of Combat System Technical Operations Manuals (CSTOM) required for maintaining own-ship combat system readiness. CSTOM provides operational forces and school commands with combat system level information concerning design, detailed intersubsystem interface data, testing, maintenance, and capabilities in fleet user-oriented language and format. The CSTOM documents functional integration of all combat subsystems in performance of the major combat system operational functions of detection and entry, tracking and identification, threat evaluation and threat-to-weapon pairing, and engagement and engagement assessment.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38748 - Microfiche; For Engineering/Technical Data, Reports, Studies and Related Data, Requirements for.
MIL-M-38784 - Manuals, Technical: General Style and Format Requirements.
MIL-P-38790 - Printing Production of Technical Manuals, General Requirements for (Photolithographic Negatives, Reproduction Assembly Sheets, Printing).

PUBLICATIONS

MILITARY

DOD 5220.22-M - Industrial Security Manual for Safeguarding Classified Information.
NAVSEA OP 1700 - Standard Fire Control Symbols.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

Y32.2 - Graphic Symbols for Electrical and Electronics Diagrams.
Y32.10 - Fluid Power Diagrams, Graphic Symbols for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Style and format. Except as otherwise specified herein, the style and format requirements for preparation of the manual and related materials shall be in accordance with MIL-M-38784.

3.1.2 Format.

3.1.2.1 Page, paragraph, illustration, and table numbering. Text pages, paragraphs, illustrations, and tables shall be numbered consecutively in Arabic numerals within each section of a chapter. Assignments shall be a three-part number: the first two numbers separated by a hyphen, and the last two numbers separated by a decimal point. The first number represents the chapter number, the second number represents the section number, and the third number is the order within the section. For example: Page 4-3.12 shall be the 12th page in the 3rd section of chapter 4. Subparagraphs shall be identified by decimal numbers. For example: Subparagraph 4-3.2.1 shall be the first subparagraph of paragraph 2.

3.1.2.2 Illustrations. Illustrations shall be in accordance with MIL-M-38784, but shall be microform compatible as specified in 3.1.5.13. Graphic symbols for electrical and fluid power distribution diagrams shall be in accordance with ANSI Y32.2 and Y32.10, respectively. Information flow on pictorial, block and signal flow diagrams shall be left to right and top to bottom. Fire control symbols shall conform to NAVSEA OP 1700.

3.1.2.3 Planned maintenance system (PMS) test references. PMS tests are identified by the PMS coded maintenance index page (MIP) and maintenance requirement card (MRC) numbers. There are two different methods for coding MIPs. They are referred to as the conventional and alternate methods. A typical MIP number for an equipment coded by the conventional method will bear a four segment number such as R-81/3-43. The first two segments, R-81, indicate the kind (radar) and type (81st registered) of equipment. The third segment, /3, indicates the particular equipment configuration. The fourth segment, -43, indicates the month and year the MIP was updated. A typical MIP number for an equipment coded in the alternate method will bear a three-segment number such as 5BAB000/2-1. The first segment, 5BAB000, indicates the equipment type to which the MIP applies. The second and third segments, /2-1, identify the equipment ORDALT configuration and the maintenance requirements revision level. When referring to PMS tests in CSTOM, the reference, either in text or in tables, should be made so that updated tests will not require a change to CSTOM. Therefore, if the MIP is coded in the conventional manner, refer only to the first two MIP number segments followed by a slash and the test periodicity and number; for example, R-81/D-1. If the MIP is coded in the alternate manner, refer to the first segment of the MIP number followed by a slash and the test periodicity and number; for example, 5BAB000/W-3.

3.1.2.4 Reproducible copy preparation methods. Reproducible copy of the final edition shall be prepared in accordance with MIL-M-38784.

3.1.2.4.1 Mechanically reproduced copy preparation methods. When specified (see 6.2.1), review manuscript, preliminary, and reproducible copy prepared by electric accounting machine, punched card, magnetic tape (MTST or MTSC), paper tape, etc., may be submitted provided that the copy conforms to other requirements for correction, mechanized assembly, and reproduction.

3.1.2.4.2 Computer output typesetting. When specified (see 6.2.1), review manuscript, preliminary, and reproducible copy may be prepared by a computer system provided the copy conforms to the format, recording medium, and reproduction requirements of MIL-M-38784 and MIL-P-38790.

3.1.2.5 Microform compatibility. Except as specified herein, technical manuals shall be prepared in accordance with MIL-M-38748.

(a) New foldouts shall be prepared in the following format:
(1) Foldout drawings shall be divided into 17-inch image area segments, except when the balance of the image area is 8-1/2 inches or less. The remaining segment will be 8-1/2 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

(2) There shall be no drawing data, other than horizontal lines, placed closer than 1/8-inch from the image area limit.
(3) Lines shall not be vertically displaced during the transition from one image area to the next.

(b) Except if specified in the contract (see 6.2.1), halftone artwork shall not be used.
(c) Minimum printed size for text shall be 8 point. Nomenclature, callouts, tabular material and symbols on illustrations shall be upper case, with 6 point minimum printed size.
(d) Style and format shall be compatible for filming onto microfiche at a reduction ratio of 24 x 1 (98 images).

3.1.2.6 Binder, paper stock, and printing requirements.

3.1.2.6.1 Cover and backbone stock. Covers and backbones shall be printed on stock designated by the procuring activity and in accordance with MIL-P-38790.

3.1.2.6.2 Paper stock. Paper stock used for printing of title pages, text, and fold-out pages shall be in accordance with MIL-P-38790.

3.1.2.6.3 Printing. Printing will be in accordance with MIL-P-38790, as specified.

3.1.3 Technical data. The contractor shall prepare the following technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2), and as specified in 3.1.3.1 through 3.1.3.9.

3.1.3.1 Data research and analysis source data. When specified in the contract (see 6.2.1), a technical manual data research analysis of existing documentation shall be provided in accordance with the data ordering document (see 6.2.2).

3.1.3.2 Technical manual outline. When specified in the contract (see 6.2.1), a CSTOM technical manual outline shall be provided in accordance with the data ordering document (see 6.2.2).

3.1.3.3 Manuscript copy. When specified in the contract (see 6.2.1), a manuscript shall be provided for Government review in accordance with the data ordering document (see 6.2.2).

3.1.3.4 Preliminary manual. When specified in the contract (see 6.2.1), preliminary manuals for interim use shall be provided in accordance with the data ordering document (see 6.2.2). Copy shall be prepared in the same style and content as the manuscript, and shall be typed double column, single spaced, and back-up printed.

3.1.3.5 Standard (final) manual. Reproducible copy for the final manual shall be in accordance with the data ordering document (see 6.2.2). Copy shall reflect the results of verification with all valid Government comments incorporated.

3.1.3.6 Changes. When specified in the contract (see 6.2.1), changes shall be provided in accordance with the data ordering document (see 6.2.2).

3.1.3.7 Revisions. When specified in the contract (see 6.2.1), revisions shall be provided in accordance with the data ordering document (see 6.2.2).

3.1.3.8 Supplements. When specified in the contract (see 6.2.1), supplements shall be provided in accordance with the data ordering document (see 6.2.2).

3.1.3.9 Status reports. When specified in the contract (see 6.2.1), the contractor shall supply a status report (see 6.2.2) at a frequency determined by the contract, prepared in narrative format, and shall indicate the following information, as applicable:

(a) Contract number.
(b) Technical manual identification number.
(c) Ship name and identification number.
(d) Title of manual.
(e) Degree of progress:
(1) Title listing each element of proposed manual and indicating the degree of completion.
(2) Required and actual completion percentage of the deliverable data item at the time of report.
(f) Problems and delays that may affect the development and timely delivery of the deliverable data items.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

(g) Scheduled delivery dates and explanation of deviations thereto.
(h) Remarks.

3.1.4 Manual arrangement. Each CSTOM shall be arranged in the order listed:

(a) Front matter.
(b) Technical content.
(c) Glossary (volume 1 only).
(d) Appendixes (when required by contract (see 6.2.1)).
(e) Alphabetical index (when required by contract (see 6.2.1)).
(f) Supplement (when required by contract (see 6.2.1)).

3.1.5 Security classification. CSTOM volumes and parts shall normally be classified "Confidential" or higher and shall be handled and marked in accordance with DOD-5220.22-M. Contractors shall certify when a separately bound volume or part does not contain classified information and request Government verification for preparing an "Unclassified" part.

3.2 Front matter. The front matter for each volume, supplement, or for part 1 of multi-part volumes shall have the following contents in the order listed. Parts other than part 1 shall include covers, backbone, title page, and table of contents, list of illustrations, and list of tables applicable to the part.

(a) Cover (including back cover and backbone).
(b) Publication change guide (applicable only to changes).
(c) Title page.
(d) List of effective pages.
(e) Change record.
(f) Foreword.
(g) Table of contents.
(h) List of illustrations.
(i) List of tables.
(j) Safety summary.
(k) Frontispiece (volume 1 only).

3.2.1 Front cover. A front cover, similar to figure 1, shall contain the following:

(a) Technical manual identification number.
(b) Volume, part, supplement, and revision number, as applicable.
(c) Technical manual prime preparing contractor and contract number.
(d) The manual title ("COMBAT SYSTEMS TECHNICAL OPERATIONS MANUAL FOR _____") and subtitle (volume title).
(e) NAVSEA seal.
(f) Ship name (or class) and identification number.
(g) Distribution statement.
(h) The words "PUBLISHED BY DIRECTION OF COMMANDER, NAVAL SEA SYSTEMS COMMAND".

If the manual is classified, a front cover similar to figure 2 shall contain the security classification, espionage notice, and automatic time-phase downgrading and declassification information. When the manual is a revision, or reissue of an earlier publication, a supersedure notice shall be shown on the cover. If the manual is classified, the parenthetical note "(This page is UNCLASSIFIED)" shall be placed beneath the classification marking at the bottom of the cover.

3.2.2 Back cover. The back cover, similar to figure 3, for each separately bound publication shall contain the technical manual identification number and, as applicable, the volume, part, supplement, and revision number. If the publication is classified, the back cover shall also contain the security classification in a manner similar to figure 4. When the manual is less than 1-inch thick, the back cover shall also contain the title and subtitle of the manual in vertical arrangement in a manner similar to figure 5.

3.2.3 Backbone. Backbones shall be used for publications that are 1-inch or more in thickness. Backbone information shall be arranged as in figure 6. The backbone shall contain the technical manual identification number; the volume, part, supplement, and revision number, as applicable; the manual title and subtitle; and the security classification, if applicable. This information shall appear vertically on the backbone of a 1- to 1-1/2-inch thick publication and horizontally on the backbone of a 2- to 2-1/2-inch thick manual.

3.2.4 Title page. A title page, similar to figure 7, shall be included in each manual. It shall be a facsimile of the front cover, but shall also include the date of basic issue

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

or revision in accordance with MIL-M-38784. For a change, a new title page similar to figure 8, containing the change number and date of the change, shall be issued. For a change to a multipart volume, a new title page shall be issued for part 1 only. Title pages for parts other than part 1 shall not be changed. For a multipart volume, a blocked notice "SEE PART 1 OF THIS VOLUME FOR FRONT MATTER" shall be shown on the title pages of parts other than part 1. Figures 9 and 10 are samples of title pages for revisions and supplements, respectively.

3.2.5 Foreword. A foreword, similar to figure 11, shall be included in each manual, supplement, volume, or in part 1 of a multipart volume. The foreword shall briefly describe the scope and content of the manual and shall relate the structure of the entire CSTOM by chapter. The foreword shall include applicable classification guides for each subsystem documented. The foreword shall start as a right-hand page immediately following the change record.

3.3 Structure of CSTOM. The manual shall comprise four volumes. The manual shall be arranged in chapters and sections and shall be in accordance with the data ordering document (sec 6.2.2) and the Government accepted manual outline. Technical information shall be correct in every detail. CSTOM will not normally include operating test, or fault isolation procedural steps provided by other documentation. However, when a deficiency in subsystem and equipment documentation exists, and bridge-the-gap data (see 6.3.1) is provided in an appendix or supplement, warnings and cautions shall be included in all operation, test, and fault isolation procedures, prior to the step that, if incorrectly performed, might lead to injury or death of personnel or damage to equipment. The individual volumes shall be divided into individually bound parts when required. Chapter and section material content shall be in accordance with the volume arrangement provided in 3.3.3. Section and subsection content and location may vary within a particular chapter to accommodate necessary or unique data coverage as specified by the contract. For a description of the detail provided by each level of data coverage, see 3.3.1.

3.3.1 Data level descriptions. Data presented shall be concise and to the depth necessary for supporting combat system objectives. Descriptive portions of CSTOM will identify principal subsystem equipment with a textual synopsis for each. A reference to the subsystem documentation shall be made for detailed information. The data in CSTOM shall consist of levels of narrative and illustration coverage as follows:

LEVEL

1 DETAILED INTRODUCTION: Block or pictorial drawings with associated text, depicting system, subsystem, and equipment interfaces, and operational or testing dependencies or capabilities.

2 FUNCTIONAL AND OPERATIONAL DESCRIPTIONS: Block or pictorial drawings with associated text defining principal system, subsystem, or equipment functions, operations, actions, capabilities, or fault isolation methods and tools.

3 DETAILED DESCRIPTION: Signal-level drawings defining signal characteristics and flow or test procedures, and detailed narrative.

3.3.2 Minimum data requirements. The text, illustrations and tables prescribed in this specification reflect minimum CSTOM requirements. Where the combat system configuration provides additional weapon subsystem capabilities, operating modes or functions, other than covered herein, equal coverage of text, illustrations and tabulated data shall be provided.

3.3.3 Volume arrangement. The chapter and section arrangement for each CSTOM volume shall be as follows:

VOLUME 1 - COMBAT SYSTEM DESCRIPTION.

Chapter 1 - INTRODUCTION.
Sections - Introduction to CSTOM.
Ship and Combat System Description.
Location of Combat System Components.

Chapter 2 - COMBAT SUBSYSTEM DESCRIPTIONS.
Sections - Introduction to the Combat System.
Search Radar Subsystem.
Tactical Data Subsystem.
Electronic Warfare Subsystem.
Gun Weapon Subsystem.
Underwater Weapon Subsystem.

Source: https://assist.dla.mil -- Downloaded: 202640-19T23:02Z
Check the source to verify that this is the current version before use.

Missile Weapon Subsystem.
Light-Airborne Multi-Purpose System (LAMPS) Subsystem.
External Communication Subsystem.
External Navigation Subsystem.
Support Subsystem.

Chapter 3 - COMBAT SYSTEM OPERATIONAL DESCRIPTION.
Sections - Introduction to Combat System Functions.
Detection and Entry.
Tracking and Identification.
Threat Evaluation and Threat-To-Weapon Pairing.
Engagement and Engagement Assessment.

Glossary

VOLUME 2 - OPERATIONAL SEQUENCES.

Chapter 4 - OPERATIONAL SEQUENCES.
Sections - Purpose and Use of the Operational Mode Network and Operational Sequence Networks.
Detection and Entry.
Tracking and Identification.
Threat Evaluation and Threat-to-Weapon Pairing.
Electronic Warfare Engagement and Engagement Assessment.
Gun Engagement and Engagement Assessment.
Underwater Engagement and Engagement Assessment.
Missile Engagement and Engagement Assessment.
Controlled Aircraft Engagement and Engagement Assessment.
LAMPS Engagement and Engagement Assessment.
Reduced Capability Conditions.
Condition III Considerations.

VOLUME 3 - COMBAT SYSTEM READINESS.

Chapter 5 - READINESS ASSESSMENT.
Sections - Introduction to Readiness Assessment.
Factors Contributing to Readiness Assessment.
Ship Electronic Readiness Team.
Planned Maintenance System Concept.
Training and Maintenance Support Elements.
Readiness Assessment Diagrams and Synoptic Test Descriptions.

Chapter 6 - FAULT DETECTION AND IMPACT EVALUATION.
Sections - Introduction to Fault Detection and Impact Evaluation.
Operational Fault Directories.
Impact Evaluation.

Chapter 7 - FAULT ISOLATION.
Sections - Introduction to Fault Isolation.
Search Radar Subsystem Fault Isolation.
Tactical Data Subsystem Fault Isolation.
Electronic Warfare Subsystem Fault Isolation.
Gun Weapon Subsystem Fault Isolation.
Underwater Weapon Subsystem Fault Isolation.
Missile Weapon Subsystem Fault Isolation.
LAMPS Subsystem Fault Isolation.
External Communication Subsystem Fault Isolation.
External Navigation Subsystem Fault Isolation.
Support Subsystem Fault Isolation.
Test Fault Directories.
Intersubsystem Interface Data.
Bibliography.

VOLUME 4 - CAPABILITIES AND LIMITATIONS.

Chapter 8 - SHIP MISSION CAPABILITIES AND LIMITATIONS.
Sections - Introduction to the Combat System Capabilities and Limitations.
Threat Categories.
Operating Environments.
Combat System Operational Modes.

Source: https://assist.dla.mil -- Downloaded: 20260-19T23:02Z
Check the source to verify that this is the current version before use.

Ship Characteristics.
AAW Mission.
ASW Mission.
SUW Mission.
Shore Mission.
Tactical Data Subsystem.
Reduced Capability Configurations.
Summary of Combat System Capabilities and Limitations.
Bibliography.

Chapter 9 - DETECTION CAPABILITIES AND LIMITATIONS.
Sections - Introduction to Combat System Detection Capabilities and Limitations.
Search Radar Subsystem Detection.
IFF Detection.
Electronic Warfare Subsystem Support Detection.
Gun Weapon Subsystem Radar Detection.
Optical Detection.
Underwater Weapon Subsystem Sonar Detection.
Missile Weapon Subsystem Radar Detection.
LAMPS Detection.
Tactical Data Subsystem Remote Detection.
Summary of Combat System Detection Capabilities and Limitations.
Bibliography.

Chapter 10 - THREAT PROCESSING CAPABILITIES AND LIMITATIONS.
Sections - Introduction to Combat System Threat Processing Capabilities and Limitations.
Accuracy of Entry Data.
Tactical Data Subsystem Tracking and Correlation.
Weapon Direction System Entry and Tracking.
Identification and Classification.
Summary of Combat System Threat Processing Capabilities and Limitations.
Bibliography.

Chapter 11 - THREAT EVALUATION AND THREAT-TO-WEAPON PAIRING CAPABILITIES AND LIMITATIONS.
Sections - Introduction to Combat System Threat Evaluation and Threat-to-Weapon Pairing Capabilities and Limitations.
Tactical Data Subsystem Threat Evaluation.
Weapon Direction System Threat Evaluation.
Electronics Warfare Threat Evaluation.
Weapon Subsystem Threat Evaluation.
Threat-to-Weapon Pairing.
Summary of Combat System Threat Evaluation and Threat-to-Weapon Pairing Capabilities and Limitations.
Bibliography.

Chapter 12 - ENGAGEMENT AND ENGAGEMENT ASSESSMENT.
Sections - Introduction to Engagement and Engagement Assessment.
Electronic Warfare ECM Engagement and Engagement Assessment.
Gun Weapon Subsystem Engagement and Engagement Assessment.
Underwater Weapon Subsystem Engagement and Engagement Assessment.
Missile Weapon Subsystem Engagement and Engagement Assessment.
Controlled Aircraft Engagement and Engagement Assessment.
LAMPS ASW/ECM Engagement and Engagement Assessment.
Search Radar Subsystem Engagement Assessment.
Link 11/14/4A Engagement Assessment.
Optical Sensors Engagement Assessment.
Tactical Data Subsystem Engagement Assessment.
Summary of Combat System Engagement and Engagement Assessment Capabilities and Limitations.
Bibliography.

Chapter 13 - EMISSION CONTROL.
Sections - Introduction to EMCON.
Cover and Deception.
EMCON Plans.
Radiation Characteristics.
EMCON Monitoring.
Summary of Combat System EMCON Capabilities and Limitations.
Bibliography.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

Chapter 14 - COMMUNICATIONS CAPABILITIES AND LIMITATIONS.
Sections - Introduction to Communications Capabilities and Limitations.
Internal Communications.
External Communications.
Bibliography.

Chapter 15 - NAVIGATION CAPABILITIES AND LIMITATIONS.
Sections - Introduction to Navigation Capabilities and Limitations.
Satellite Navigation System.
OMEGA Navigation System.
Radio Direction Finders.
TACAN.
Bibliography.

Appendix A - SUPPLEMENTARY DATA.
Glossary

3.4 Volume 1 - combat system description. Volume 1 shall provide information for combat system personnel that summarizes ship physical and functional characteristics, missions, capabilities, and the interrelationships between combat system subsystems. This volume shall provide technical personnel with a description and analysis of the principal functions of each subsystem, the interfaces between subsystems, and the dependency of the combat system mission on the subsystems. Subsystem descriptive material shall not be unnecessarily duplicated. A reference to the appropriate subsystem technical manual shall be provided for each subsystem description.

3.4.1 Chapter 1 - introduction. This chapter shall introduce the CSTOM and provide a physical and functional description of the ship. It shall provide an overview of the ship mission and capabilities, and shall identify the subsystem equipment comprising the combat system, and the physical location of selected subsystem equipment. A general description shall be provided for each of the subsystems, together with the functional role each has in supporting the ship combat mission.

3.4.1.1 Introduction to CSTOM. This section shall briefly describe the purpose, scope, and content of CSTOM. This section shall also include the following paragraph:

"Ships, training activities, supply points, depots, Naval shipyards, and supervisors of shipbuilding are requested to arrange for the maximum practical use and evaluation of NAVSEA technical manuals. All errors, omissions, discrepancies, and suggestions for improvements to NAVSEA technical manuals shall be reported to Naval Sea Data Support Activity (NSDSA), Naval Ship Weapon Systems Engineering Station (Code 5740), Port Hueneme, CA 93043 on NAVSEA Technical Manual Deficiency/Evaluation Report (TMDER), NAVSEA Form 5600/2. To facilitate such reporting, three copies of NAVSEA Form 5600/2 are included at the end of each unclassified bound part of this manual. All feedback comments will be thoroughly investigated and originators will be advised of action resulting therefrom. Extra copies of NAVSEA Form 5600/2 may be requisitioned from the Naval Publications and Forms Center, Philadelphia, PA 19120."

3.4.1.2 Ship and combat system description. This section shall provide a description of ship missions, capabilities, and physical characteristics; include a level 1 description of all ship combat subsystems and related armament; viz, search radar subsystem, tactical data subsystem, electronic warfare subsystem, gun weapon subsystem, underwater weapon subsystem, missile weapon subsystem, light-airborne multi-purpose system (LAMPS) subsystem, external navigation subsystem, external communication subsystem and support subsystem. The primary ship combat missions of antiair warfare, antisubmarine warfare, surface and shore bombardment shall be described. An introduction to combat system operation in the manual, semiautomatic and automatic, and casualty modes shall be provided. Major combat system functions of detection and entry, tracking and identification, threat evaluation and threat-to-weapon pairing, and engagement and engagement assessment shall be described. A level 1 block diagram, similar to figure 12, shall be provided. The effective engagement envelope for each weapon system shall be illustrated.

3.4.1.3 Location of combat system components. This section shall describe the ship combat system compartments and areas. A ship profile drawing, similar to figure 13, shall pictorially identify external subsystem components and area locations. A three-dimensional plan view, similar to figure 14, shall depict strategically located important combat system compartments and areas. A two-dimensional room arrangement drawing, similar to figure 15, shall identify and locate major subsystem equipment found in various compartments and areas. Except for CSTOMs prepared for a single ship configuration, these drawings shall be typical and shall represent the lead ship of a class. The preparing activity shall recommend the compartments to be pictorially represented in the manual outline. Figure references shall be provided and areas shall be shaded, similar to figures 13 and 14, to identify compartments and areas for which equipment room arrangement diagrams are provided.

Source: https://assist.dla.mil -- Downloaded: 202640-19T23:02Z
Check the source to verify that this is the current version before use.

3.4.2 Chapter 2 - combat subsystem descriptions. Chapter 2 shall provide an abridged description of the physical and functional characteristics and, as applicable, a brief description and flow diagrams of operational programs for each subsystem. A pictorial of the combat system, similar to figure 16, shall be provided. Related text shall describe subsystem and equipment functions; functions shall be described to the depth necessary to enable the reader to understand each subsystem role in the combat system. The physical and functional characteristics of the combat system subsystems shall be described to levels 1 and 2. Pictorial art, similar to figures 17 and 18, shall be used to depict the physical characteristics and define the functional interface between subsystem elements, and between equipment of the subsystem element. A separate table for each subsystem shall list major equipment. The tables shall include colloquial name or abbreviation, nomenclature, quantity, and location. A bibliography for each subsystem shall list all source documentation.

3.4.2.1 Introduction to the combat system. This section shall introduce the subsystems that comprise the combat system and will provide the user with an overview of the chapter 2 content.

3.4.2.2 Search radar subsystem. This section shall describe and illustrate components of the sea and air search radar subsystem, including the radar recognition, and radar distribution systems; all major components shall be described.

3.4.2.3 Tactical data subsystem (see 6.3.14). This section shall describe and illustrate all components of the tactical data subsystem. Areas of interface, other than with the ship support system, shall be addressed to the extent necessary for user understanding of tactical data subsystem role in the ship combat system. Block diagrams, similar to figure 19, shall illustrate the tactical data subsystem and weapon subsystem computer operational module interfaces. When applicable, tactical data subsystem equipment for monitoring subsystem performance, and for testing combat system interfaces shall be described. Identification of operating stations for each mode configuration shall be provided.

3.4.2.4 Electronic warfare subsystem. This section shall describe and illustrate all components of the electronic warfare subsystem, including electronic support measures, electronic countermeasures, and ancillary equipment.

3.4.2.5 Gun weapon subsystem. This section shall describe and illustrate all components of the gun weapon subsystem, including gun ordnance and ancillary equipment.

3.4.2.6 Underwater weapon subsystem. This section shall describe and illustrate all components of the underwater weapon subsystem, including antisubmarine ordnance and ancillary equipment.

3.4.2.7 Missile weapon subsystem. This section shall describe and illustrate all components of the missile weapon subsystem, including operational and training missile ordnance and ancillary equipment.

3.4.2.8 Light-airborne multi-purpose system (LAMPS) subsystem. This section shall describe and illustrate all components of the LAMPS subsystem, including all airborne LAMPS equipment, ordnance, and shipboard LAMPS equipment.

3.4.2.9 External communication subsystem. This section shall describe and illustrate all external communications systems and related components. Coverage shall be provided for, but not limited to, crypto devices, secure voice and non-secure voice links, satellite communication, teletype circuits, underwater radio (UHF and VHF), tactical data link, and ancillary equipment.

3.4.2.10 External navigation subsystem. This section shall describe and illustrate all components of the navigation subsystem. Coverage shall be provided for, but not limited to, satellite navigation, OMEGA navigation, radio direction finder, tactical air navigation (TACAN), and ancillary equipment.

3.4.2.11 Support subsystem. This section shall describe and illustrate the support equipment associated with the combat system. Simplified block diagrams, similar to figures 20 and 21, shall depict interfaces between the ship support subsystems and the combat system. As a minimum, coverage for the following support subsystems shall be provided:

(a) Air and nitrogen pressurization systems.
(b) Liquid cooling and heating systems.
(c) Interior-ship communications.
(d) Air conditioning.
(e) Bridge ship-control system.

Source: https://assist.dla.mil -- Downloaded: 202640-19T23:02Z
Check the source to verify that this is the current version before use.

(f) Ship parameters.
(g) Power distribution.
(h) Test equipment repair and calibration facility, when applicable.

3.4.3 Chapter 3 - combat system operational description. Chapter 3 shall describe major combat system functions used in all modes of combat system operations. Include combat system functions of detection and entry, tracking and identification, threat evaluation and threat-to-weapon pairing, and engagement and engagement assessment. The major functions of the combat system shall be defined and summarized, and each function shall be described in detail in subsequent sections. Level 1 combat system configurations, including normal, alternate, and backup features available through combat system switching, that relate to the major combat system functions shall be described. A level 2 function flow pictorial, similar to figure 22, shall be used to illustrate each combat system function or principal subfunction described in chapter 3. The functional flow pictorial drawings in chapter 3 shall be divided into segments that are compatible for the production of vugraphs. Each segment shall have an image area of 7-1/2-inches (vertical) by 8-inches (horizontal). In order for these foldout pictorials to also be microfilm compatible (see 3.1.2.5), there shall be two segments per 17-inch microfiche image area.

3.4.3.1 Introduction to combat system functions. This section shall introduce and define combat system functions, and subsystem functions pertinent to the modes of combat system operation, and explain data transmission between subsystems. The general description shall prepare the reader for more detailed functional descriptions in subsequent sections.

3.4.3.2 Detection and entry. This section shall describe the initial track, detection and entry functions, and functional flow between subsystems performed for all radar, sonar, visual, or remote contacts to include air, surface, or subsurface threats entered manually or automatically into the combat system.

3.4.3.3 Tracking and identification. This section shall describe tactical data subsystem tracking and identification functions and functional flow between subsystems occuring for any track entered into the combat system, either locally by own-ship sensors or remotely by any participating unit within the operational area of the fleet.

3.4.3.4 Threat evaluation and threat-to-weapon pairing. This section shall describe threat evaluation and threat-to-weapon pairing function, including weapon assignment. Data flow between subsystems to accomplish threat evaluation and weapon pairing for tracks considered hostile to own ship and to fleet security shall be described.

3.4.3.5 Engagement and engagement assessment. This section shall describe target engagement and engagement assessment functions, and functional flow between subsystems occuring for targets assigned to the electronics warfare subsystem, gun weapon subsystem, underwater weapon subsystem, missile weapon subsystem, the interceptor aircraft, or LAMPS.

3.5 Volume 2 - operational sequences. Volume 2 shall present the sequential functional operations performed by combat system personnel and equipment during all major combat missions. Volume 2 shall provide coverage of total combat system operational sequences for antiair warfare, antisubmarine warfare, surface warfare, and shore operations. Volume 2 shall assume readiness condition I manning. Information provided shall address what is done and not how it is done. The user shall be referred to subsystem technical manuals for detailed step-by-step equipment operations. The operations required of each subsystem of the combat system shall be presented in relation to their contribution to the combat system functions of detection and entry, tracking and identification, threat evaluation and threat-to-weapon pairing, and engagement and engagement assessment. An operational mode network similar to figure 23, will provide the user with an overview of the combat system functional requirements. The operational mode network (see 6.3.10) shall reference combat system operational sequence network (see 6.3.11) drawings, similar to figure 24, for explanation of the operational sequences required for each function identified on the operational mode network. The operational mode network and each combat system operational sequence network drawing shall be supported by a textual description.

3.5.1 Chapter 4 - operational sequences. Chapter 4 shall consist of an operational mode network (OMN), combat system operational sequence network (OSN) drawings, and supporting text describing the sequential functional operations of the combat system in conduct of ship combat missions.

3.5.1.1 Purpose and use of the operational mode network and operational sequence network. This section shall describe the purpose and use of the drawings contained in chapter 4 to depict combat system operational sequences. The relationship of the OMN and the combat system OSN drawings to each other and to other volumes of the CSTOM shall be

Source: https://assist.dla.mil -- Downloaded: 202640-19T23:02Z
Check the source to verify that this is the currentversion before use.

explained. The OMN diagram shall define combat system operating sequences. The OMN (figure 25), an overall summary and reference diagram, shall provide for entry into more detailed operational functions depicted on second-level OSN drawings, similar to figure 24. The OMN and OSNs shall be prepared on the basis of the major combat system operational descriptions specified in 3.4.3. OMN drawings shall identify all major operational functions. Entry into the second-level OSN diagrams shall be by figure reference from the first-level OMN. Each OSN shall subsequently cover a specific operational function. Where operational functions refer to an operator action, reference shall be made to the appropriate subsystem manual or system operator manual. The OMN drawing shall be reverse-side printed with an apron to permit opening the OMN to the left while viewing individual OSN drawings to the right. Each OSN shall specify the initial setup conditions and configuration requirements upon which the OSN is predicated.

3.5.1.2 Detection and entry. This section shall describe the combat system detection and entry operational sequence for the initial response to any air, surface, or subsurface detection while operating in condition I (see 6.3.7). Narrative description in this section shall be supported by the OMN, and OSNs for each combat system subsystem, as appropriate.

3.5.1.3 Tracking and identification. This section shall describe the tracking and identification operational sequence and procedures for the initial response to any normally expected threat that may occur when operating in condition I. Narrative description in this section shall be supported by the OMN, and OSNs for each combat system subsystem as appropriate.

3.5.1.4 Threat evaluation and threat-to-weapon pairing. This section shall describe the threat evaluation and threat-to-weapon pairing operational sequence for the initial response to any normally expected threat that may occur when operating in condition I. Include effects of the doctrine management function as it provides for manual insertion of resource-management doctrine into the combat system to adjust and select automated system reactions to threats and the tactical environments. Narrative description in this section shall be supported by the OMN, and OSNs for each combat system subsystem as appropriate.

3.5.1.5 Electronic warfare engagement and engagement assessment. This section shall describe the electronic warfare engagement operational sequence and procedures for the initial response to any normally expected threat that may occur when operating in condition I. Narrative description in this section shall be supported by the OMN, and OSNs for each combat system subsystem as appropriate.

3.5.1.6 Gun engagement and engagement assessment. This section shall describe the gun engagement operational sequence and procedures for the initial response to any normally expected threat that may occur when operating in condition I. Narrative description in this section shall be supported by the OMN, and OSNs for each combat system subsystem as appropriate.

3.5.1.7 Underwater engagement and engagement assessment. This section shall describe the underwater engagement operational sequence and procedures for the initial response to any normally expected threat that may occur when operating in condition I. Narrative description in this section shall be supported by the OMN, and OSNs for each combat system subsystem as appropriate. OSNs shall be provided, as necessary, to depict torpedo countermeasure and torpedo evasion operational sequences.

3.5.1.8 Missile engagement and engagement assessment. This section shall describe the missile engagement operational sequence and procedures for the initial response to any normally expected threat that may occur when operating in condition I. Narrative description in this section shall be supported by the OMN, and OSNs for each combat system subsystem as appropriate.

3.5.1.9 Controlled aircraft engagement and engagement assessment. This section shall describe the controlled aircraft engagement operational sequence and procedures for the initial response to any normally expected threat that may occur when operating in condition I. Narrative description in this section shall be supported by the OMN, and OSNs for each combat system subsystem as appropriate.

3.5.1.10 LAMPS engagement and engagement assessment. This section shall describe the LAMPS engagement operational sequence and procedures for the initial response to any normally expected threat that may occur when operating in condition I. Narrative description in this section shall be supported by an OMN, and OSNs for each combat system subsystem as appropriate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

3.5.1.11 Reduced capability conditions. This section shall describe the operational sequence and procedure for continued operations under reduced capabilities based on selected casualties that may occur when operating in condition I. Casualties selected shall be those that impact the overall operation of the combat system, such as those within the TDS, i.e., a central processor or WDS rather than casualties that impact only one of the engagement subsystems, e.g., missiles, guns, etc. Narrative description in this section shall be supported by the OMN, and OSNs for each combat system-subsystem as appropriate.

3.5.1.12 Condition III considerations. The last section of chapter 4 shall provide an explanation of the effect of reduced combat system manning, as would be experienced in readiness condition III (see 6.3.8), on combat system operational sequences. This section shall provide combat system manning requirements for various readiness conditions during combat system operational sequences for ASW, AAW, SUW, and shore missions.

3.6 Volume 3 - combat system readiness. Volume 3 shall provide guidance to the ship electronic readiness team (SERT) (see 6.3.13) for evaluating and improving combat system material and operational readiness. Volume 3 shall primarily address organizational level maintenance within the capability of ship senior maintenance personnel.

3.6.1 Chapter 5 - readiness assessment (see 6.3.12). Chapter 5 shall describe the principles and objectives of combat system readiness assessment using the SERT concept, and shall define the means for determining system readiness within ship organizational constraints. Details of the planned maintenance system (PMS), with reference to the PMS record shall be discussed.

3.6.1.1 Introduction to readiness assessment. This section shall contain an overview of the combat system readiness assessment, fault detection, fault isolation, and impact evaluation tools that shall be provided in volume 3. A drawing similar to figure 25 shall be used to introduce the content of chapters 5, 6, and 7, and to explain the relationship between the chapters.

3.6.1.2 Factors contributing to readiness assessment. The section shall define the principles and objectives of readiness assessment. The description shall cover the scope and organization of readiness assessment, and identify major factors and processes contributing to combat system readiness. A drawing similar to figure 26, shall illustrate the dependency of readiness assessment on fault detection, fault isolation, and impact evaluation.

3.6.1.3 Ship electronic readiness team. This section shall explain SERT organization and responsibilities. Emphasis shall be given to SERT activities such as test, training, materiel, and personnel management; preventive and corrective maintenance management; and status reporting. Techniques to generate optimum test schedules based on personnel and material assets and the ship's operating schedule will be presented. Recommended forms, similar to figures 27 and 28, for preparing mission summary reports and combat system daily fault reports shall be provided and described.

3.6.1.4 Planned maintenance system concept. This section shall describe the principles and elements of integrated testing, and explain the concepts and documentation available for testing. The major objectives of fault isolation shall be defined, and the various types of fault isolation tools that personnel may select for fault isolation shall be described. Each type of fault isolation tool, i.e., directories, computer programs, fault logic diagrams, etc., shall be sufficiently described to facilitate its understanding and use. A figure, similar to figure 29, shall depict the planned maintenance system. The integration of combat system, subsystem, and equipment documentation for establishing maintenance requirements, test scheduling, fault isolation, and corrective action verification shall be depicted by a drawing similar to figure 30.

3.6.1.5 Training and maintenance support elements. This section shall identify and describe the training and maintenance support elements available to, or resident in, the applicable ship class, and shall evaluate their functions as tools to increase combat system personnel and equipment readiness. Tables similar to figures 31 and 32 shall be developed to:

(a) Identify the subsystems of the combat system challenged by each of the exercises in fleet exercise publications, FXP2 and FXP3.
(b) Identify on a system or equipment level all exercises that challenge the subsystem and identify the major subsystem functions, such as detection, entry, identification and tracking, that are challenged by each exercise.
(c) Identify the training operational programs and procedures resident on the ship; determine the subsystem challenged by each one, and provide a brief description.

Source: https://assist.dla.mil -- Downloaded: 202640-19T23:02Z
Check the source to verify that this is the currentversion before use.

3.6.1.5.1 Ship test equipment repair facility. When the ship configuration includes a ship test equipment repair facility (STERF) (phase A-1, A-2, and C-1, C-2), a list shall be provided of test equipment calibration requirements that show parameters to be calibrated, calibration periodicity, and references to calibration procedures. When the ship configuration does not include a STERF, documentation shall be specified that prescribes calibration requirements and outlines procedures for obtaining off-ship calibration services.

3.6.1.6 Readiness assessment diagrams and synoptic test descriptions. This section shall identify the top level combat system, subsystem, and equipment level tests required to determine combat system readiness for antiair warfare (AAW), antisubmarine warfare (ASW), surface warfare (SUW), and shore missions. A table of tests, with fault isolation pictorial and fault impact evaluation references, similar to figure 33, and a readiness assessment diagram (RAD), similar to figure 34, shall be provided for each combat mission (AAW, ASW, SUW, and shore bombardment). The table and its associated RAD shall be provided for each major combat system function, including detection and entry, tracking and identification, threat evaluation and threat-to-weapon pairing, and engagement and engagement assessment. The RAD shall identify required functional interfaces applicable to the specific mission and major combat system function, and shall provide a reference to the table that lists the highest level test (combat system level, subsystem level, or equipment level) that verifies each function identified. The table shall immediately precede the applicable RAD, so that the table can be referred to while reviewing the RAD. Detailed, comprehensive, user instructions that explain the use of the RAD and its associated table shall be provided. A synoptic description of each combat system, subsystem, and equipment test included on all the tables shall be provided. The test descriptions shall state test objectives, restrictions and specify significant parameters that evaluate readiness.

3.6.2 Chapter 6 - fault detection and impact evaluation. Chapter 6 shall describe methods of detecting materiel and personnel faults encountered during testing and operation and shall provide guidance to the SERT in evaluating the impact of detected faults on combat system readiness. The purpose of this chapter is to support the SERT in carrying out their corrective maintenance management responsibilities of coordinating fault isolation efforts and discerning impact evaluation.

3.6.2.1 Introduction to fault detection and impact evaluation. This section shall explain the use of impact evaluation tables and operational fault directories provided in chapter 6. The relationship of fault detection to readiness assessment shall be stressed. Primary topics shall cover the fault detection role in readiness assessment and testing, fault detection through operation, and impact evaluation.

3.6.2.2 Operational fault directories. This section shall include operational fault directories for faults observed during combat system operation. Operational fault directories shall identify hardware and software fault isolation tools employed with each subsystem to isolate operationally related fault symptoms. Operational fault directories for each mission and major combat system functions shall be provided. Faults occurring as a result of loss or degradation of intersubsystem functions and parameters shall be provided. A narrative shall explain the role of operational fault directories in the combat system maintenance concept. See figure 35 for a typical presentation format.

3.6.2.3 Impact evaluation. This section shall provide SERT with the criteria required to determine the impact of specific faults on combat system capabilities. A table, similar to figure 36, shall list equipment functions, and the impact the loss of these functions will have on specific combat system missions. Primary topics of combat system degradation, trade-off considerations, restoration time, fault isolation constraints, and alternative combat system configurations shall be related.

3.6.3 Chapter 7 - fault isolation. Chapter 7 shall identify and describe fault isolation tools and procedures, and their effective use. Interface data for subsystem-to-subsystem functions shall be provided.

3.6.3.1 Introduction to fault isolation. This section shall introduce and describe a typical fault isolation pictorial and its associated table of interface references. The table of interface references, similar to figure 37, shall be described. The use of a typical fault isolation pictorial, similar to figure 38, for diagnosing combat system faults shall be explained in detail. When assembled in CSTOM, the table of interface references shall immediately precede its associated test and fault isolation pictorial, so that the tabulation can be referred to while viewing the test and fault isolation pictorial, which will normally be a foldout.

Source: https://assist.dla.mil -- Downloaded: 202640-19T23:02Z
Check the source to verify that this is the current version before use.

3.6.3.2 Search radar subsystem fault isolation. This section shall describe search radar subsystem fault isolation tools and their effective use. Descriptions shall encompass such factors as when and under what circumstances a particular fault isolation tool should be used, and the degree of reduced capabilities incurred as a result of using a particular fault isolation tool. More detailed aspects of search radar subsystem fault isolation shall be addressed by referencing existing fault logic diagrams contained in search radar subsystem technical manuals. A synoptic description of fault isolation techniques presented in the highest level subsystem technical manual shall be provided.

3.6.3.3 Additional sections. Additional sections shall be added to describe fault isolation to the same level as required by 3.6.3.2 for the following subsystems, as appropriate:

(a) Tactical data subsystem.
(b) Electronic warfare subsystem.
(c) Missile weapon subsystem.
(d) Gun weapon subsystem.
(e) Underwater weapon subsystem.
(f) Light-airborne multi-purpose subsystem.
(g) External communication subsystem.
(h) External navigation subsystem.

3.6.3.4 Support subsystem fault isolation. This section shall address fault isolation techniques for the various systems that comprise the support subsystem. Descriptions shall be to level 2, and involve only basic methods of fault isolation. Fault isolation material to support this section shall include system schematics similar to figures 39 and 40; a figure index, similar to figure 41, showing the applicability of support systems to individual subsystems of the combat system; tables of fault symptoms and probable causes, similar to figure 42; and a table of audiovisual alarms similar to figure 43. The table of audiovisual alarms, in addition to specifying alarm and alarm sensor locations, shall prescribe action to be taken when the alarms are activated. The action, in most cases, will involve preliminary checks to verify and assess the abnormal condition, and a reference to a system publication for detailed fault isolation procedures. The primary topics of this section shall support, as applicable, fault isolation of the following support subsystem elements:

(a) Central dry air.
(b) High pressure air.
(c) High pressure nitrogen.
(d) Electronic salt water liquid coolant.
(e) Chilled water liquid coolant.
(f) Air conditioning.
(g) Steam heating.
(h) Salt water firemain.
(i) Anti-icing.
(j) Power distribution.
(k) Ship parameters.

When detailed procedures are required, reference shall be made to appropriate documentation included in the bibliography.

3.6.3.5 Test fault directories. This section shall provide test fault directories similar to figure 44. The test fault directories shall list each interface function by equipment and shall identify the test that verifies the function. There shall be a test fault directory for each subsystem of the combat system except the support subsystem. The purpose and content of test fault directories, and the limitations of the test fault directories caused by non-correlation between PMS tests and referenced subsystem and equipment documentation, shall be addressed.

3.6.3.6 Intersubsystem interface data. This section shall provide subsystem-to-subsystem interface data that serves as a tool to isolate faults associated with intersubsystem functions. All analog and digital intersubsystem interfaces shall be identified, and the following minimum data prepared:

(a) Intersubsystem interface function list. This list shall include all analog and digital functions that interface between the combat system subsystems. A narrative description of each function shall be provided. The description shall specify the origin of the function, its flow path, and its termination. Each digital function shall be keyed by an interface reference number, to a digital intersubsystem interface diagram. Each analog function shall be keyed by an interface reference number, to an analog intersubsystem interface block diagram.

Source: https://assist.dla.mil -- Downloaded: 202640-19T23:02Z
Check the source to verify that this is the current version before use.

(b) Digital intersubsystem interface diagram. One or more digital intersubsystem interface diagrams, and associated digital interface cable lists, shall be provided. A digital interface cable list, similar to figure 45, shall immediately precede a digital intersubsystem interface diagram, similar to figure 46. Each interface on the digital intersubsystem interface diagram shall refer to a digital interface cable list that identifies each interface cable, its associated connectors, the equipment it interconnects and a reference for defining the digital data.

(c) Analog intersubsystem interface diagram. One or more analog intersubsystem interface diagrams, and associated interconnection tabulation sheets shall be provided. The interconnection tabulation sheets, similar to figure 47, shall immediately precede an associated analog intersubsystem interface diagram similar to figure 48.

3.6.3.7 Combat system alinement. This section shall provide methods, tools, and support information necessary to determine when alinement is required. Combat system alinement structure, the dependent nature of various alinements in the structure, and alinement verification shall be defined and discussed. Information shall comprehensively reference subsystem and equipment documentation, as required to identify all relative alinement tolerances and procedures.

3.6.3.8 Bibliography. This section shall provide a list of documents used in preparing chapter 7.

3.7 Volume 4 - capabilities and limitations. Volume 4 shall provide senior ship combat system personnel with a compact presentation of technical data that effectively describes the capabilities and limitations of the combat system to counter expected threats under various environmental conditions and states of readiness. Data shall encompass only technical aspects of the combat system and shall not imply or recommend deployment or specific utilization of the combat system in tactical situations except as may be required to demonstrate system capabilities and designed-in responses to specific threats or other tactical stimuli. Level 1 and 2 descriptions shall be used in volume 4. Chapter 8, a composite description of the overall combat system capabilities and limitations against air, surface, and subsurface threats, shall be covered at level 1. Chapters 9 through 13 shall consist of detailed descriptions of combat system capabilities and limitations and shall be presented at level 2. Volume 4 shall provide combat system personnel with information useful in preparation of ship doctrine, with emphasis on combat system capabilities and limitations; procedural considerations to assist combat decision makers; and documentation directly usable by school commands for combat system officer training. Maximum use shall be made of illustrations, charts, graphs, and matrices to improve understanding. Volume 4 text shall include detailed explanations of all charts, graphs and matrices used to illustrate combat system capabilities and limitations. Capabilities and limitations in the manual, semiautomatic or automatic operation mode shall be described for the principle ship missions: AAW, ASW, SUW, and shore bombardment. Reduced capabilities shall be defined for degraded readiness and environment, and multimission activities.

3.7.1 Chapter 8 - ship mission capabilities and limitations. Chapter 8 shall provide information to assist supervisory personnel in developing ships doctrine and to aid combat system coordinators in performing evaluations before making operational decisions. Chapter 8 shall describe the capabilities and limitations of the combat system that result from integrating the independent capabilities and limitations of the systems and associated threat processes. The capabilities and limitations for each AAW, ASW, SUW, and shore engagement mission shall be included.

3.7.1.1 Introduction to the combat system capabilities and limitations. This section shall provide an overview to chapter 8, describing the purpose, content, and method of presentation.

3.7.1.2 Threat categories. This section shall generically categorize air, surface, and subsurface threat profiles, using available threat data. Threat categories shall be further grouped according to threat-to-mission application. Threat-mission groups shall be the basis upon which combat system capabilities and limitations shall be described throughout this volume.

3.7.1.3 Operating environments. This section shall describe the environmental conditions in which the ship may be required to operate to execute the assigned missions. The conditions shall include weather, visibility, sea state, land background, and electromagnetic emissions. The specified conditions shall be used as bases for the combat system capabilities and limitations described throughout this volume.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

3.7.1.4 Combat system operational modes. This section shall describe the tactical data subsystem manual operational mode basic to most of the material to be presented in this volume. Alternate automatic and semiautomatic operational modes, when applicable, shall also be described. Changes in capabilities and limitations associated with the alternate modes shall be described in this volume.

3.7.1.5 Ship characteristics. This section shall describe the ship characteristics during adverse sea and wind conditions. The capability to maintain speed and course under the weather and sea-state conditions specified in this chapter shall be described. Ship maneuverability shall be described for specified speeds under the same environmental conditions.

3.7.1.6 AAW mission. This section shall describe the combat system capabilities and limitations to encounter, in the primary operating mode, each AAW threat group, in clear and cluttered environments, from the functional aspects of detection and entry, tracking and identification, threat evaluation and weapon pairing, engagement and engagement assessment. The AAW capabilities and limitations in the automatic and semiautomatic modes shall also be described. The capabilities and limitations of intercept aircraft, when available, in supporting the AAW mission shall be addressed.

3.7.1.7 ASW mission. This section shall provide the same information as required by 3.7.1.6, but addressing the ASW mission. The capabilities and limitations during both own-ship and LAMPS engagements shall be addressed.

3.7.1.8 SUW mission. This section shall provide the same information as required by 3.7.1.6, but addressing the SUW mission.

3.7.1.9 Shore mission. This section shall describe the combat system capabilities and limitations to engage shore targets with all applicable fire control systems and ordnance. The range limitations, accuracies, modes of firing and navigational aids and off-ship support requirements shall be addressed.

3.7.1.10 Tactical data subsystem. This section shall describe the capabilities and limitations of the tactical data subsystem, in the principal operating modes, to support own-ship combatant missions, including simultaneous multi-mission engagements, and to support or coordinate task force and aircraft operations. This section shall also describe condition I constraints caused by personnel fatigue, equipment failure due to excessive operation, and access restriction imposed by setting condition Zebra. Reduced capability conditions caused by tactical data subsystem equipment failure shall be addressed.

3.7.1.11 Reduced capability configurations. This section shall describe specific combat system configurations that will reflect the inoperability of selected systems or equipment. Reduced capabilities of the combat system shall be described for each configuration in applicable portions of this volume.

3.7.1.12 Summary of combat system capabilities and limitations. This section shall present a summary of AAW, ASW, SUW, and shore mission capabilities and limitations. The summary shall indicate the influence of all factors considered within this volume.

3.7.1.13 Bibliography. This section shall provide a list of documents used in preparing chapter 8.

3.7.2 Chapter 9 - detection capabilities and limitations. Chapter 9 shall describe combat system capabilities and limitations with regard to threat detection. Normal operation and reduced capabilities in degraded environments shall be described. Illustrations and tables shall be developed to support description of pertinent combat system detection capabilities of search radars, electronic support measures, gun weapon subsystem radars, sonar, missile weapon subsystem radars, optical detection devices, LAMPS, and data links.

3.7.2.1 Introduction to combat system detection capabilities and limitations. This section shall describe the content, purpose, and method of presentation of the material in chapter 9.

3.7.2.2 Search radar subsystem detection. This section shall describe search radar capabilities and limitations relative to threat detection in a clear, ECM, and non-ECM degraded environments. Illustrations and tables, similar to figures 49, 50, and 51 shall depict the detection volume and characteristics of search radars. Description and illustrations covering search radar capabilities relative to threat detection in an ECM environment to include passive and active jamming. Description and illustrations shall also describe search radar detection capabilities in a non-ECM environment degraded by weather, sea clutter, and land mass background.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

3.7.2.2.1 IFF detection. This section shall describe own-ship identification friend or foe (IFF) detection capabilities. The description shall identify range, altitude, and bearing accuracy and limitations, and available modes of operations. The maximum number of friendly tracks the system may process shall be described. Primary topics of operational modes, detection sectors, and track overloads shall be discussed.

3.7.2.3 Electronic warfare subsystem support measures detection. This section shall describe overall detection capabilities of the electronic support measures suite and the individual equipment capabilities during normal operation. Primary topics shall include electronic support measures countermeasures receiving and direction finding set capabilities and limitations.

3.7.2.4 Gun weapon subsystem radar detection. This section shall provide a description of the capabilities and limitations of the gun weapon subsystem radars in performance of initial threat detection without designated target position quantities supplied. Coverage shall include detection of air and surface targets in clear, ECM, and non-ECM degraded environments. Level and type of coverage shall be similar to that in 3.7.2.2.

3.7.2.4.1 Optical detection. This section shall describe hardware and operator capabilities and limitations of optical devices, including the target designation transmitter and television cameras. The detection ranges, field of vision, and the accuracy of the transmitted data shall be included. Primary topics of detection ranges, clear bearing sectors, accuracy (bearing and elevation), and operator proficiency shall be discussed.

3.7.2.5 Underwater weapon subsystem sonar detection. This section shall provide a description of the capabilities and limitations of the sonar relative to threat detection in active and passive modes. Drawings similar to figure 52 shall be supplied to indicate expected detection ranges in a variety of water conditions resulting from optimum and degraded environments.

3.7.2.6 Missile weapon subsystem radar detection. This section shall provide a description of the capabilities and limitations of the missile weapon subsystem radars in performance of initial threat detection without designated target position quantitites supplied. Coverage shall include detection of air and surface targets in clear, ECM, and non-ECM degraded environments. Level and type of coverage shall be similar to that in 3.7.2.2.

3.7.2.7 LAMPS detection. This section shall describe the capabilities and limitations of the LAMPS for detecting air, surface and subsurface targets. The detection capabilities of LAMPS devices including LAMPS search radar, ESM, sonobuoy, and magnetic anomoly detector (MAD) shall be described. The constraints on normal ASW operations while employing one or more LAMPS detection devices shall be described.

3.7.2.8 Tactical data subsystem remote detection. This section shall describe remote combat system detection capabilities for link 11, link 14, and link 4A. Limitations of the data links as a function of altitude, range, weather, ECM, and sea clutter shall be described. The accuracy of received data and the data update rate shall also be addressed.

3.7.2.9 Summary of combat system detection capabilities and limitations. This section shall provide, in a figure similar to figure 53, a summary of detection capabilities and limitations for supporting each of the AAW, ASW, SUW and shore missions. Primary topics shall describe the impact of detection on AAW, ASW, SUW, and shore mission success.

3.7.2.10 Bibliography. This section shall provide a list of documents used in preparing chapter 9.

3.7.3 Chapter 10 - threat processing capabilities and limitations. Chapter 10 shall discuss the capabilities and limitations of the combat system with regard to target entry, tracking, identification, and classification. Limitations on position accuracy due to source and induced errors in track bearing, range, and altitude shall be described. The capabilities and limitations of the combat system to accomplish threat identification and classification in relation to specific threat categories shall also be described.

3.7.3.1 Introduction to combat system threat processing capabilities and limitations. This section shall describe the purpose, content, and method of presentation of material in chapter 10.

3.7.3.2 Accuracy of entry data. This section shall describe the data sources identified in chapter 9 according to the accuracy and the conditions under which entries from one source are more accurate than those from another. The limitations imposed on the subsystems when using incomplete or unreliable data shall be discussed. Threat entries shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

described in terms of software configuration when applicable; for example, automatic mode versus manual mode. The accuracies of radar data, optical data, sonar data, ESM data, IFF data and tactical data system data links shall be provided in a table similar to figure 54.

3.7.3.3 Tactical data subsystem tracking and correlation. This section shall address the maximum number of tracks capable of being simultaneously processed by the tactical data system. Operator problems shall be addressed relative to radar distribution failures and console failures which require one operator to search sectors normally searched by two or more operators. The maximum and minimum processing time needed to establish a firm track under the manual and automatic modes of operation shall be described. The comparative effect of tracking accuracies on engagements by the different weapon subsystems shall be addressed. The reduced tracking and correlation capabilities resulting from tactical data subsystem equipment malfunctions and software configurations shall be described. Primary topics shall include track capacity, manual and automatic tracking and correlation rates, tracking accuracy, reduction of hardware and software capabilities caused by equipment casualties, and effect of degraded environment on tracking accuracy. A table similar to figure 55 shall specify tactical data subsystem track capacities.

3.7.3.4 Weapon direction system entry and tracking. This section shall describe the same basic subject matter as that specified in 3.7.3.3. Additionally, the effect of weapon direction system casualties shall be discussed. Primary topics shall include maximum number of tracks, tracking accuracy in clear environment and degraded environment, and capability reductions resulting from equipment casualties.

3.7.3.5 Identification and classification. This section shall describe the merits of various identification sources. These sources shall include IFF and selective identification feature (SIF), intelligence reports, ESM receivers, ordered maneuver responses, and operational orders. Typical time spans required for identification and classification using each of the identifying means shall be tabulated. Primary topics shall include automatic threat identification, manual threat identification, ESM threat identification (when not a part of manual or automatic threat identification), and sonar threat identification.

3.7.3.6 Summary of combat system threat processing capabilities and limitations. This section shall summarize the combat system threat processing capabilities and limitations for supporting AAW, ASW, SUW and shore missions. Each primary topic shall describe the impact threat processing has on mission success.

~~3.7.3.7 Bibliography. This section shall provide a list of documents used in preparin chapter 10.~~

3.7.4 Chapter 11 - threat evaluation and threat-to-weapon pairing capabilities and limitations. Chapter 11 shall describe the combat system threat evaluation and threat-to-weapon pairing capabilities and limitations during normal and reduced capability operations. All factors involved in automatic and manual threat evaluation shall be described, including threat position relative to the defended zone; threat identification; and threat course, speed, and altitude.

3.7.4.1 Introduction to combat system threat evaluation and threat-to-weapon pairing capabilities and limitations. This section shall describe the purpose, content, and method of presentation of the material in chapter 11.

3.7.4.2 Tactical data subsystem threat evaluation. This section shall describe the capabilities and limitations of the tactical data subsystem to perform both manual and automatic threat evaluation. The capabilities and limitations inherent in the methods and algorithms used for tactical data subsystem threat evaluation shall be described. The impact of a computer failure on tactical data subsystem threat evaluations shall be described. Primary topics shall include threat value and relative threat calculations, threat evaluation displays, and reduced capabilities.

3.7.4.3 Weapon direction system threat evaluation. This section shall describe the weapon direction subsystem capabilities and limitations for threat evaluation while operating under normal and reduced capabilities including evaluation of threats detected by the target designation transmitter. The display of threat parameters relative to the accuracy achieved and the proficiency of operators in evaluating these displays shall be described. Primary topics of display, operator judgement (experience and training), and reduced capability shall be addressed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

3.7.4.4 Electronics warfare threat evaluation. This section shall describe threat evaluation capabilities and limitations of the ESM suite. Threat evaluation performed by the EW supervisor shall be addressed with regard to available display parameters under normal and reduced capabilities. Primary topics shall include types of displays, determination of display parameters, and evaluation of display parameters.

3.7.4.5 Weapon subsystem threat evaluation. This section shall describe threat evaluation capabilities and limitations of the gun weapon subsystem, underwater weapon subsystem, and missile weapon subsystem. Display capabilities relative to threat parameters, the number of threats that can be processed, and the limitations imposed by operator decisions shall be described. Primary topics of display (normal and reduced capability), operator judgement, and effects of environmental considerations shall be addressed.

3.7.4.6 Threat-to-weapon pairing. This section shall address combat system facilities for threat-to-weapon pairing, which is the process by which a threat is assigned for engagement by a particular weapon subsystem. This section shall describe the combat system manual, semi-automatic, and automatic capabilities for threat-to-weapon pairing. These capabilities shall include subsystem reaction times, engagement envelopes, non-firing and unclear zones for applicable weapons subsystems. The combat system displays and computer recommendations available to operators, and the limitations associated with these facilities shall be addressed. Primary topics of manual threat-to-weapon pairing, semi-automatic threat-to-weapon pairing, automatic threat-to-weapon pairing, and threat-to-weapon pairing under casualty conditions shall be addressed.

3.7.4.7 Summary of combat system threat evaluation and threat-to-weapon pairing capabilities and limitations. This section shall provide a summary of threat evaluation and threat-to-weapon pairing capabilities and limitations supporting the AAW, ASW, SUW and shore mission. Included under each primary topic shall be a description of the impact threat evaluation and threat-to-weapon pairing has on mission success.

3.7.4.8 Bibliography. This section shall provide a list of documents used in preparing chapter 11.

3.7.5 Chapter 12 - engagement and engagement assessment. Chapter 12 shall describe combat system capabilities and limitations with regard to threat engagement and engagement assessment (threat destruction or threat deception). The combat system capabilities against air, surface, subsurface, and shore threats shall be assessed by examining capabilities of weapon subsystems against specific threat categories. Illustrations shall be provided to depict reaction time, trial intercept displays and engagement envelopes against specific threat categories by each applicable weapon subsystem.

3.7.5.1 Introduction to engagement and engagement assessment. This section shall describe the purpose, content, and method of presentation of material in chapter 12.

3.7.5.2 Electronic warfare ECM engagement and engagement assessment. This section shall describe the capabilities and limitations of the ECM subsystem in performing active and passive engagement and engagement assessment. The combat system capabilities for an ECM engagement and engagement assessment shall first be described for a typical, non-degraded condition (AAW mission, manual mode, non-ECM, clean environment, without emission control restrictions or multimission conflicts). The impact of modifying considerations on ECM engagement and engagement assessment shall then be described and include the following:

(a) Alternate type of mission (ASW, SUW, or shore).
(b) Automatic or semiautomatic mode.
(c) Combat system casualties.
(d) Alternate threat categories.
(e) Multimission activity.
(f) Degraded environment (weather and ECM).
(g) EMCON requirements.
(h) Reduced manning level.

3.7.5.3 Additional sections. Additional sections shall be used to describe engagement and engagement assessement capabilities and limitations for other applicable weapon subsystems or subsystem controlled functions. Where applicable, the weapon subsystem engagement capabilities shall include a description of the available subsystem firepower. The additional sections shall describe engagement and engagement assessment to the same level as in 3.7.5.2 for the following:

(a) Gun weapon subsystem engagement and engagement assessment.
(b) Underwater weapon subsystem engagement and engagement assessment (includes tables and illustrations similar to figures 56 through 58).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

(c) Missile weapon subsystem engagement and engagement assessment (includes illustration similar to figure 59).
(d) Controlled aircraft engagement and engagement assessment.
(e) LAMPS ASW and ECM engagement and engagement assessment.
(f) Search radar engagement assessment.
(g) Link 11, 14, and 4A engagement assessment.
(h) Optical sensors engagement assessment.
(i) Tactical data subsystem engagement assessment.

3.7.5.4 Summary of combat system engagement and engagement assessment capabilities and limitations. This section shall provide a summary of engagement and engagement assessment capabilities and limitations that support the AAW, ASW, SUW, and shore missions. Included under each primary topic shall be a description of the impact of engagement and engagement assessment on mission success. This section shall include discussions and illustrations of target cross-range engageability envelope versus target altitude, weapon subsystem clear firing zone limits, target engageability versus target speed, and firepower against a radial air attack.

3.7.5.5 Bibliography. This section shall provide a list of documents used in preparing chapter 12.

3.7.6 Chapter 13 - emission control. Chapter 13 shall describe available ship features that minimize electronic detection by hostile forces, the relative effectiveness of these techniques, and methods for monitoring own-ship emissions. The effect of emission control (EMCON) plans on combat system readiness shall be described with regard to reduced capabilities for detection and engagement during an EMCON condition.

3.7.6.1 Introduction to EMCON. This section shall introduce the topic EMCON and shall describe the purpose, content, and method of presentation of material in chapter 13.

3.7.6.2 Cover and deception. This section shall describe technical capabilities of ship systems and equipment to provide cover and deception. Topics shall include the means available for active and passive cover and deception.

3.7.6.3 EMCON plans. This section shall discuss the existing ship EMCON plans relative to the reduced combat system capabilities that are imposed. The primary topic shall be effects of EMCON plans on combat system readiness. This section shall include tabular presentation of radiating equipment and effects on combat system readiness.

3.7.6.4 Radiation characteristics. This section shall define radiation characteristics of own-ship equipment, such as frequency and power directly related to the risk of having emissions intercepted by the enemy. Emission characteristics of all significant electromagnetic sources as defined by the procuring activity, shall be included in a matrix similar to figure 60, listing the emission characteristics, radiation power levels, transmission frequency, beam pattern, and silence features.

3.7.6.5 EMCON monitoring. This section shall detail ship capabilities for monitoring emitter status (on, off, dummy load). The relative effectiveness of internal and external emission monitoring features as they affect detection avoidance shall be discussed.

3.7.6.6 Summary of combat system EMCON capabilities and limitations. This section shall provide a summary of EMCON capabilities and limitations that support AAW, ASW, SUW and shore missions. Included under each primary topic shall be a description of the impact of EMCON on mission effectiveness.

3.7.6.7 Bibliography. This section shall provide a list of documents used in preparing chapter 13.

3.7.7 Chapter 14 - communications capabilities and limitations. Chapter 14 shall describe capabilities and limitations of ship internal and external communication systems. Primary and alternate communication systems integral to conduct of ship combat missions shall be addressed.

3.7.7.1 Introduction to communications capabilities and limitations. This section shall introduce the topic of communications and shall describe the purpose, content, and method of presentation of material in chapter 14.

3.7.7.2 Interior communications. This section shall describe the capabilities and limitations of all ship interior communication systems and related components. Coverage shall include tactical data subsystem intercommunications, sound powered telephone, announcing circuits, and dial telephone circuit.

Source: https://assist.dla.mil -- Downloaded: 202640-19T23:02Z
Check the source to verify that this is the current version before use.

3.7.7.3 External communications. This section shall describe the capabilities and limitations of all ship external communication systems and related components. Coverage shall include satellite communication, crypto devices, secure voice and non-secure voice link, teletype circuits, underwater radio (UHF and VHF), and tactical data subsystem links 11, 14, and 4A.

3.7.7.4 Bibliography. This section shall provide a list of documents used in preparing chapter 14.

3.7.8 Chapter 15 - navigation capabilities and limitations. Chapter 15 shall describe all ship navigation subsystem capabilities and limitations with respect to conduct of ship combat missions. Coverage shall include satellite navigation, OMEGA, direction finder sets, and TACAN.

3.7.8.1 Introduction to navigation capabilities and limitations. This section shall introduce the topic of navigation and shall describe the purpose, content and method of presentation of material in chapter 15.

3.7.8.2 Satellite navigation system. This section shall describe the capabilities and limitations of the satellite navigation system. Accuracy and reliability of the satellite navigation system in clear and degraded weather, and effects of temperature variations and sunspot activity shall be addressed. Reduced capabilities resulting from equipment casualties shall also be described.

3.7.8.3 OMEGA navigation system. This section shall describe the capabilities and limitations of the OMEGA navigation system. Accuracy and reliability of the OMEGA navigation system in clear and degraded weather, and effects of daily and seasonal ionospheric variations shall be addressed. Reduced capabilities resulting from equipment casualties shall also be described.

3.7.8.4 Radio direction finders. This section shall describe the capabilities and limitations of shipboard radio direction finders (RDF). Accuracy and reliability of RDF readings in clear and degraded weather, and effects of daily and seasonal ionospheric variations and sunspot activity shall be addressed.

3.7.8.5 TACAN. This section shall describe the capabilities and limitations of the shipboard TACAN navigation system. Reduced capabilities for airborne navigation resulting from equipment casualties shall be addressed.

3.7.8.6 Bibliography. This section shall provide a list of documents used in preparing chapter 15.

3.8 Appendix A - supplementary data. When specified in the contract (see 6.2.1), an appendix shall provide test procedures, fault isolation tools, or other maintenance data either unavailable or inadequately covered in subsystem or equipment documentation. An engineering analysis of subsystem documentation shall determine the need for the appendix. The appendix shall be sectionalized by subsystem. Text shall be developed to the same style and format as that provided by the respective subsystem technical manual. Diagrams, schematics, tables and line art developed as part of the appendix shall be prepared to the same format, density and depth as equivalent material contained in the respective subsystem technical manual. The method of promulgating the appendix, whether as an addendum to volume 3 or a separately bound document, shall be specified in the Government contract (see 6.2.1).

3.9 Glossary. A glossary, prepared in accordance with MIL-M-38784, shall be provided. The glossary shall provide a comprehensive list of definitions and names (excluding equipment), functions, and modes of operation. It shall define terms not adequately defined in the text or in a standard dictionary. Uncommon abbreviations and acronyms shall be listed and explained. The glossary shall immediately follow the last chapter of volume 1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

4.1.1 Quality assurance. Except as otherwise specified herein, quality assurance provisions shall be in accordance with the data ordering document (see 6.2.2).

4.2 Validation plan. When specified in the contract (see 6.2.1), the contractor shall document and maintain a validation plan (see 6.2.2) to assure that the CSTOM will accurately and adequately reflect the total combat system in a presentation suitable for the manual intended use and user. Acceptance of the plan shall be obtained from the procuring activity or its designated representative before commencement of any validation effort. The plan shall assure comprehensive control of data collection and development of the manual. The plan shall include arrangements for incorporating corrections and changes resulting from Government reviews and verification.

4.2.1 Validation plan (general). The validation plan shall be in accordance with the following:

(a) Effective management to promote quality assurance shall be clearly described. Positions responsible for assuring performance of quality functions shall have well defined responsibility and authority. The status and adequacy of the program shall be reviewed on a regularly scheduled basis.
(b) Initial planning shall include the development, maintenance, and implementation of methods and procedures to assure quality of manual reviews, and actions to assure compatibility of preparation, inspection, and documentation.
(c) The validation plan shall assure there is a system for inspection, validation, and correction of the manual. Validation shall provide a measure of the overall quality of the CSTOM. When the Government performs an inspection at a contractor's plant, such inspection shall not be used by the contractor as evidence of effective control of quality. When revisions or corrections are required after any inspection, validation, or review, there shall be reinspection, revalidation, or re-review by the contractor, and if necessary, by the Government of all data affected.

4.2.2 Validation plan (specific). The validation plan shall include, but not be limited to the following:

(a) Designation of authority, functions, and duties of personnel responsible for preparation, inspection, and validation of the manual.
(b) Coordination with equipment or system design and production activities to assure the use of the latest technical data and information, including operation and maintenance procedures, drawings, and illustrations for the preparation of the manual.
(c) Establishment of in-process inspection points and development, maintenance, and implementation of inspection standards to control the adequacy during the development stages of the manual.
(d) Identification of Government-furnished material or information required in the performance of validation.
(e) Validation by comparing physical and functional descriptions, and test and fault isolation references in the manual against current Government-approved combat subsystem documentation.
(f) Detailed review of final manuscript to ensure that this material is identical to the authorized, validated, verified, corrected, and accepted preliminary manuals and complies with legibility and reproducibility requirements.
(g) Designation of books, logs, etc., that will be maintained for quality assurance and validation records that are suitable for Government inspection and in-process review.

4.3 Validation. The object of validation is to assure that the contractor has expended the best effort to provide accurate and adequate technical manual content for support of the combat system. A technical manual shall not be considered validated until contractor's engineering review has been completed, and information reflects the combat system configuration specified by the contract (see 6.2.1).

4.4 Rejection criteria. Failure of contractor to meet all contractual requirements, including those specified in this and other referenced specifications, shall be cause for rejection of the deliverable items. The following requirements shall also apply to rejection criteria:

(a) Failure to conform with the Government accepted technical manual outline.
(b) Failure of all information to exactly represent the combat system being described and to be consistent with descriptive terms.

Source: https://assist.dla.mil -- Downloaded: 202640-19T23:02Z
Check the source to verify that this is the current version before use.

(c) Omission of any applicable data elements required by this and other referenced specifications.
(d) Failure to provide adequate evidence of validation.
(e) Failure to include changes or corrections contractually required as conditions of acceptance.
(f) Failure to conform with the terms of the contract.

4.5 Inspection of preparation for delivery. Preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Unless otherwise specified in the contract (see 6.2.1), the packaging, packing, and marking requirements for delivery of CSTOM outlines, manuscript copies, preliminary manuals, original art work, and reproducible copy (basic issue) shall be in accordance with MIL-M-38784. Printed copies shall be prepared for shipment in accordance with MIL-P-38790.

6. NOTES

6.1 Intended use. Technical manuals prepared in accordance with this specification are intended for use by the Navy in operation and maintenance of ship combat system at organizational and management levels. The manuals are also intended for use in personnel training.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of issue; basic, change, revision, or supplement.
(c) Technical manual identification number.
(d) Material to be delivered and delivery schedules.
(e) Security classification (DD Form 254).
(f) Government-furnished information and disposition instructions.
(g) When mechanized reproduction or computer typesetting may be used (see 3.1.2.4.1 and 3.1.2.4.2).
(h) When halftone artwork is used (see 3.1.2.5(b)).
(i) When a technical manual data research analysis is required (see 3.1.3.1).
(j) When a technical manual outline is required (see 3.1.3.2).
(k) When a manuscript copy is required (see 3.1.3.3).
(l) When preliminary manuals are required (see 3.1.3.4).
(m) When changes are required (see 3.1.3.6).
(n) When revisions are required (see 3.1.3.7).
(o) When supplements are required (see 3.1.3.8 and 3.1.4(f)).
(p) When status reports are required (see 3.1.3.9).
(q) When appendix is required (see 3.1.4(d) and 3.8).
(r) When alphabetical index is required (see 3.1.4(e)).
(s) Method of promulgating appendix, if required (see 3.8).
(t) When validation plan is required (see 4.2).
(u) Combat system configuration (see 4.3).
(v) Packaging, packing, and marking requirements, if other than as specified in MIL-M-38784 (see 5.1).

6.2.2 Data requirements. When this specification is used in a procurement which invokes the provision of the "Requirements for Data" of the Armed Services Procurement Regulations (ASPR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the ASPR are not invoked in a procurement, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

Paragraph	Data requirements	Applicable DID	Option
3.1.3.1	Technical manual data research and analysis source data	DI-M-6158	------
3.1.3.2	Technical manual outline book plan	DI-M-2041	MIL-M-38784
3.1.3.3	Technical manual manuscript copy	DI-M-2042	MIL-M-38784
3.1.3.4	Manual, technical, preliminary	DI-M-2043	MIL-M-38784
3.1.3.5	Manual, technical, standard	DI-M-2044	MIL-M-38784
3.1.3.6	Technical, manual permanent change pages	DI-M-2046	MIL-M-38784
3.1.3.7	Manual, technical-update revision	DI-M-2047	MIL-M-38784
	Manual, technical-complete revision	DI-M-2048	MIL-M-38784
3.1.3.8	Manual, technical-supplement	DI-M-2049	MIL-M-38784
3.1.3.9	Technical manual status report	DI-M-2052	------
4.1.1 and 4.2	Technical manual quality assurance data	DI-M-2051	MIL-M-81203

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the procuring/purchasing activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item procured to this specification. This does not apply to specific data which may be required for each procurement regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Definitions.

6.3.1 Bridge-the-gap data. Data that is identified by engineering analysis as being required to complete subsystem documentation. Bridge-the-gap data will be initially included in a CSTOM appendix until such time as subsystem documentation is updated to eliminate the deficiency.

6.3.2 Combat subsystem. A combat subsystem is a grouping of equipment that comprise a selected portion of the combat system; it is a physically limited quantity of associated equipment whose combined functions result in achieving a predetermined objective. Combat subsystems may be functional, organizational, or mission-oriented and are identified as unique systems when not considered as part of a combat system. For example, a missile weapon subsystem may be comprised of a weapons direction system, a guided missile fire control system, and a guided missile launching system. Each of these systems may be independently identified as systems when being discussed as functional or architectural entities.

6.3.3 Combat system. The composite of those elements of a platform whose functions include surveillance, detection, processing, and actively or passively engaging enemy forces. The functions performed by the combat system are accomplished by an integrated assemblage of systems, methods, trained personnel, and documentation having regular interaction and interdependence.

6.3.4 Combat system baseline. The level at which subsystems are interfaced to form the foundation of the combat system. It represents documentation necessary for integrated testing and fault isolation of combat subsystems.

6.3.5 Combat system functions. The combat system functions referred to herein are those major operational functions that include detection and entry, tracking and identification, threat evaluation and threat-to-weapon pairing, engagement and engagement assessment.

6.3.6 Combat system maintenance. Combat system maintenance, as provided in CSTOM, does not entail any peculiar scheduled or unscheduled maintenance procedures. However, combat system maintenance does identify PMS subsystem and equipment tests that verify combat system functions (detection and entry, tracking and identification, threat evaluation and threat-to-weapon pairing, and engagement and engagement assessment) associated with each of the combat system missions (AAW, ASW, SUW, and shore). The results of these tests are used to assess degraded readiness for specific combat system missions, and to aid in combat system intersubsystem fault isolation.

6.3.7 Condition watch I. Defines ship wartime cruising during periods when active combat is imminent, and the complete battle complement at stations is required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

6.3.8 Condition watch III. Defines ship normal war-time cruising during periods when active combat is not imminent, and about one-third of the battle complement at stations is required. This provides optimum balance between combat readiness and personnel efficiency for extended cruising periods.

6.3.9 Level of coverage. A unit of data description that represents depth of content beginning with an overview and progressing to more detailed coverage. Levels of data description 1 through 3 are applicable to CSTOM as defined in 3.3.1.

6.3.10 Operational mode network. A drawing that specifically defines combat system operating sequences and provides the link between subsystem functional descriptions and operational sequence network drawings.

6.3.11 Operational sequence network. Narrative and graphic representations which specifically identify operations that are sequentially performed when operating the combat system in a tactical environment.

6.3.12 Readiness assessment. A method to assess a level of confidence in the combat system to meet an assigned mission, and to evaluate the level of degradation of combat system effectiveness when a fault occurs.

6.3.13 Ship electronic readiness team (SERT). Senior maintenance personnel assigned within ship organization whose responsibilities encompass maintenance personnel training, combat system testing, and fault isolation. This includes systematic conduct of testing to reveal combat system degradation and fault isolation with the goal of maintaining maximum operational readiness.

6.3.14 Tactical data subsystem. As used in this specification, the tactical data subsystem refers to any or all elements of the most basic to the most sophisticated tactical data collecting, correlation, processing, and distribution systems installed in a modern Navy ship. As a part of the combat system, the tactical data subsystem may include the naval tactical data system, command and control system, sensor interface data system, command and decision system, remote data links, or any other similar system installed on particular ships.

6.3.15 Validation. Validation, as applied to CSTOM, is a process the preparing activity uses to test a technical manual for accuracy and adequacy. This is accomplished by verifying that the information conforms to the baseline of the Government furnished source data and that it has been subjected to in-process reviews to ensure compliance with governing specifications for technical content, style and format.

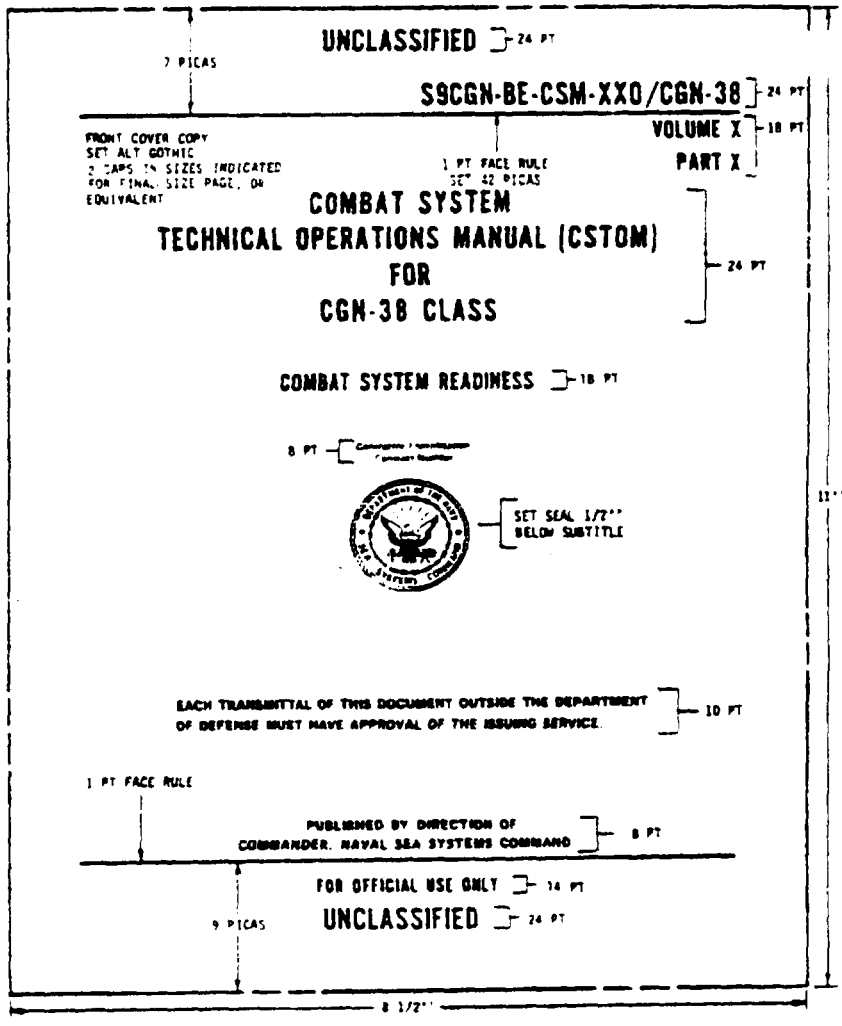
6.3.16 Verification. Verification, as applied to CSTOM, is the process whereby Navy personnel verify the adequacy and accuracy of the subsystem equipment descriptions (location and applicability), the combat system operating sequences, and the references to test and fault isolation procedure, and other maintenance data. A manual shall not be accepted from the preparing activity in final form until all deficiencies found during the verification process have been corrected.

6.4 Supersession data. This specification supersedes XWS-17917A, dated 15 May 1977.

Preparing activity:
Navy - SH
(Project TMSS-N115)

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.2.1 for textual requirements.

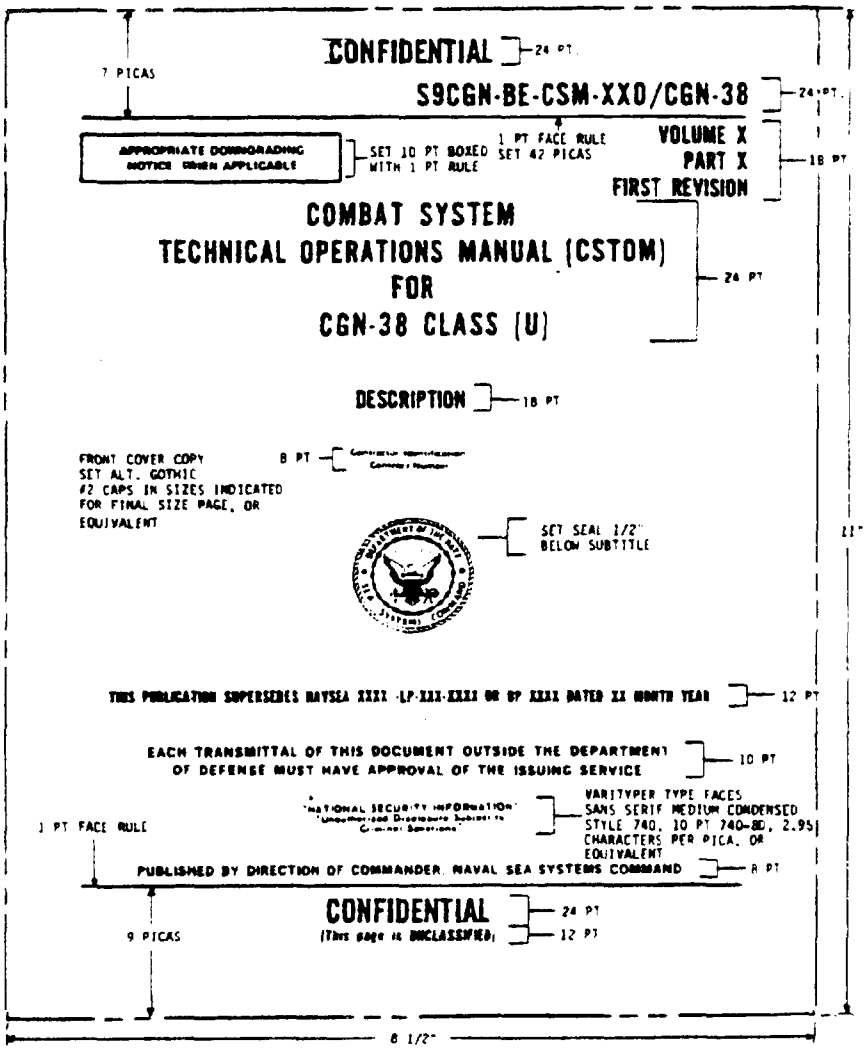


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 1. Sample front cover, unclassified.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.2.1 for textual requirements.

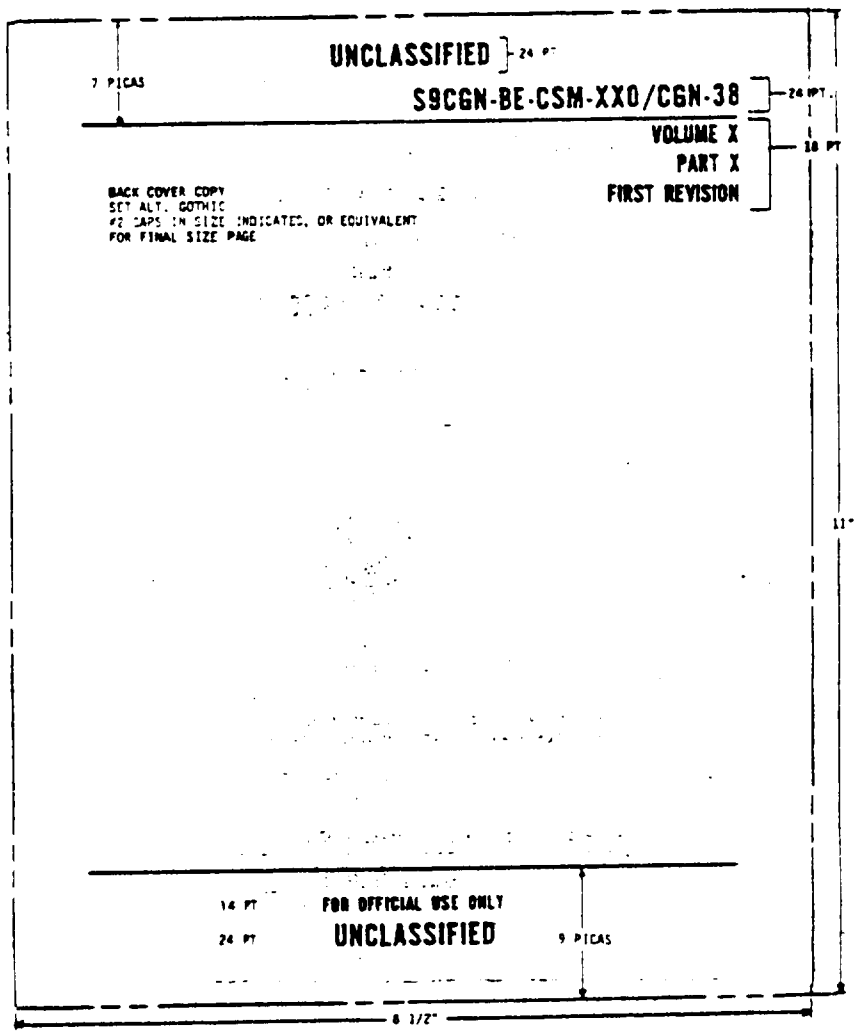


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 2. Sample front cover, classified.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.2.2 for textual requirements.

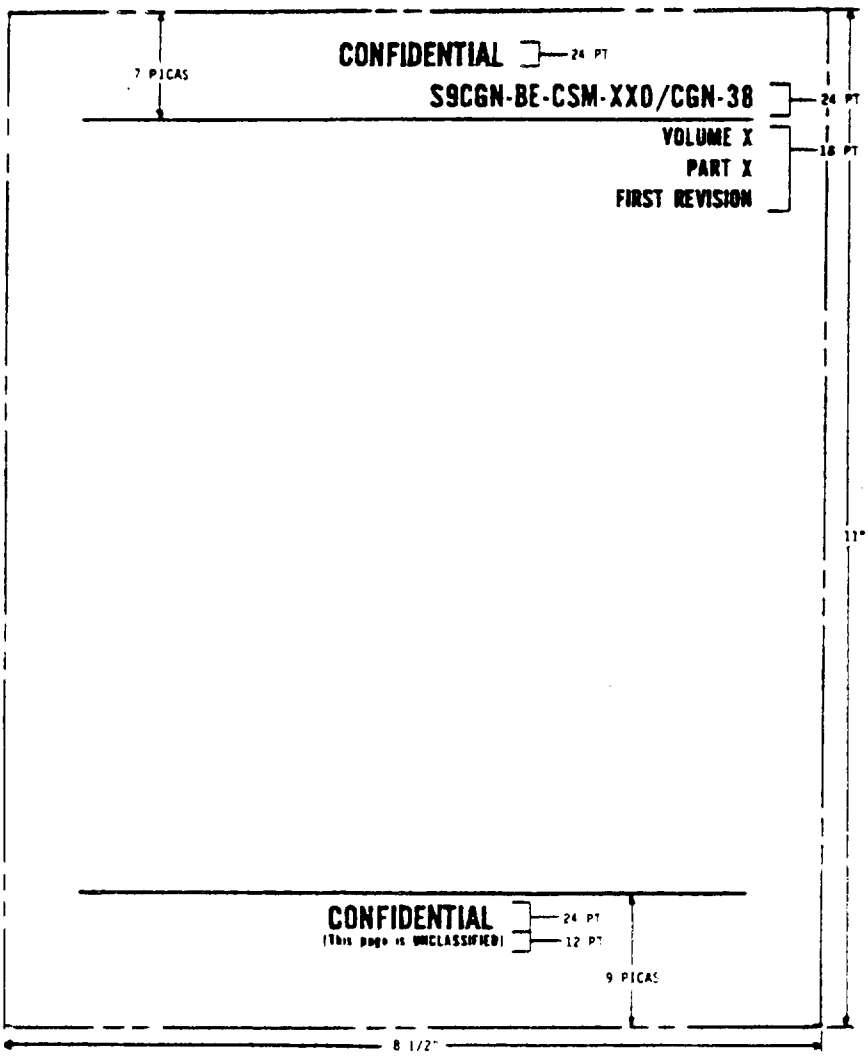


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 3. Sample back cover, unclassified.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.2.2 for textual requirements.

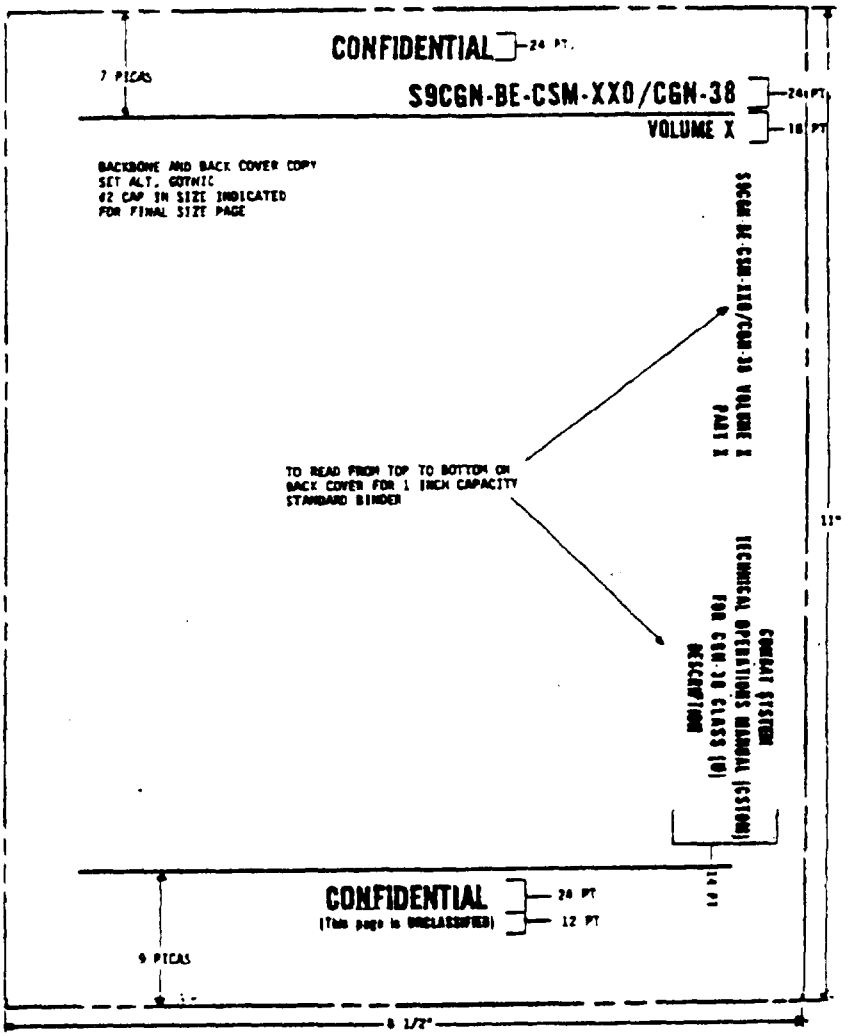


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 4. Sample back cover, classified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.2.2 for textual requirements.

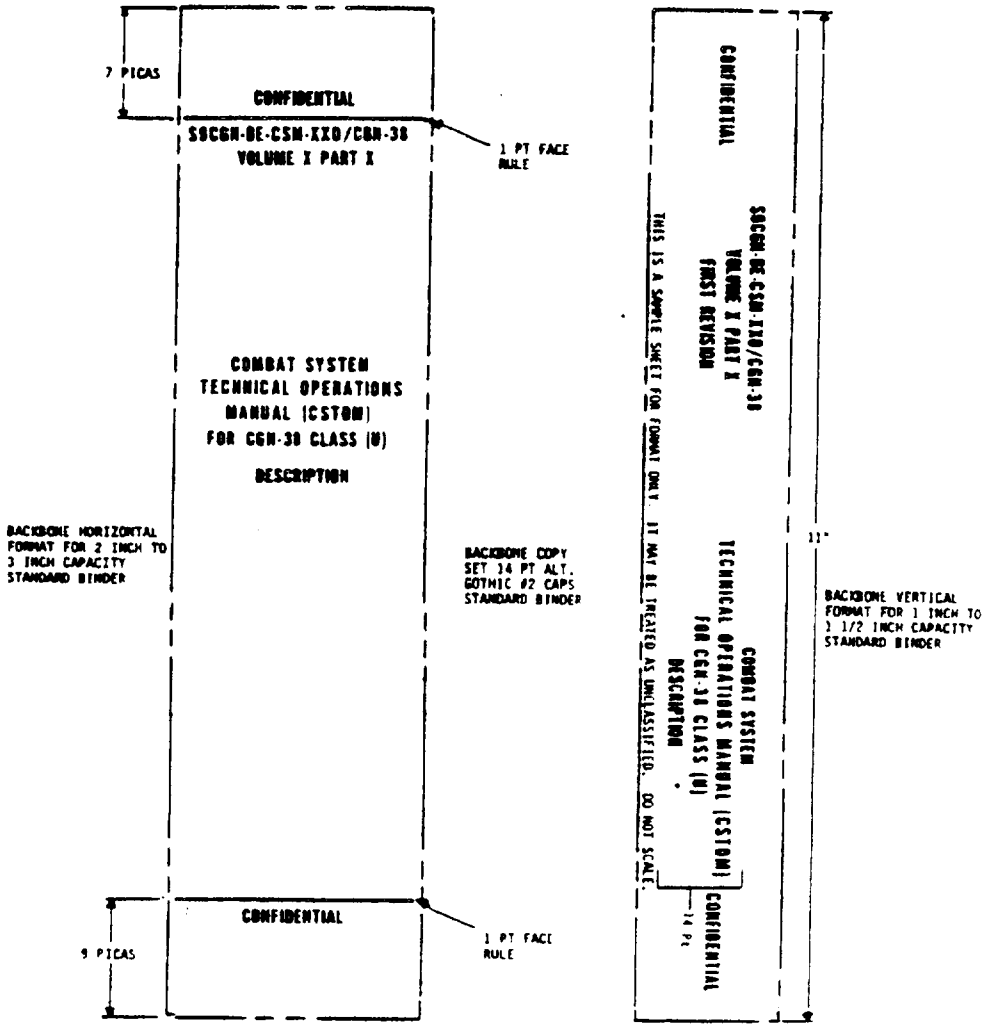


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 5. Sample back cover for volume without backbone.

Source: https://assist.dla.mil -- Downloaded: 202640-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.2.3 for textual requirements.

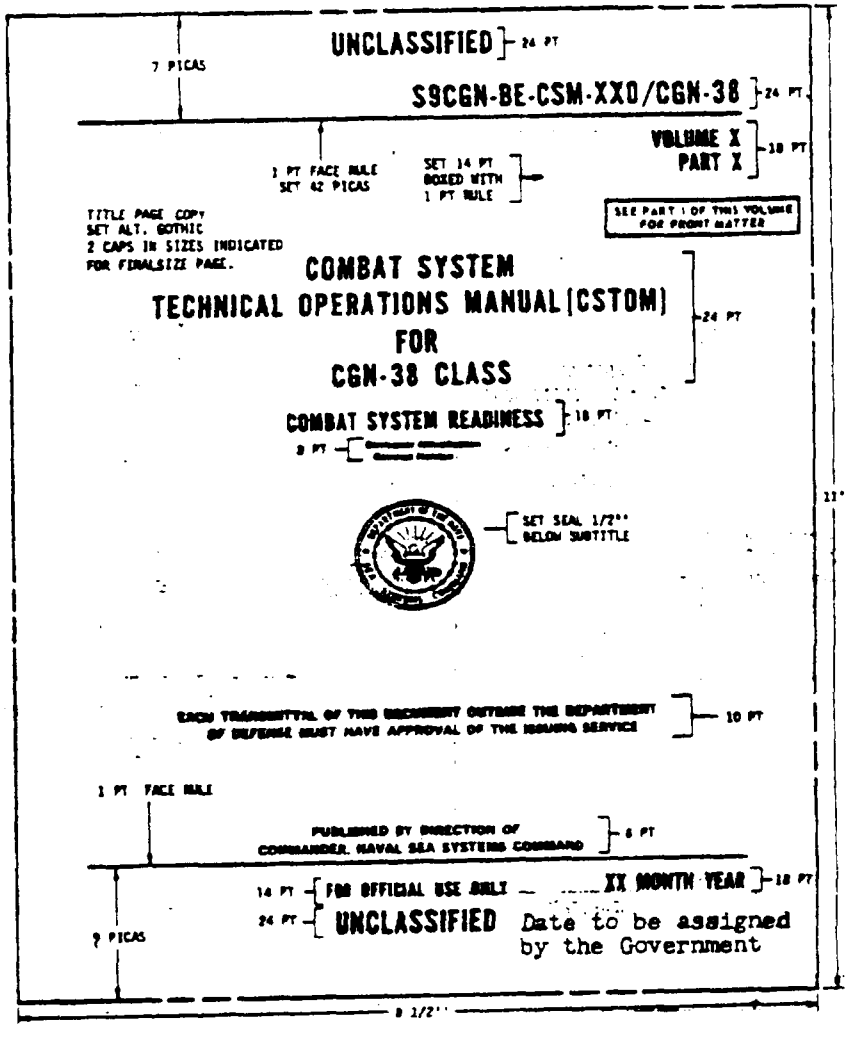


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 6. Sample backbone.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.2.4 for textual requirements.

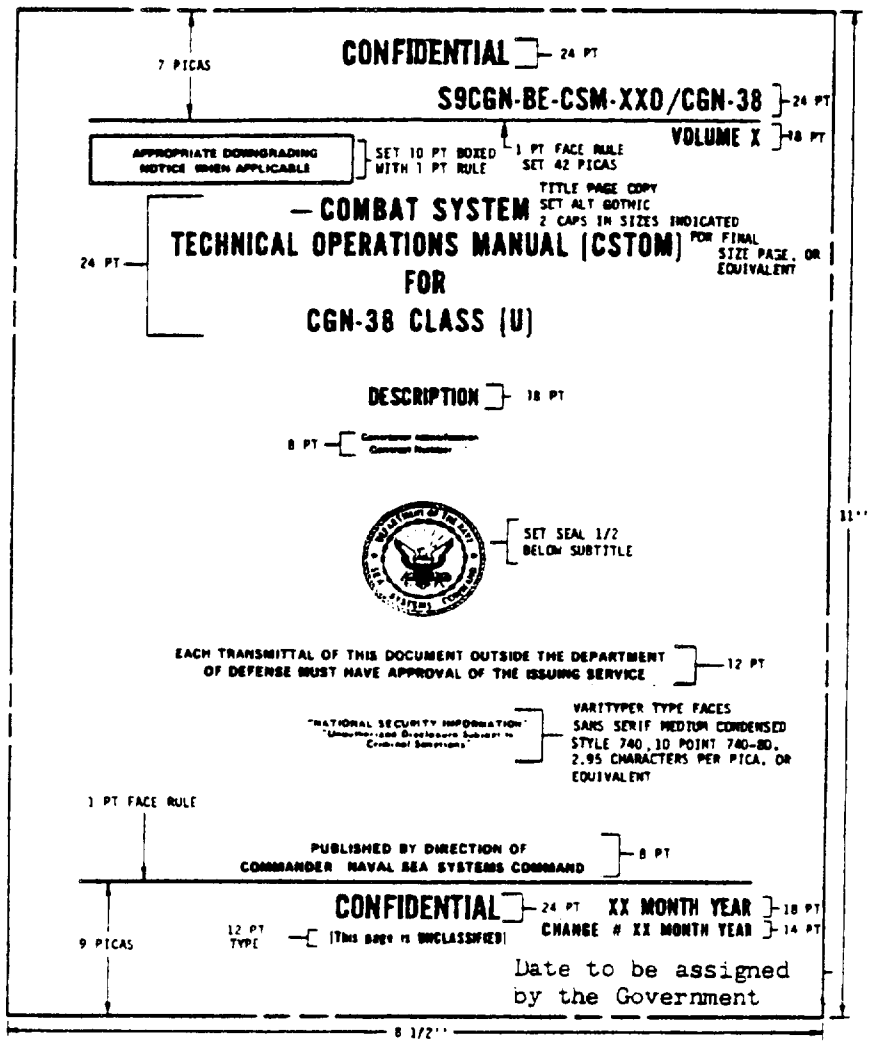


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 7. Sample title page for basic issue, unclassified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.2.4 for textual requirements.

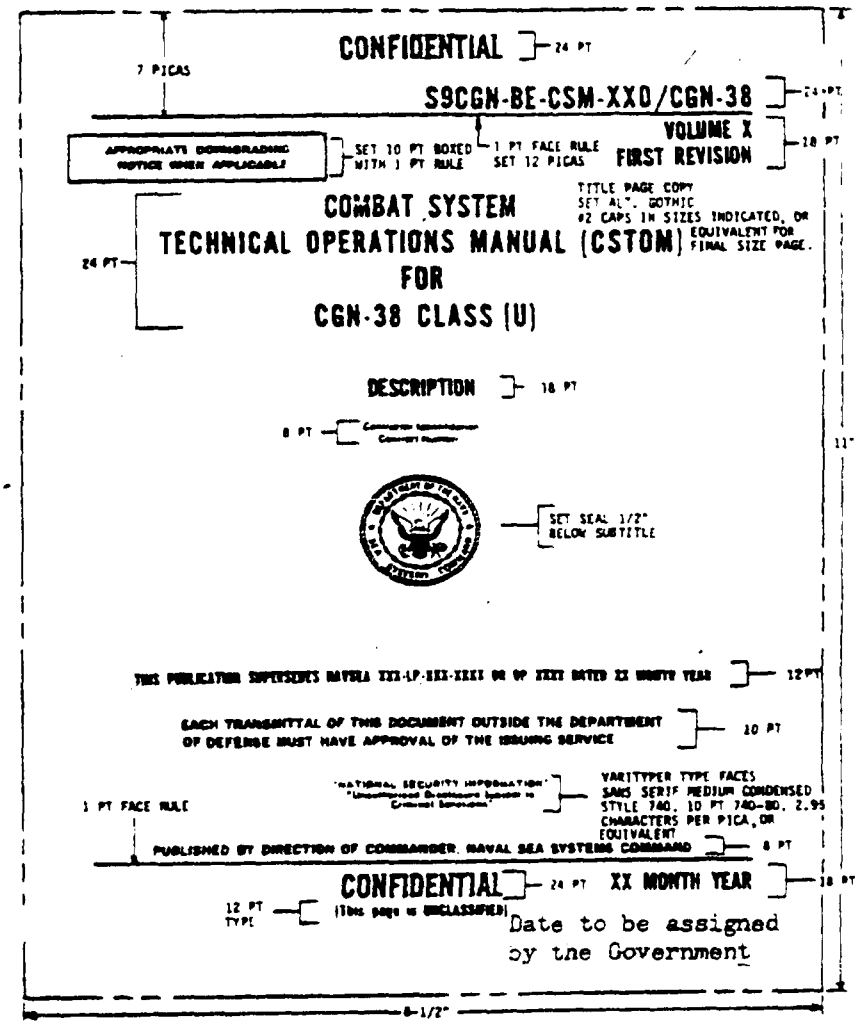


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 8. Sample title page for a change, classified.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.2.4 for textual requirements.

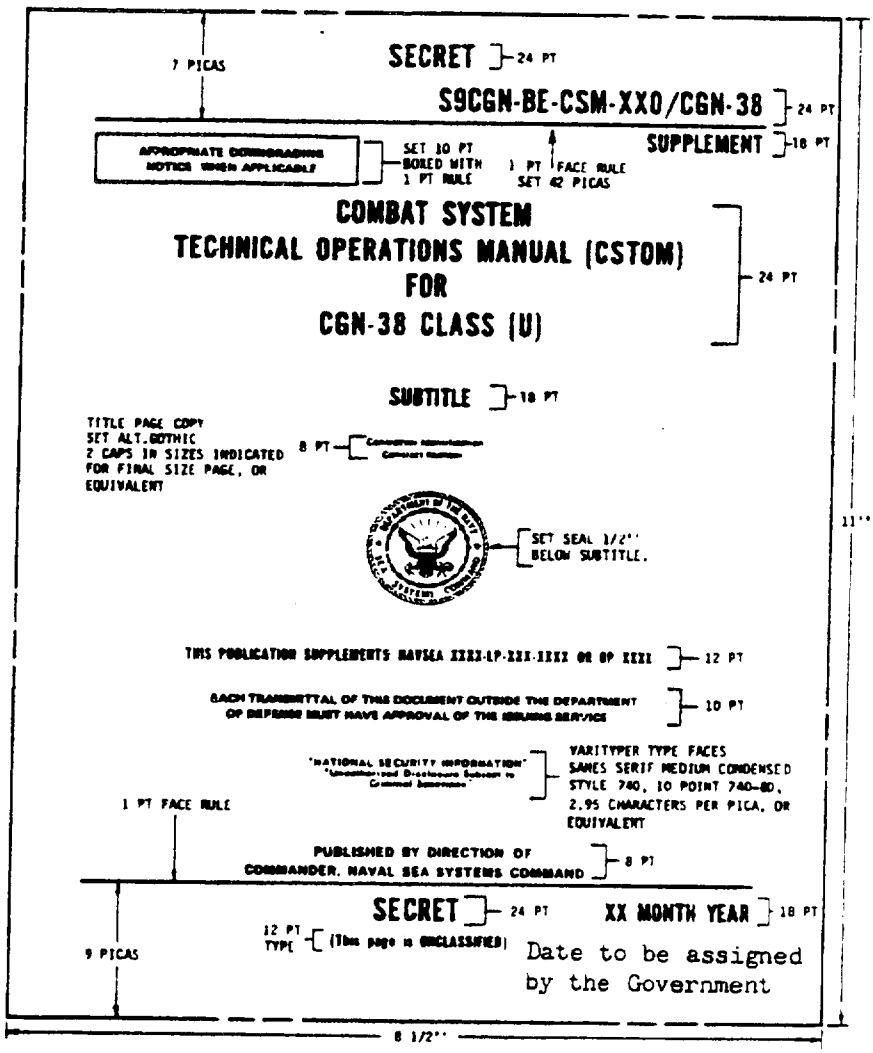


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 9. Sample title page for a revision, classified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.2.4 for textual requirements.



Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 10. Sample title page for a supplement, classified secret.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.2.5 for textual requirements.

FOREWORD

This Combat System Technical Operations Manual (CSTOM) is applicable to the CGN-38 class ships. This CSTOM organizes the technical data needed by shipboard personnel to operate and maintain the integrated combat system during combat operations, to maintain material and personnel readiness of the combat system, and to define significant capabilities and limitations of the combat system. The combat system is considered in relation to its ability to integrate all subsystems for performing the basic system functions (detection and entry, tracking and identification, threat evaluation and threat-to-weapon pairing, and engagement and engagement assessment).

This CSTOM is also intended for use in classroom training and self-instruction. Text and art are designed for classroom training to make specially prepared student handouts unnecessary. Pictorial diagrams, in lieu of conventional block diagrams, are used throughout the CSTOM to facilitate an appreciative recognition of physical and functional relationships of equipment and subsystems.

The copy freeze date for material contained in this publication is XX Month 19XX.

This publication is arranged in a volume and chapter structure as follows:

Volume 1 - COMBAT SYSTEM DESCRIPTION

Chapter 1 - Introduction
Chapter 2 - Combat Subsystem Descriptions
Chapter 3 - Combat System Operational Description

Volume 2 - OPERATIONAL SEQUENCES

Chapter 4 - Operational Sequences

Volume 3 - COMBAT SYSTEM READINESS

Chapter 5 - Readiness Assessment
Chapter 6 - Fault Detection and Impact Evaluation
Chapter 7 - Fault Isolation

Volume 4 - CAPABILITIES AND LIMITATIONS

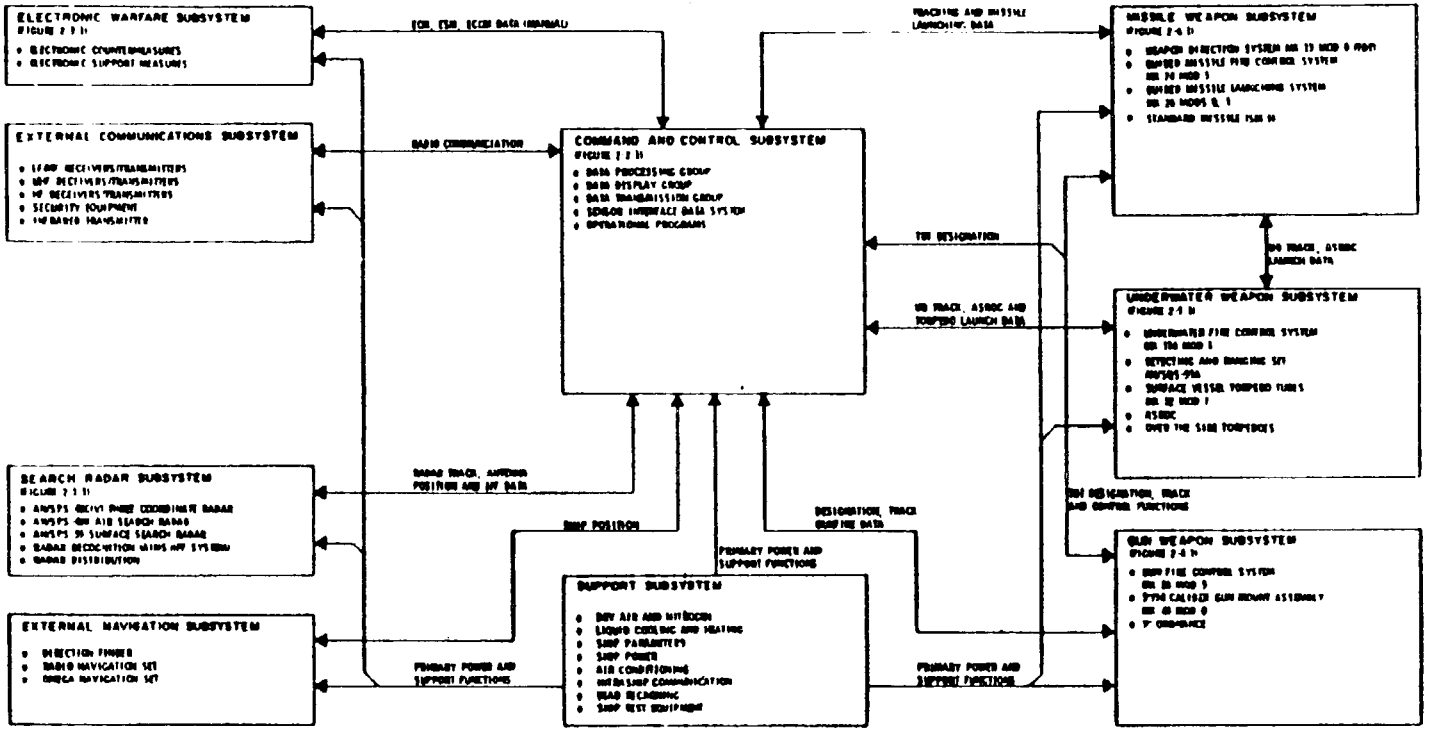
Chapter 8 - Ship Mission Capabilities and Limitations
Chapter 9 - Detection Capabilities and Limitations
Chapter 10 - Threat Processing Capabilities and Limitations
Chapter 11 - Threat Evaluation and Threat-to-Weapon Pairing Capabilities and Limitations
Chapter 12 - Engagement and Engagement Assessment

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 11. Sample foreword.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.4.1.2 for textual requirements.

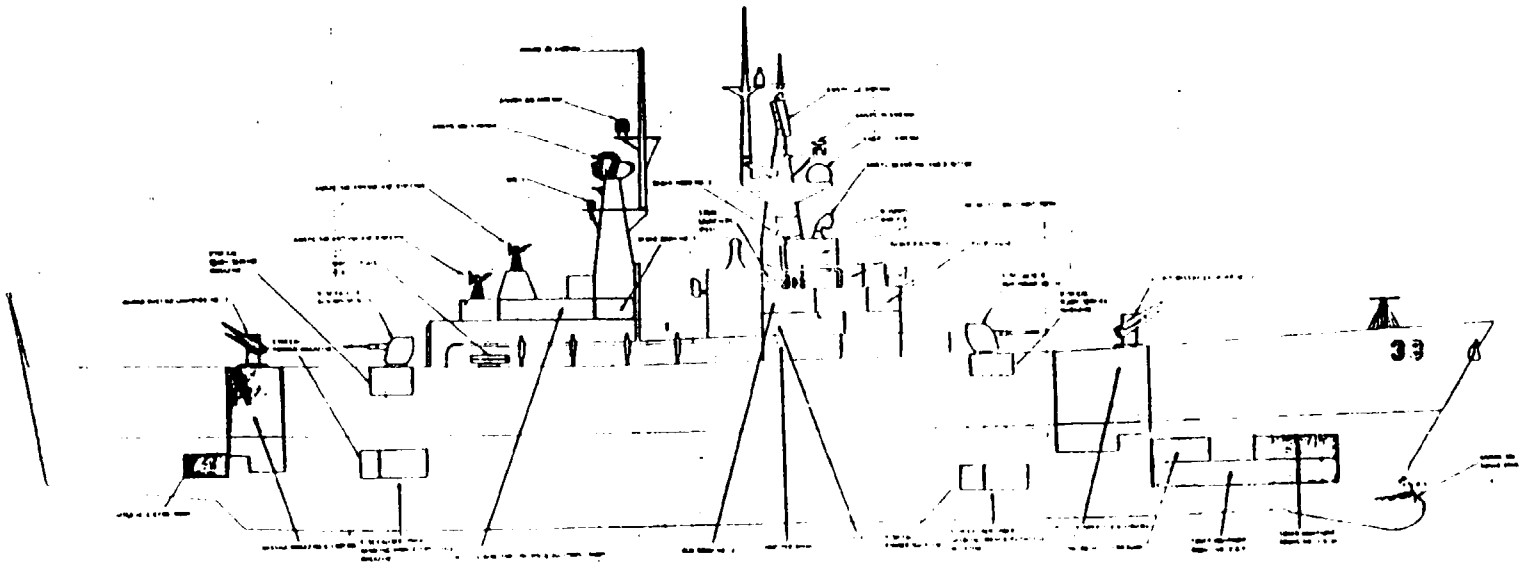


Sample arrangement only. Size, legibility, and microfiche requirements donot necessarily conform to minimum specification requirements.

FIGURE 12. Sample combat system level 1 block diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.4.1.3 for textual requirements.

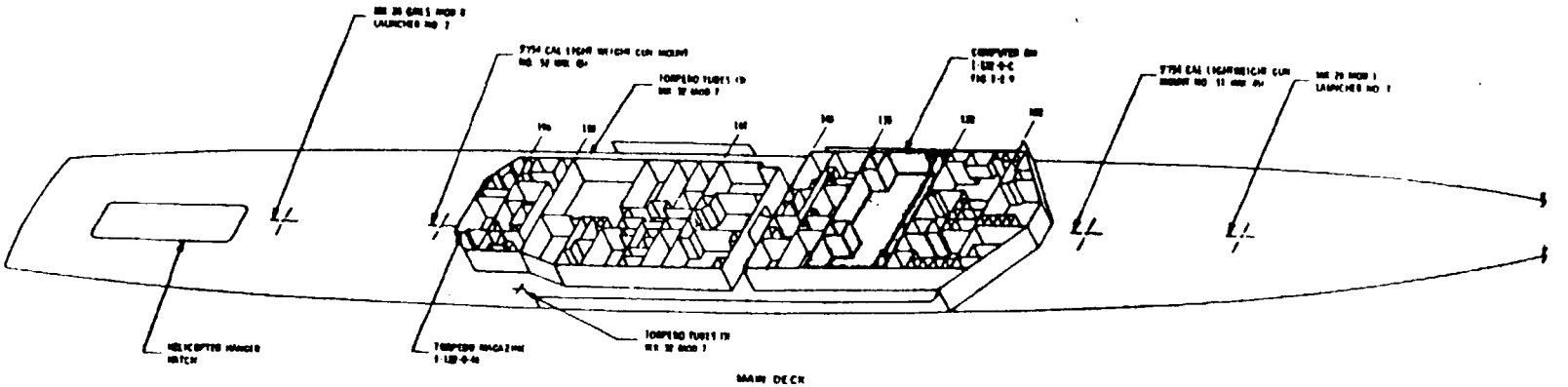


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 13. Sample ship profile and external components pictorial.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.4.1.3 for textual requirements.

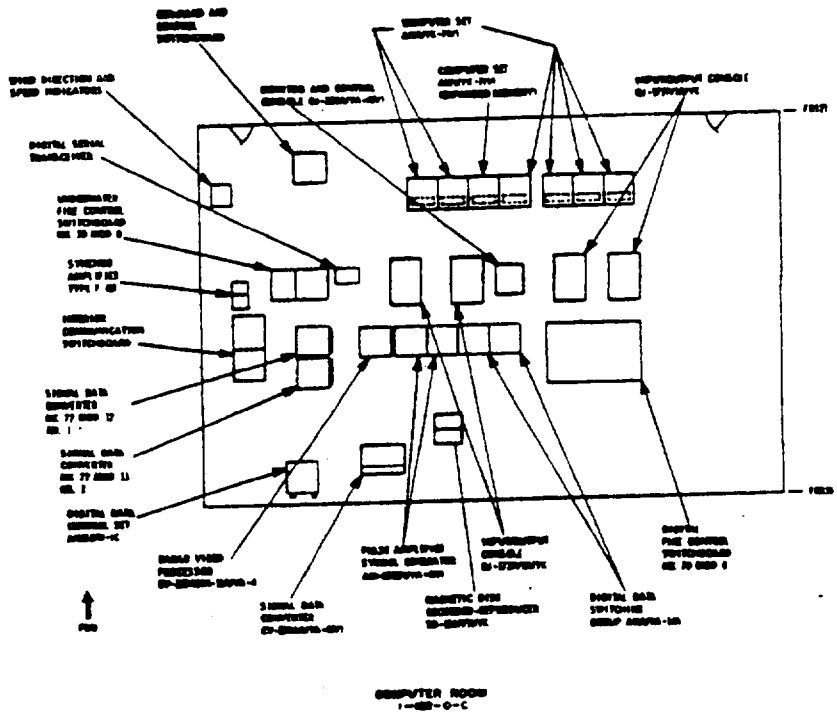


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 14. Sample three-dimensional deck and compartment arrangement pictorial.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the currentversion before use.

See 3.4.1.3 for textual requirements.

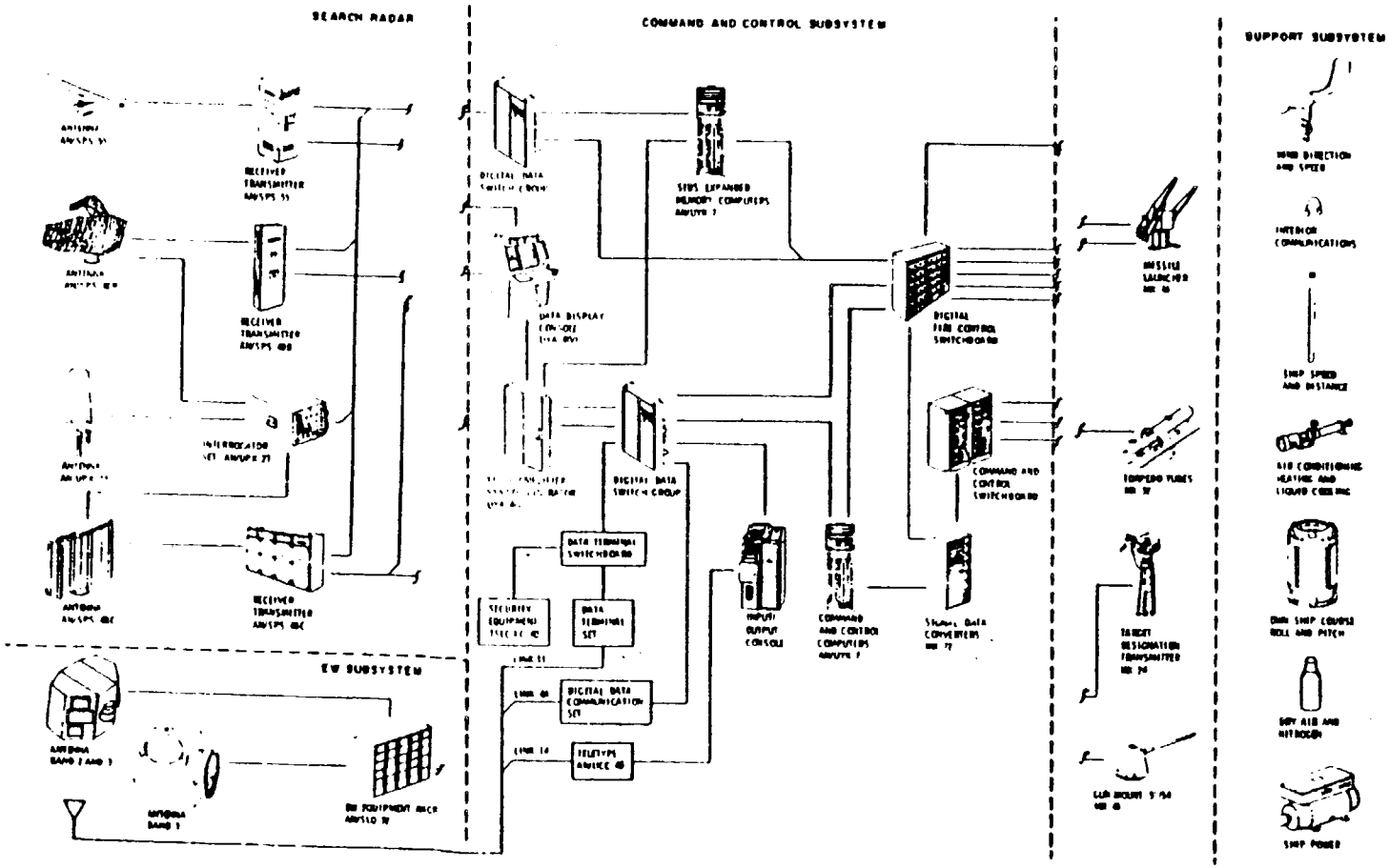


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 15. Sample room arrangement diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.4.2 for textual requirements.

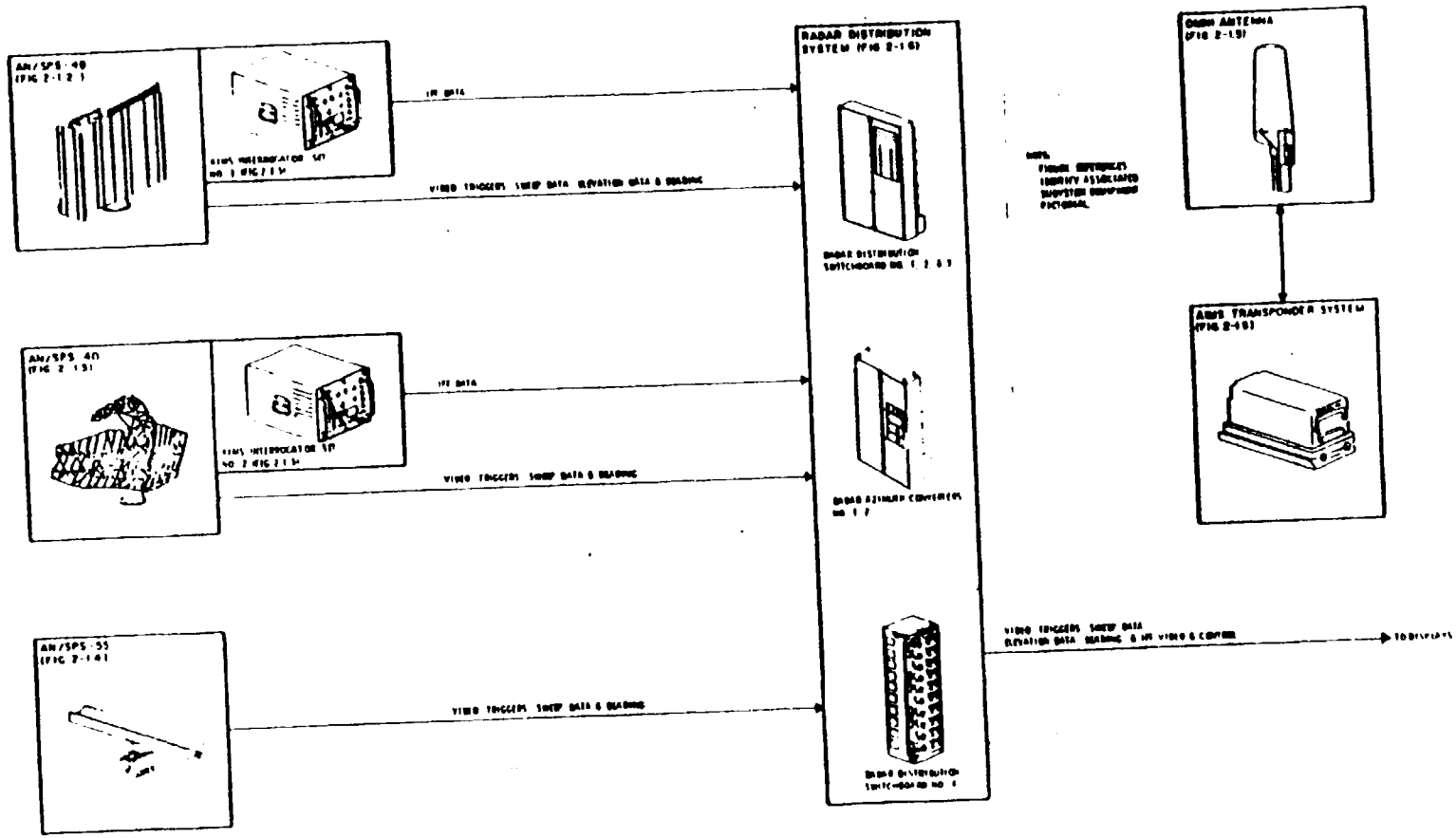


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 16. Sample combat system level 1 pictorial.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.4.2 for textual requirements.

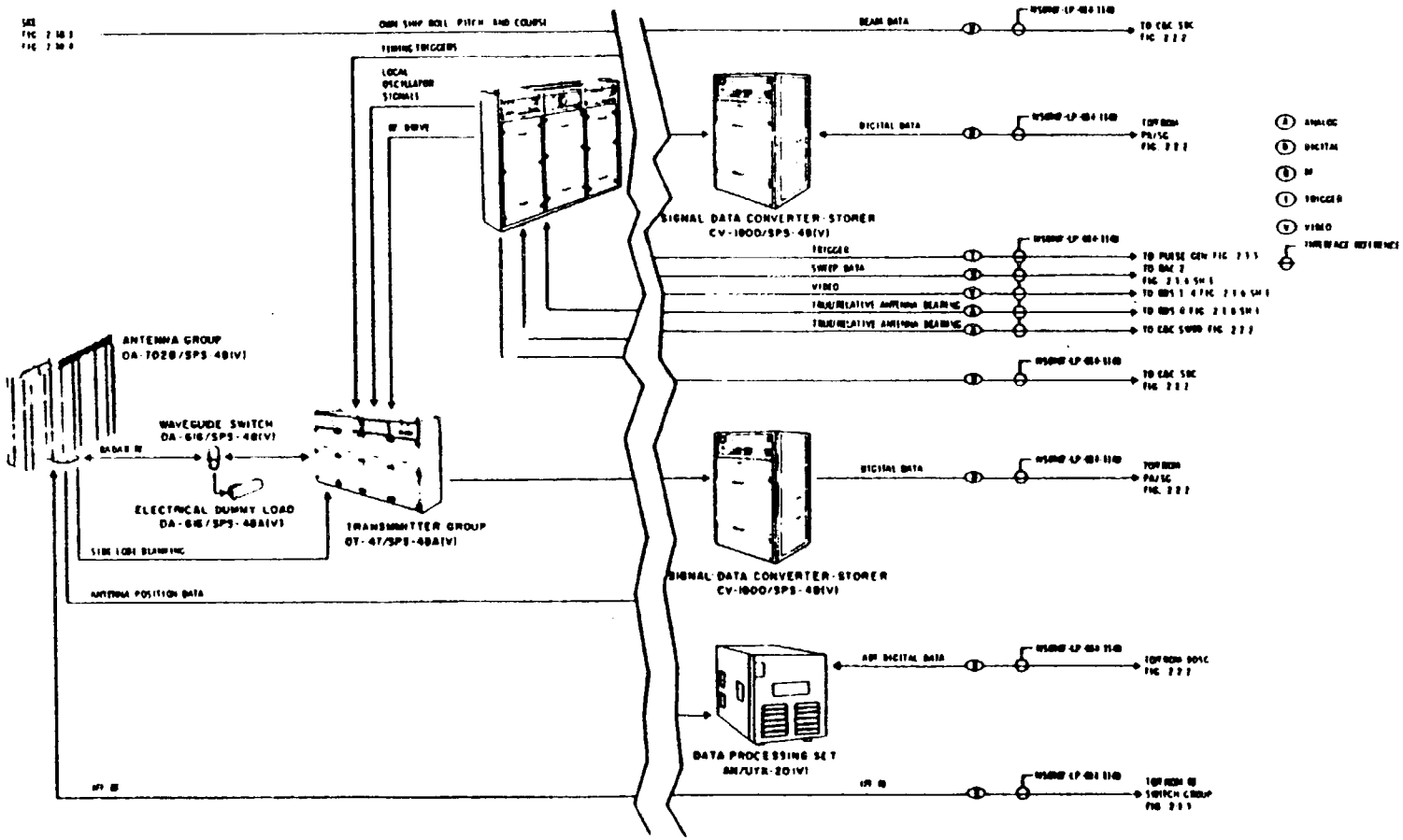


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 17. Sample subsystem level 2 pictorial.

Source: https://assist.dla.mil + Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.4.2 for textual requirements.

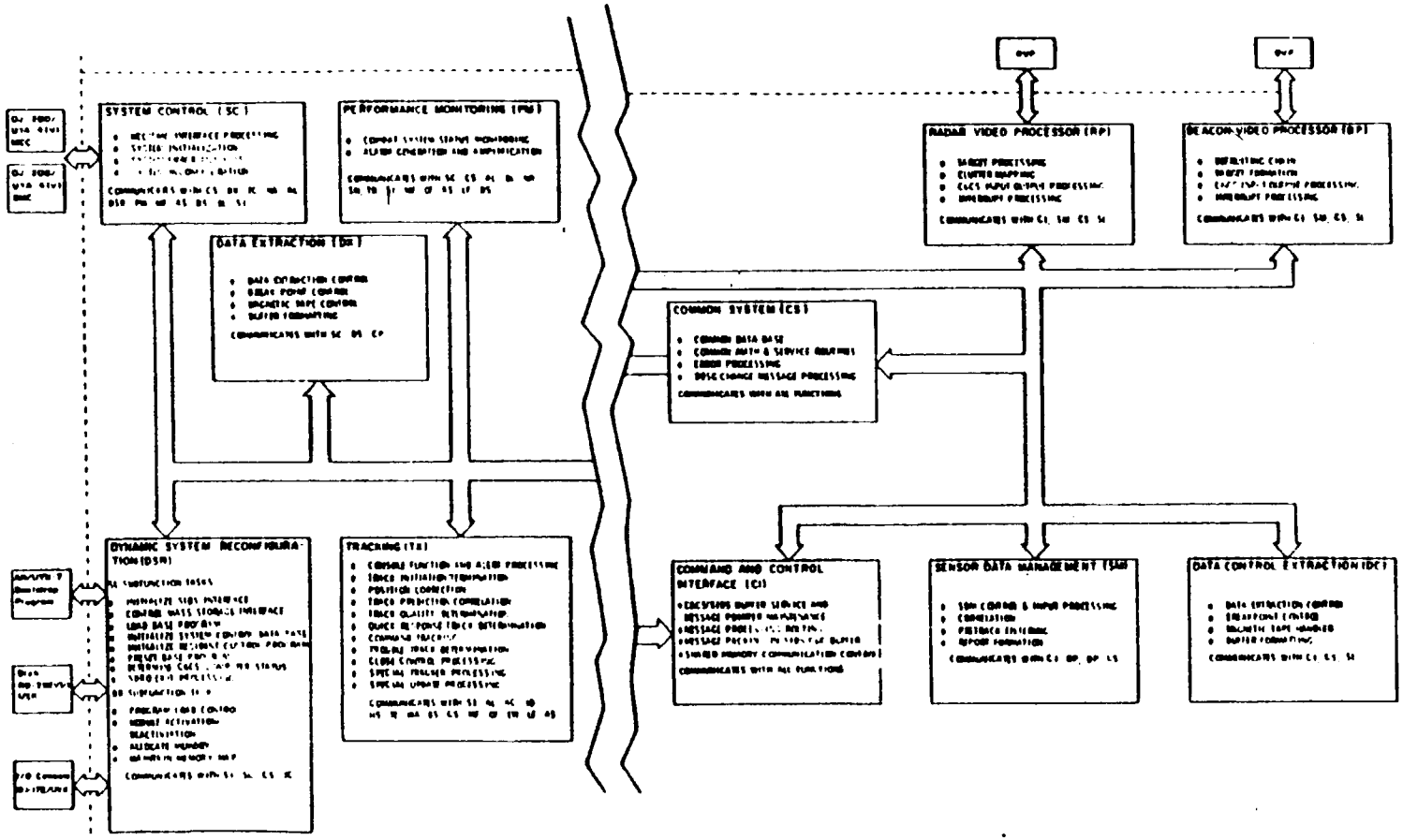


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 18. Sample equipment level 2 pictorial.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.4.2.3 for textual requirements.

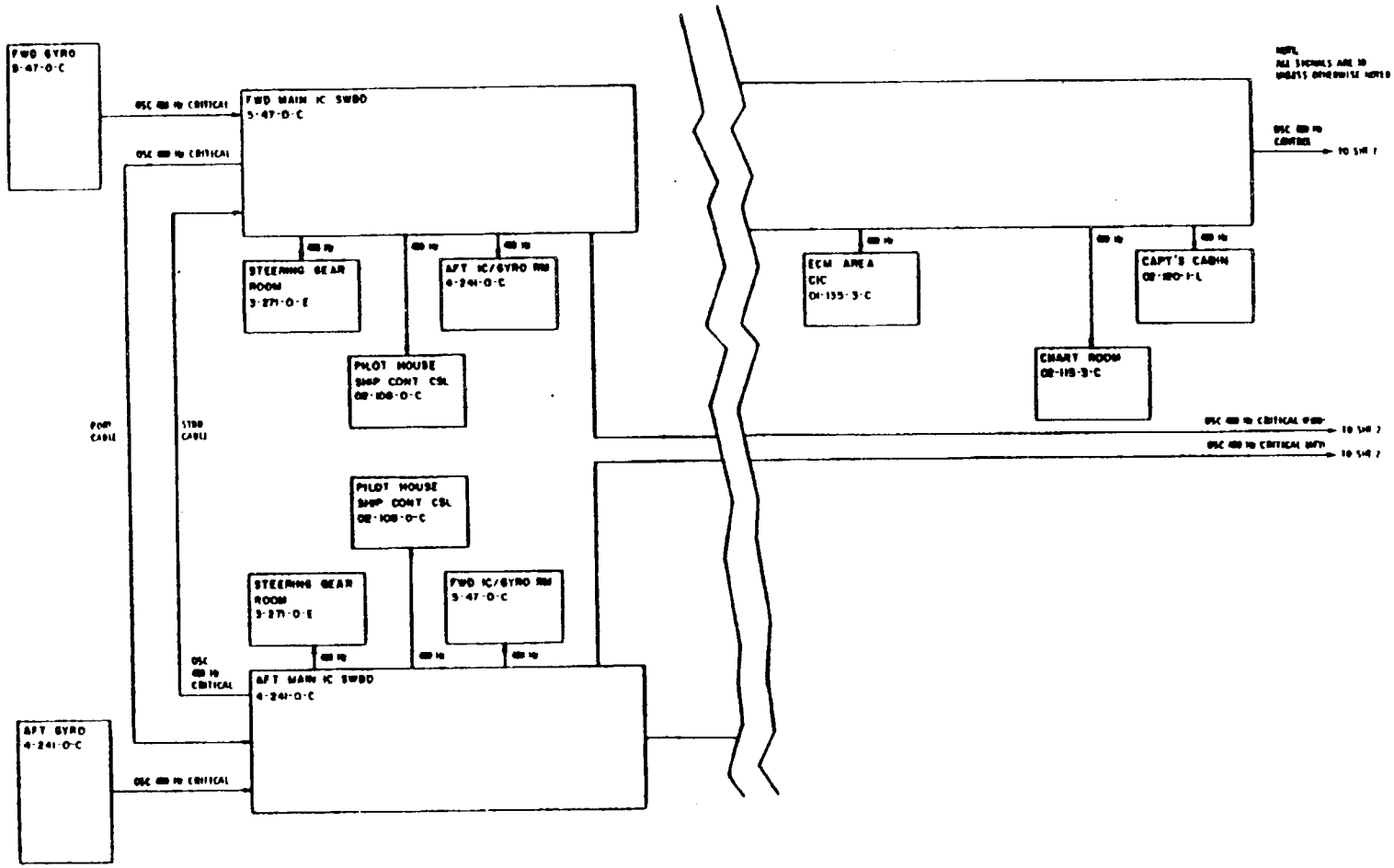


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 19. Sample subsystem operational program block diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.4.2.11 for textual requirements.

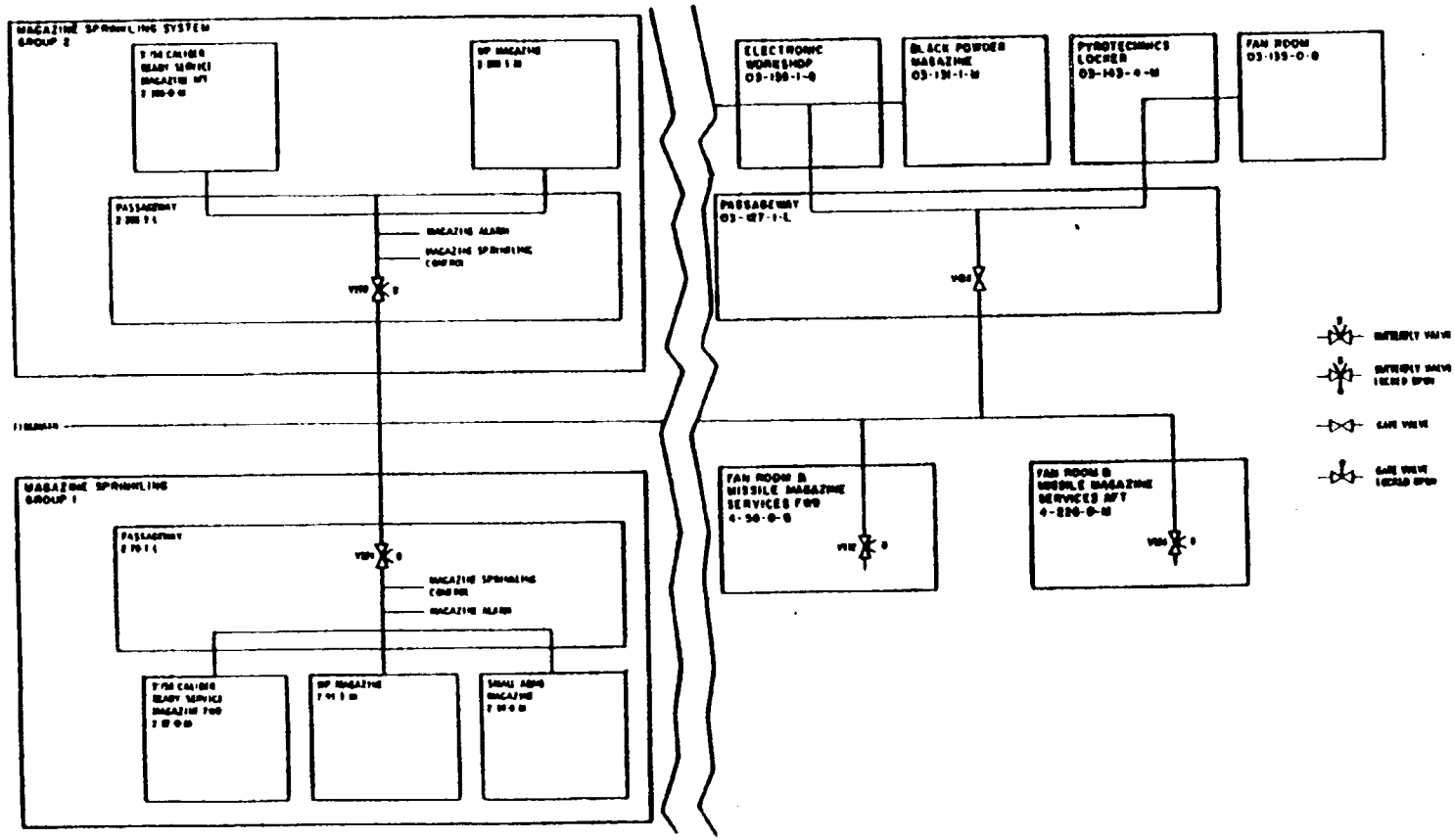


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 20. Sample support subsystem own ship heading distribution block diagram.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.4.2.11 for textual requirements.

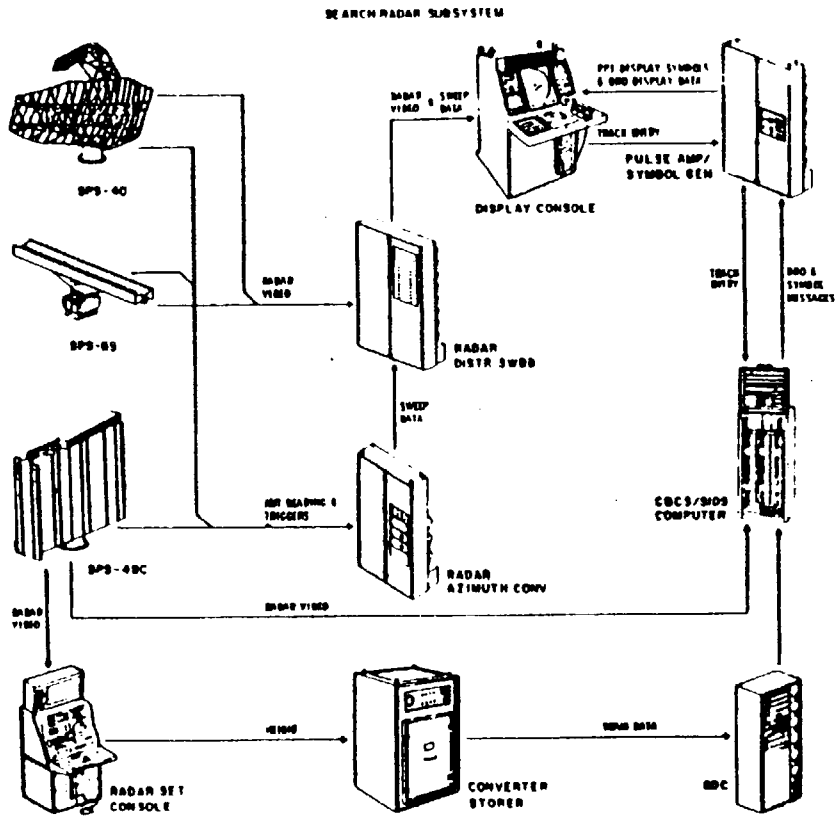


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 21. Sample support sybsystem liquid coolant, steam heating, and partial fire main block diagram.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.4.3 for textual requirements.

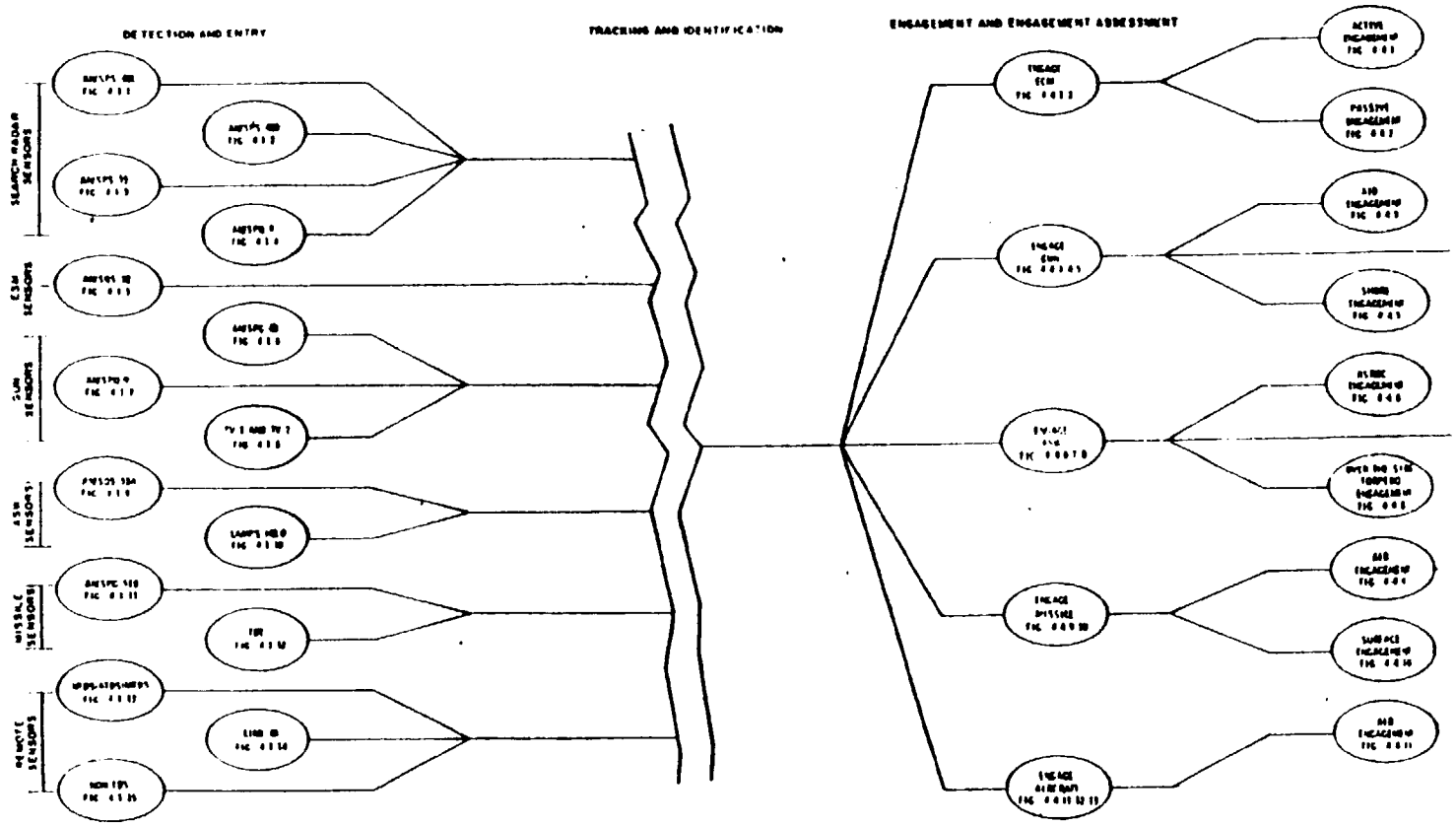


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 22. Sample level 2 functional flow pictorial.

Source: https://assist.dla.mil -- Downloaded: 2026-05-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.5 for textual requirements.

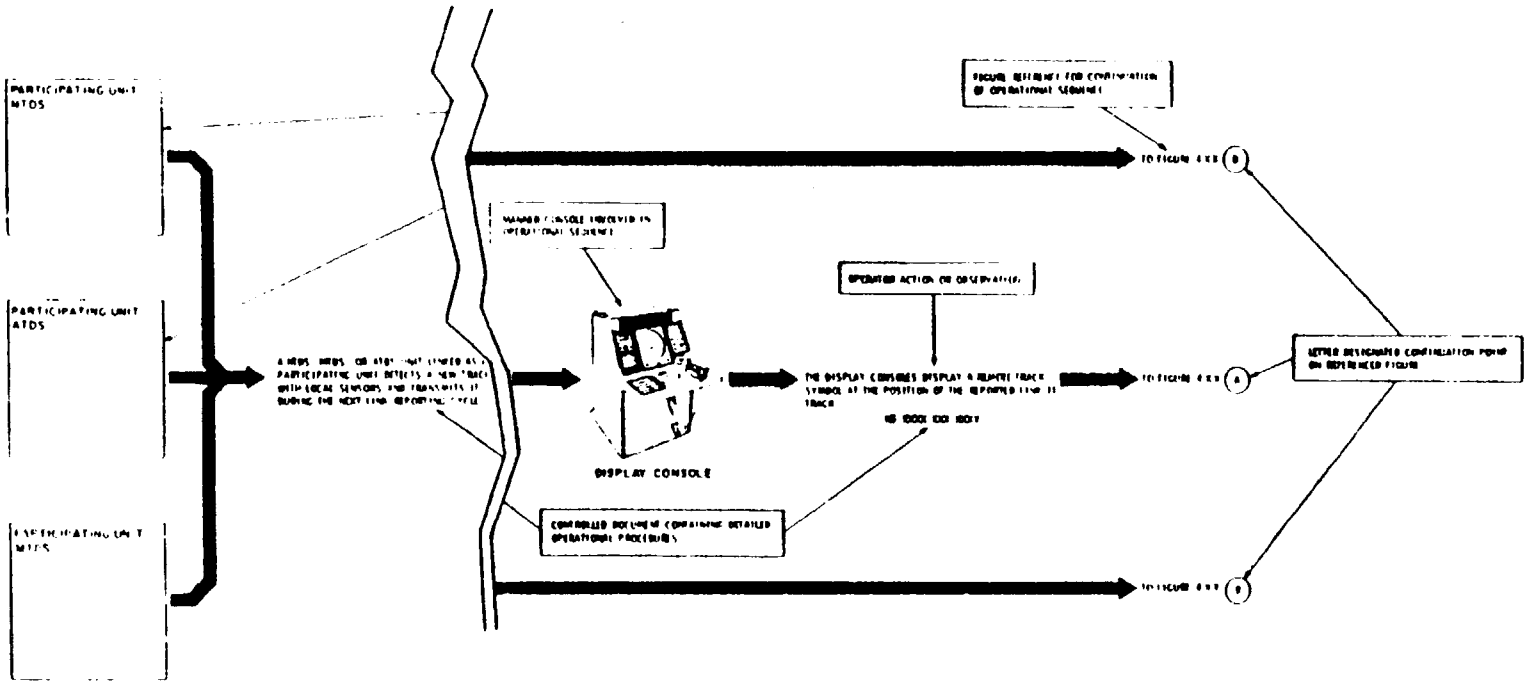


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 23. Sample operational mode network (OMN).

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.5 for textual requirements.

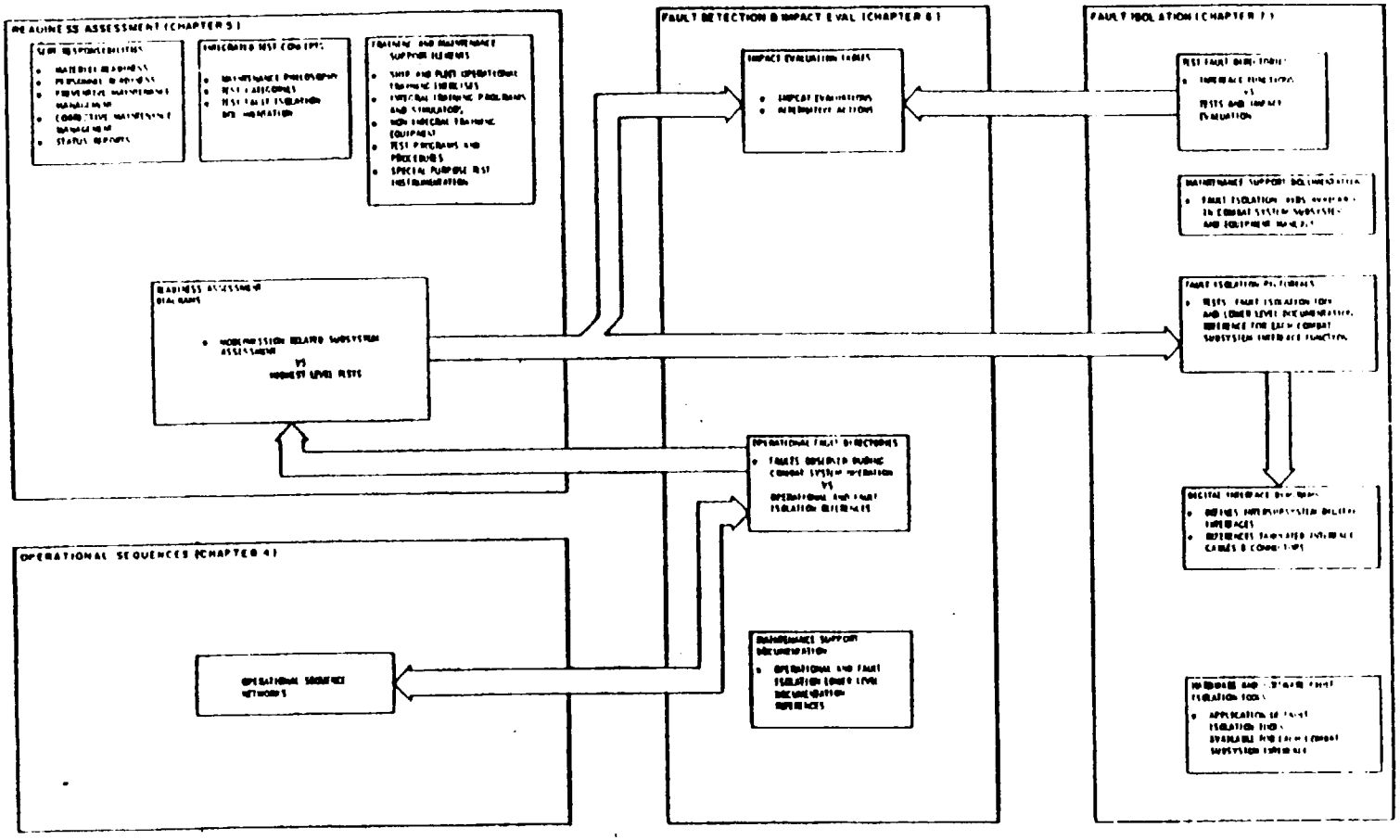


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 24. Sample operational sequence network (OSN).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.1.1 for textual requirements.

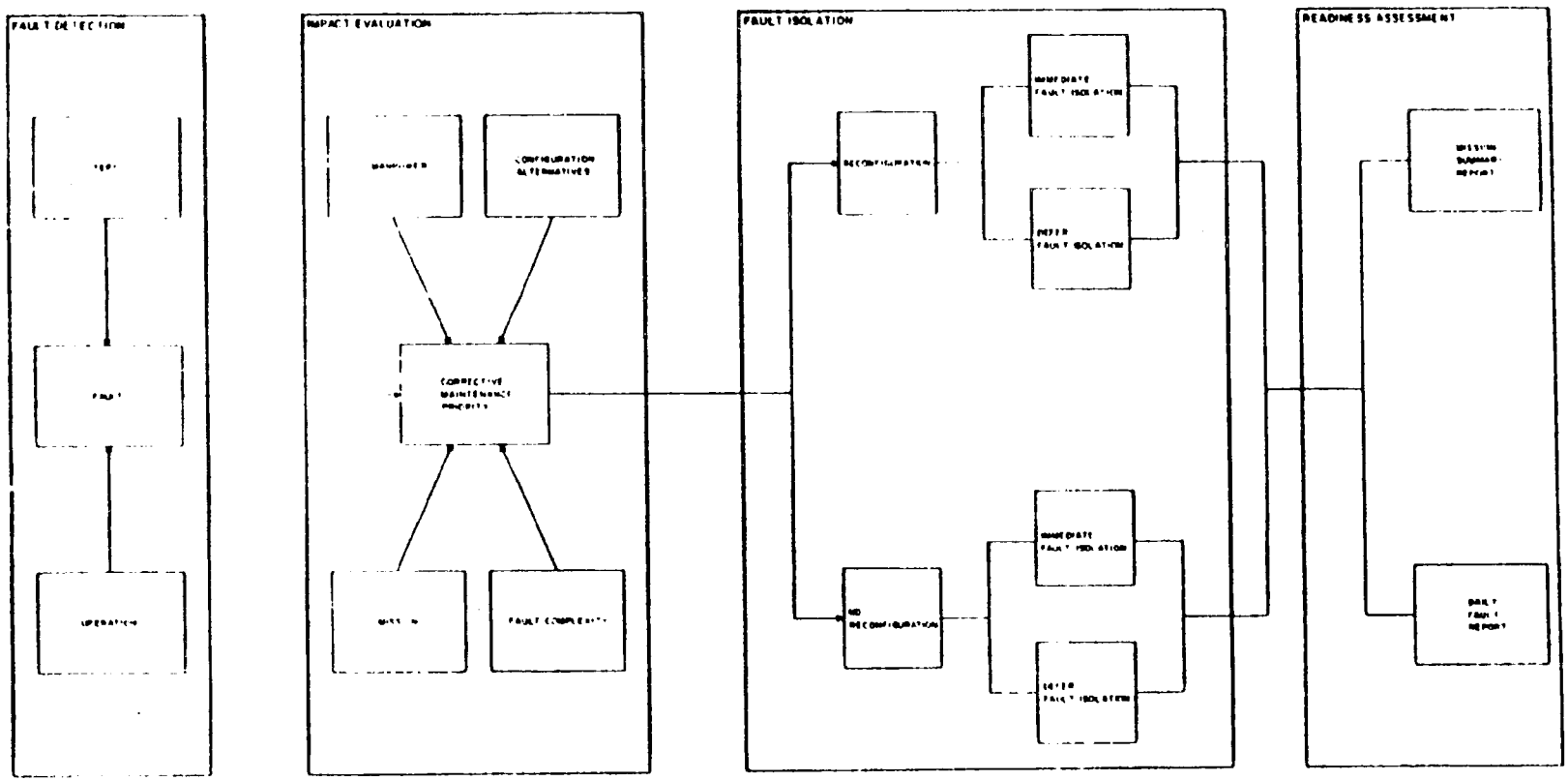


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 25. Sample readiness assessment and fault isolation tools overview.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.1.2 for textual requirements.



Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 26. Sample readiness assessment process diagram.

MIL-M-24570(SH)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.1.3 for textual requirements.

FUNCTION	DETECTION	PROCESSING EVALUATION	COMMAND DECISION	WEAPONS ASSIGNMENT	EW ENGAGEMENT	MISSILE ENGAGEMENT	GUN ENGAGEMENT	UNDERWATER ENGAGEMENT
STATUS	SCR	CR	CR	MCR	CR	MCR	MCR	NA
REMARKS	REDUCED DETECTION RANGE			MARGINAL DESIGNATION TO GFCS		SPG-49 NO.2 TRANSMITTER DOWN	MARGINAL DESIGNATION ACCURACY	

SAMPLE

LEGEND: SCR SUBSTANTIALLY COMBAT READY
CR COMBAT READY
MCR MARGINALLY COMBAT READY
NCR NOT COMBAT READY

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 27. Sample mission summary report.

Source: https://assist.dla.mil - Downloaded: 20261019T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.1.3 for textual requirements.

MAINTENANCE STATUS	FAULT	IMPACT				ALTERNATIVE	ETR	NOTE
		AAW	ASW	SURFACE	SHORE			
D	SPS-48 Low Power Output	Reduced range	NA	NA	NA	3D SPS-30 2D SPS-43	1630	-
D	Marginal designation to GFCS	Increased acquisition time	ASROC Tracking questionable	Increased acquisition time	NA	TDT or SPA-4	Sunday	Alignment
M	SPS-49 No. 2 pulse transformer	Reduced Forward Fire Power	NA	NA	NA	None		No part CASREPT

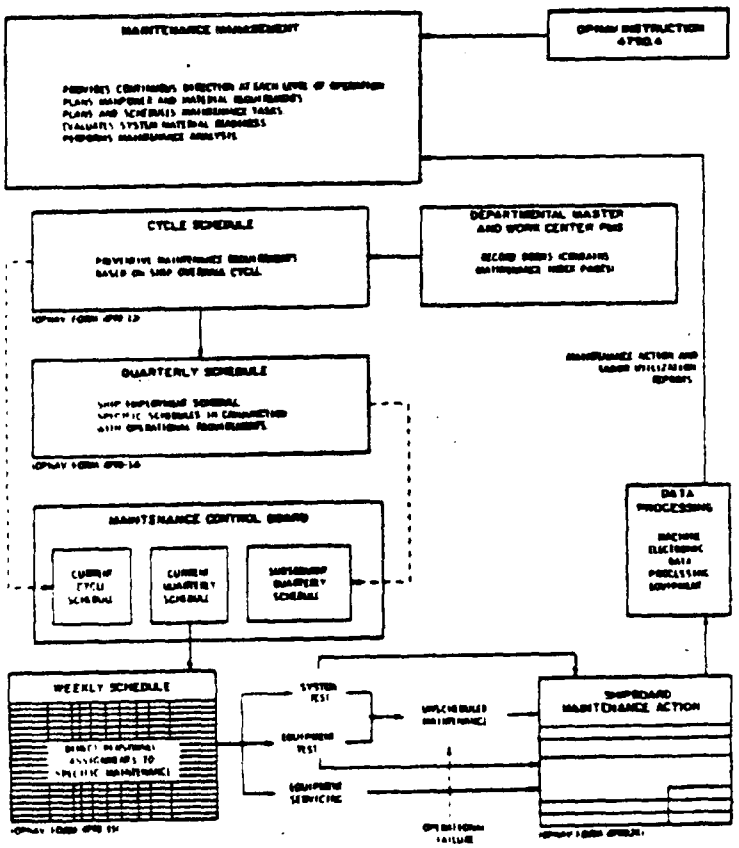
LEGEND: D = DEFERRED
M = MANDATORY

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 28. Sample combat system daily faults report.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.1.4 for textual requirements.

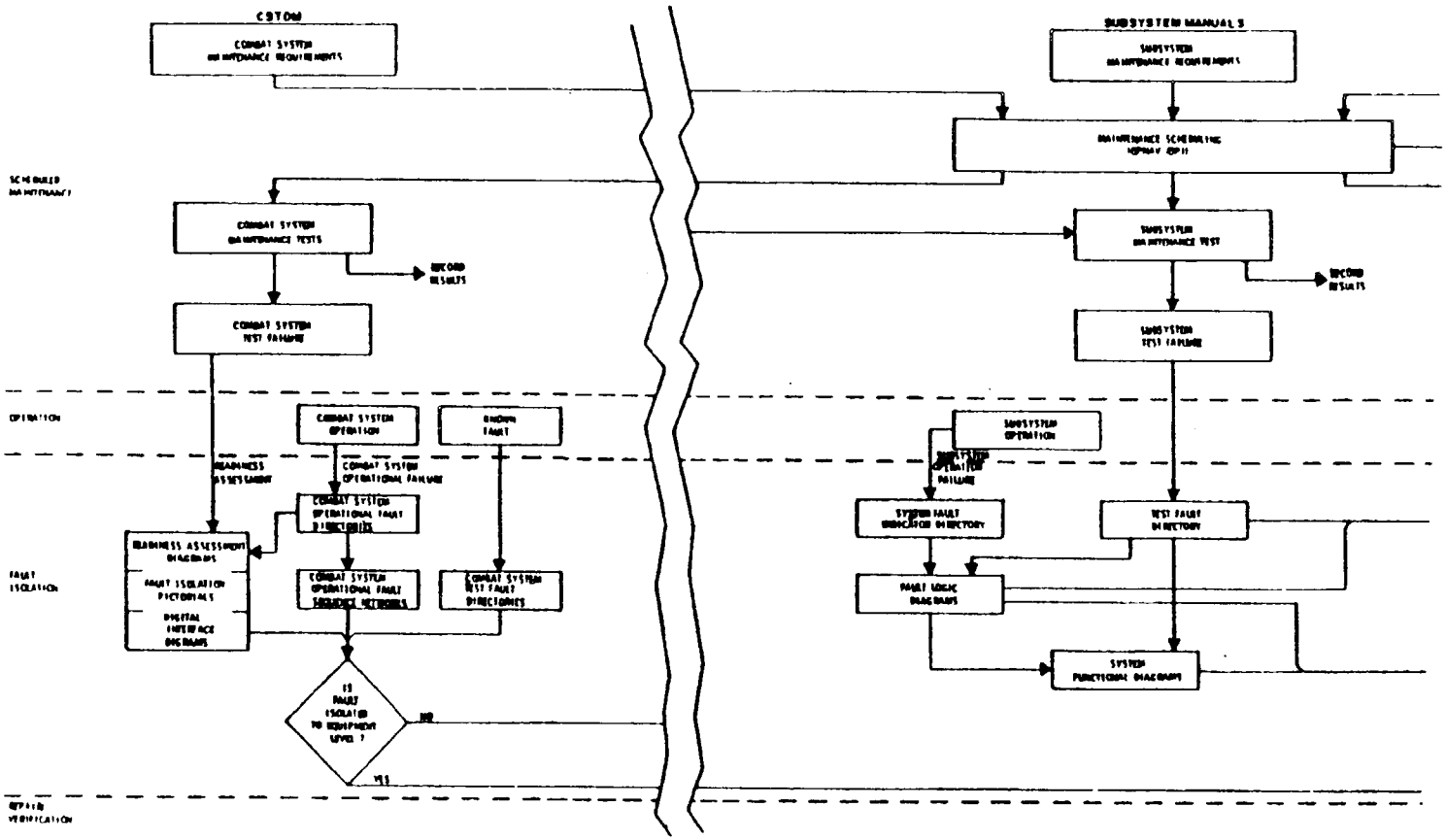


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 29. Sample planned maintenance system block diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.1.4 for textual requirements.



Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 30. Sample integrated maintenance concept block diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.1.5 for textual requirements.

FXP Series Exercise	Title	Subsystem					Operational Phase			
		Search Radar	C&CS	Missile Weapon	Gun Weapon	EW	Detection, Evaluation, FCS Assign, and Track	Engageability Evaluation	Launcher Assign and Missile Firing	Engagement Assessment
Z-1-GM	Single Target Processing	X	X	X			X	X	X	
Z-2-GM	Multiple Target Processing	X	X	X			X			
Z-3-GM	Multiple Target Processing	X	X	X			X	X		
Z-4-GM	ECCM - Mechanical Jamming	X	X	X			X			
Z-5-GM	ECCM - Cap Coordination Mechanical Jamming	X	X	X			X	X	X	
Z-6-GM	ECCM - Electromagnetic Jamming	X	X	X			X			
Z-7-GM	ECCM - Mechanical and Electro-magnetic Jamming	X	X	X			X			
Z-10-GM	Tactical AAW	X	X	X			X	X	X	
Z-11-GM	Tactical AAW - Simulated Countermeasures	X	X	X			X	X	X	
Z-12-GM	Tactical AAW - Live Countermeasures	X	X	X		X	X	X	X	
Z-13-GM	Tactical AAW - CAP and Missile Coordination	X	X	X			X	X	X	
Z-14-GM	Tactical AAW - CAP and Missile Coordination with Simulated Countermeasures	X	X	X			X	X	X	
Z-15-GM	Comprehensive Non-Firing SAM in AAW	X	X	X	X	X	X	X	X	
Z-20-GM	Coordinated Missile Employment in Fleet AAW	X	X	X			X	X	X	
Z-21-GM	Coordinated Missile Employment in Fleet AAW - Countermeasures Envir.	X	X	X			X	X	X	

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 31. Sample table of fleet training exercises subsystem applicability.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.1.5 for textual requirements.

Test	Subsystem						Purpose and description	Reference document	Time reqd	Periodicity	Scheduling responsibility
	Search radar	TDS	Missile weapon	Gun weapon	EW	Underwater weapon					
ASW CSOT (Live)	X	X		X		X	Verifies combat system ASW operability between surface-search radar, C&CS, UFCS and SONAR, and GFCS.	OD 44859 CSOT	1 hr	M	CGN-38
ASW CSOT (Simulated)		X		X		X	Verifies combat system ASW operability between C&CS, UFCS and SONAR, and GFCS. This test is similar to the live ASW CSOT, except a simulated target is entered by C&CS and sonar target acquisition is simulated.	OD 44859 CSOT	1 hr	M	CGN-38

SAMPLE

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 32. Sample table of operational test programs.

Source: https://assist.dla.mil - Downloaded: 2025-10-19 23:02
Check the source to verify that this is the current version before use.

See 3.6.1.6 for textual requirements.

Ref	Test	Test brief	Fault isolation pictorial*	Impact evaluation reference **
1		CSAT	Fig. 7-X.X	Table 6-X.X
2	R-81/R-2D	AN/SPS-48 Radar		
3	R-81/R-3D	AN/SPS-48 Radar		
4	R-81/R-6D	AN/SPS-48 Radar		
5		OCSOT		
6	TD-68/Q-1	AN/UYA-4 Data Display Group		
7	TD-141/R-1Q	Display Pofa		
8	R-13/R-2D	AN/SPS-40 Radar		
9		OCSOT		

Note: This table and the associated diagram comprise the Readiness Assessment Diagram (RAD). Each RAD shall identify the highest level test or combination of tests that will verify readiness of the combat system to perform in a particular phase of a given mission. Specific tests are tabulated and reference given to a pertinent fault isolation pictorial as well as an impact evaluation reference.

* Refers to fault isolation pictorial in chapter 7 and interface data tabulation reference on that figure.

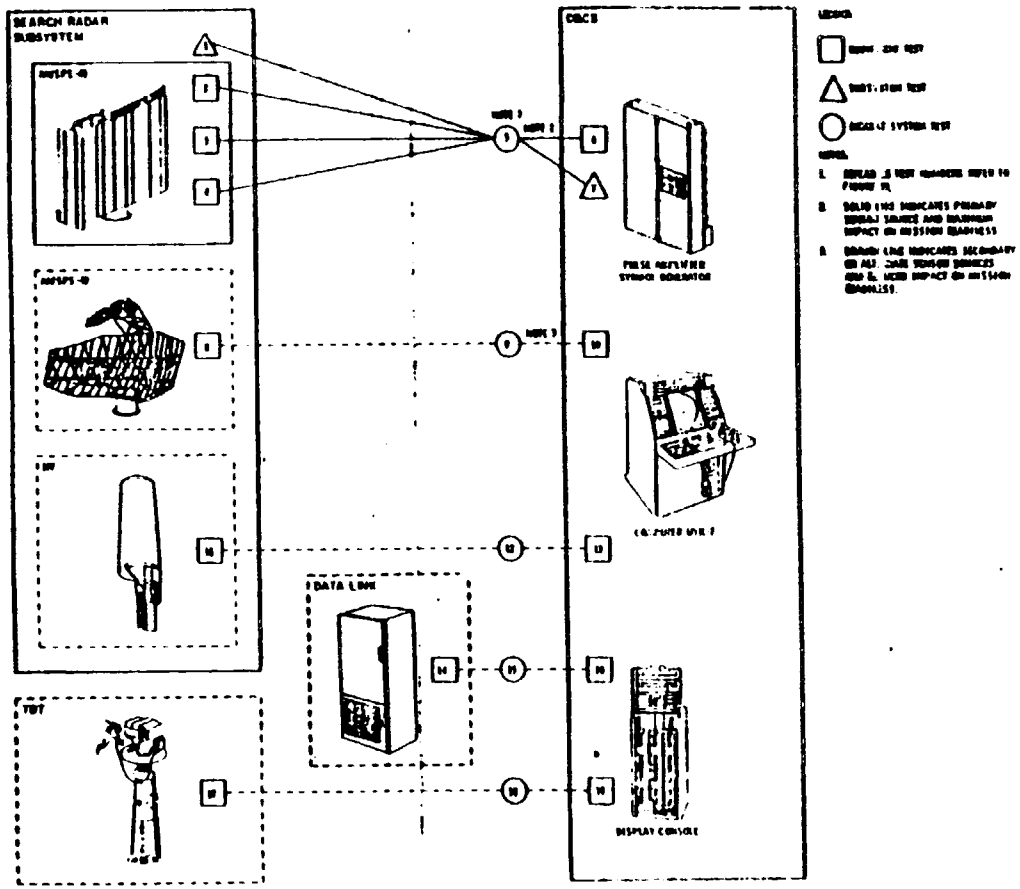
** Refers to impact evaluation table in chapter 6.

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 33. Sample table of test and fault impact evaluation references.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.1.6 for textual requirements.



Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 34. Sample readiness assessment diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.2.2 for textual requirements.

Fault symptom	Operational reference				Fault isolation reference							
Note:* F = figure	*	OSN Vol 2, F4-1.1	OSN Vol 2, F4-1.2	NS0967-(P-014-1160	RAD_F5-5.8	NS0967-441-9010	NS0967-230-7010					
SPS-48C: DETECTOR TRACKER No radar video displayed in RADAR 5 positions.	X		X		X		X					
SPS-48-C: DETECTOR TRACKER No symbology displayed when new track is entered.	X		X		X		X					
SPS-40B: DETECTOR TRACKER No radar video displayed in RADAR 4 positions.		X	X		X	X	X					
SPS-40B: DETECTOR TRACKER No symbology displayed when new track is entered.		X	X		X	X						
TRACK SUPERVISOR No radar video displayed in RADAR 5 positions.	X		X		X	X						
TRACK SUPERVISOR No symbology displayed when new track is entered.	X	X	X		X	X						
SPS-48C: RADAR SET CONSOLE Track not on SPS-48C RSC sequence list.	X				X							
TRACK SUPERVISOR No late detect alert on a late detect target.	X	X	X		X	X						

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 35. Sample operational fault directory.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.2.3 for textual requirements.

SUBSYSTEM EQUIPMENT/FUNCTION	AAW	ASW	SUW	SHORE	IMPACT	ALTERNATE
AN/SPS-48C RADAR						
1. Ro	X	X	X		1. Reduced designation capability due to loss of sweep on display consoles in radar select position 5.	Use alternate radar select position for designation inputs.
2. Rmax	X	X	X		2. Reduced designation capability due to inaccurate range data.	Same as above.
3. V_0, V_1, V_2, V_3	X	X	X		3. Reduced designation due to no video presented on display consoles in radar select position 5.	Same as above.
AN/SPS-55						
1. To			X		1. Reduced surface designation capability due to loss of sweep on display consoles in radar select position 2.	Use alternate radar select position for designation inputs.

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 36. Sample subsystem impact evaluation table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.3.1 for textual requirements.

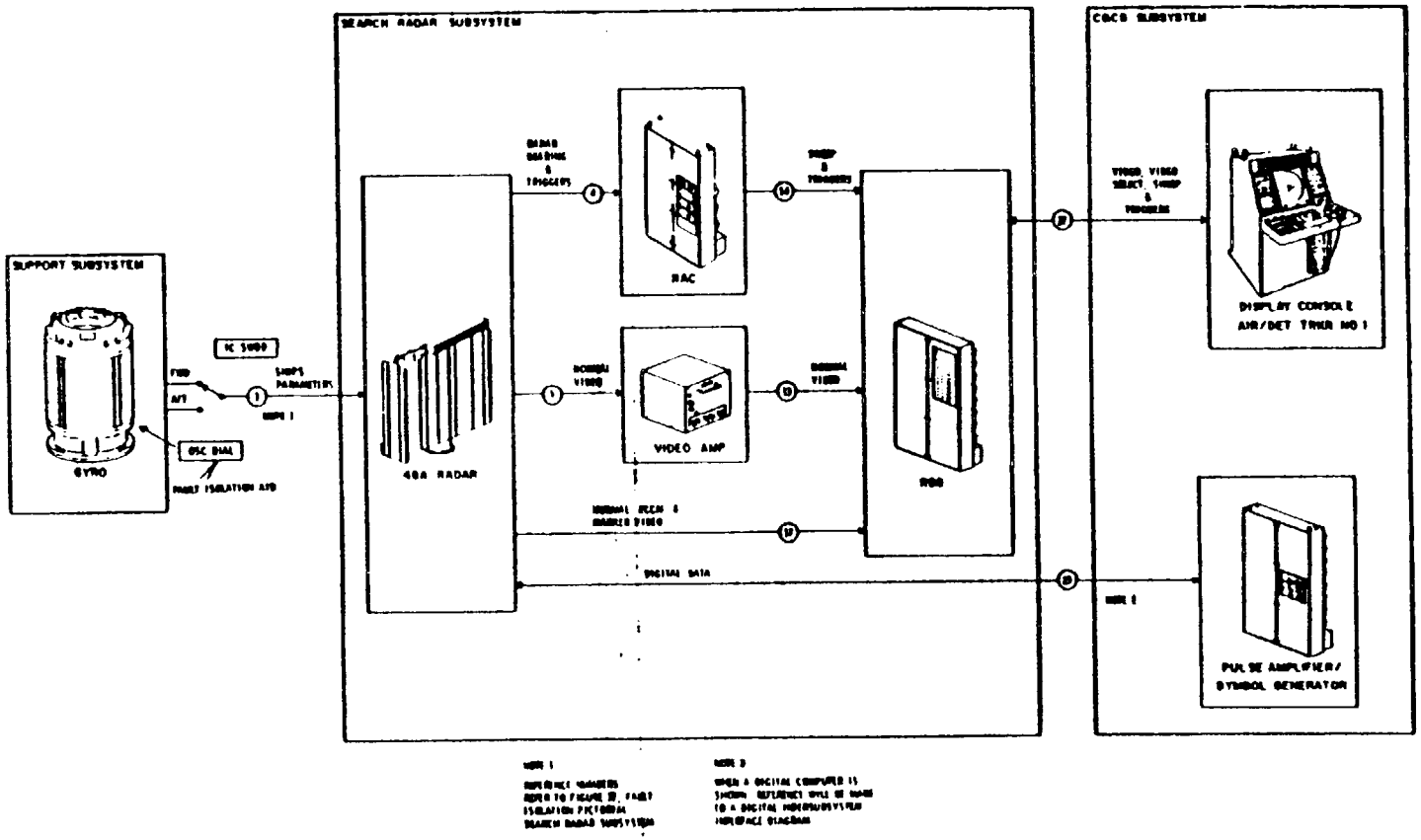
Int no.	Function name	From		To		Interface reference
		Equipment	Test	Equipment	Test	
4	Ro	SPS-48A	R81/D-3	RAC	TD141/R-1Q	NS 0967-LP-014-1140 Vol 3/Pt 2
	Rmax	SPS-48A	R81/D-3	RAC	TD141/R-1Q	NS 0967-LP-014-1140 Vol 3/Pt 2
	Rss	SPS-48A	R81/D-3	RAC	TD141/R-1Q	NS 0967-LP-014-1140 Vol 3/Pt 2
	Sin By 5	SPS-48A	R81/W-7	RAC	TD141/R-1Q	NS 0967-LP-014-1140 Vol 3/Pt 2
	Cos By 5	SPS-48A	R81/W-7	RAC	TD141/R-1Q	NS 0967-LP-014-1140 Vol 3/Pt 2
5	Vo	SPS-48A	SPS-48 Self Test	Video Amp No. 1	R25/Q-2	NS 0967-LP-014-1140 Vol 3/Pt 2
13	Vo	Video Amp No. 1	R25/Q-2	RDS	TD141/R-1Q	NS 0967-LP-014-1140 Vol 3/Pt 2
14	ΔX, ΔY	RAC	TD141/R-1Q	RDS	TD141/R-1Q	NS 0967-LP-014-1140 Vol 3/Pt 2
	Sign ΔX, ΔY	RAC	TD141/R-1Q	RDS	TD141/R-1Q	NS 0967-LP-014-1140 Vol 3/Pt 2
	θD	RAC	TD141/R-1Q	RDS	TD141/R-1Q	NS 0967-LP-014-1140 Vol 3/Pt 2
	End Sweep	RAC	TD141/R-1Q	RDS	TD141/R-1Q	NS 0967-LP-014-1140 Vol 3/Pt 2

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 37. Sample table of interface references for fault isolation pictorial.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.3.1 for textual requirements.

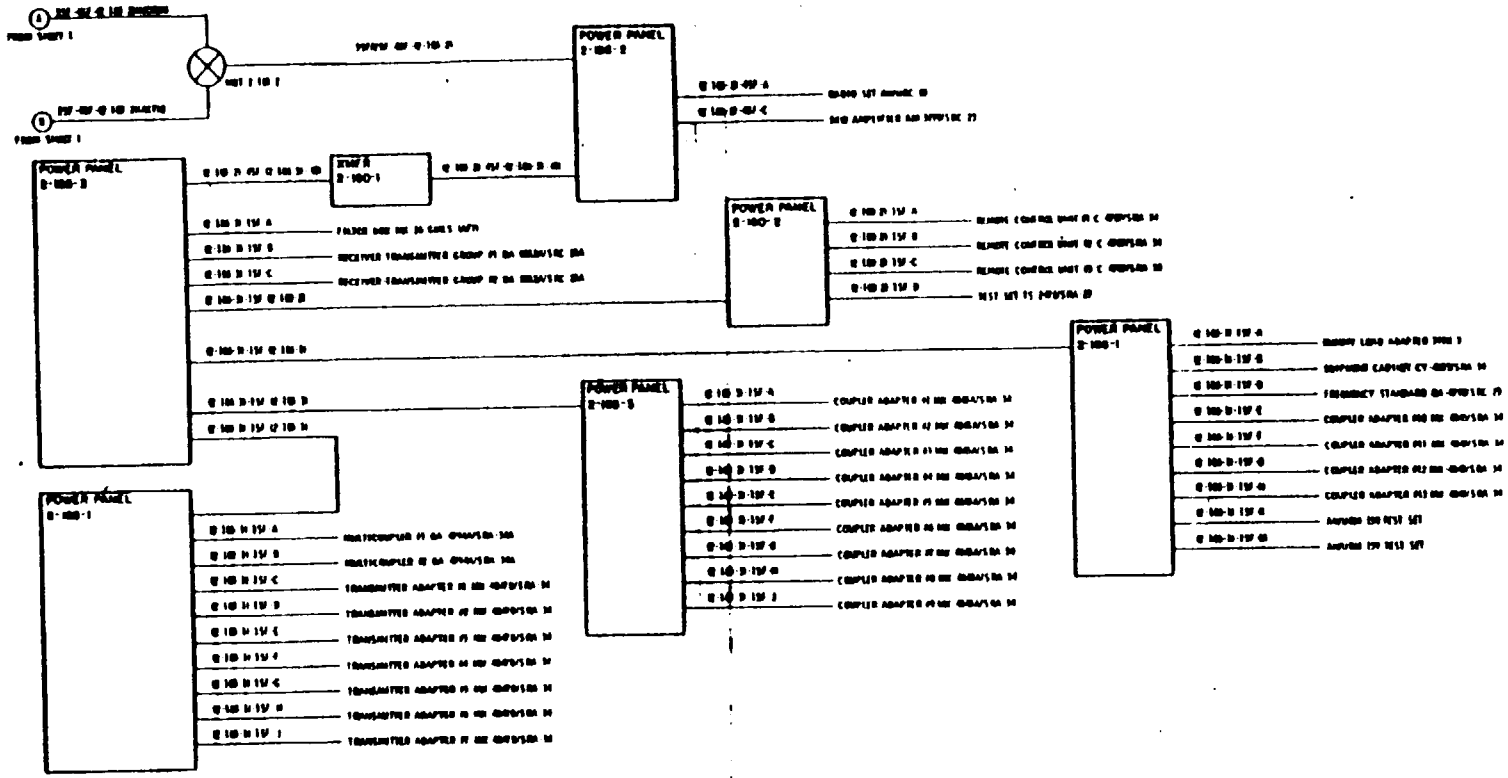


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 38. Sample fault isolation pictorial.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.3.4 for textual requirements.

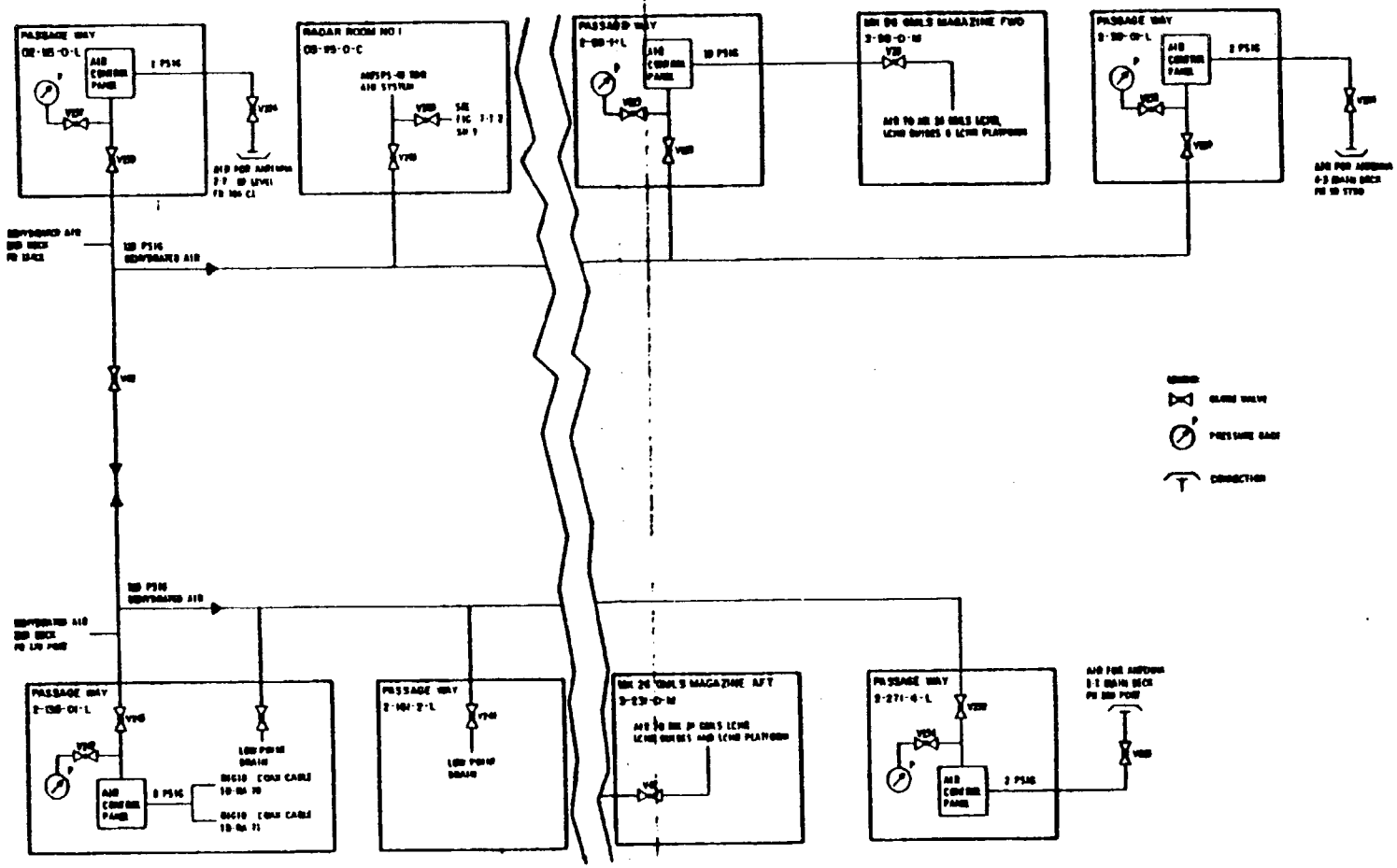


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 39. Sample support subsystem electrical power distribution diagram.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.3.4 for textual requirements.



Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 40. Sample support subsystem dry air system distribution diagram.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.3.4 for textual requirements.

SUB SYSTEMS \ SHIPS SERVICES	450V 60 Hz	120V 60 Hz	450V 400 Hz	120V 400 Hz	120V DC	600V DC	Ship Speed (Knots)	Ship Speed (Knots)	Roll & Pitch (Plus & Minus)	Wind Speed & Dir (Knots & Degrees)	Central Dry Air	H.P. Air	H.P. Nitrogen	S/W Supply	C/W Supply	Steam Supply	Emergency S/W Supply
TACTICAL DATA SYSTEM														7-7.X	7-7.X		7-7.X
DATA PROCESSING GROUP		7-7.X	7-7.X	7-7.X		7-7.X				7-7.X							
DATA DISPLAY GROUP		7-7.X		7-7.X		7-7.X											
DATA TRANSMITTING GROUP		7-7.X		7-7.X			7-7.X	7-7.X	7-7.X								
ASW																	
SONAR	7-7.X			7-7.X										7-7.X			7-7.X
ASROC	7-7.X	7-7.X		7-7.X	7-7.X		7-7.X	7-7.X	7-7.X	7-7.X		7-7.X		7-7.X		7-7.X	7-7.X
TORPEDO TUBES		7-7.X										7-7.X					
COMMUNICATION																	
AN/GRC 27A (UHF)											7-7.X						
AN/GRC 31			7-7.X														
MISSILE WEAPON																	
WDS		7-7.X	7-7.X	7-7.X		7-7.X		7-7.X									
TARTAR FCS	7-7.X	7-7.X	7-7.X	7-7.X		7-7.X		7-7.X	7-7.X								
AN/SPG 51C		7-7.X	7-7.X					7-7.X						7-7.X			7-7.X
TARTAR LCHG SYS	7-7.X					7-7.X						7-7.X	7-7.X	7-7.X		7-7.X	
TALOS FCS	7-7.X	7-7.X	7-7.X	7-7.X	7-7.X	7-7.X	7-7.X	7-7.X		7-7.X							
AN/SPW 2				7-7.X	7-7.X	7-7.X					7-7.X						
AN/SPG 49			7-7.X	7-7.X	7-7.X	7-7.X		7-7.X						7-7.X			7-7.X
TALOS LCHG SYS	7-7.X	7-7.X	7-7.X	7-7.X		7-7.X							7-7.X			7-7.X	
NAVIGATION																	
AN/SPN 6	7-7.X								7-7.X		7-7.X						
AN/URD 4		7-7.X															
IFF SYSTEM (AIMS)																	
AN/APA		7-7.X															
AN/UPX		7-7.X															
TELEMETRY																	
AN/SRQ 2				7-7.X		7-7.X											

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 41. Sample figure index showing support subsystem diagram applicability to subsystem equipment.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.3.4 for textual requirements.

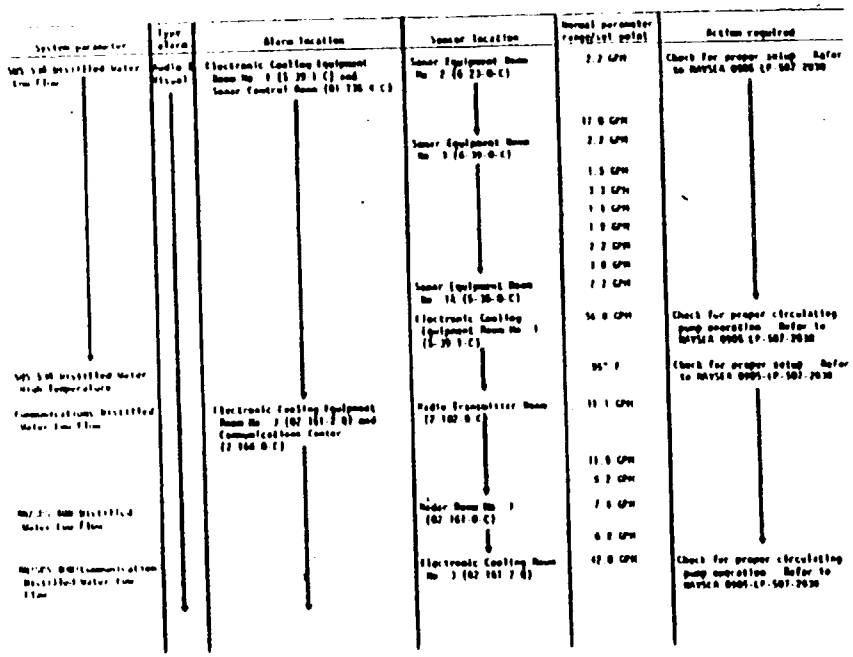
Fault cause \ Fault symptom	Pressure relief valve opens	High inlet pressure	Low inlet pressure	High outlet pressure	Low outlet pressure	Dew point alarm ON	High flow rate	Low flow rate	Pressure alarm ON
Panel outlet pressure too high	x								
Relief valve maladjustments	x								
Defective relief valve	x								
Excessive air main pressure		x							
Failure priority valves			x						x
Air leaks			x		x	x	x		x
Unauthorized uses on line			x			x			x
Pressure drop in dehydrator			x						x
Line valve partially closed			x						x
Bypass valve not closed				x					
Internal RS maladjustment				x	x				x
RS component failure				x	x				x
Excessive Moisture						x			
Improper sensor setting						x			x
Defective sensor						x			x
Line blockage								x	x

Sample arrangement only. Size, legibility, and microfiche requiremnts do not necessarily conform to minimum specification requirements.

FIGURE 42. Sample support subsystem symptom/cause fault directory.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.3.4 for textual requirements.



Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 43. Sample table of audio-visual alarms.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.3.5 for textual requirements.

Equipment/function	Test: R-81/R-250	R-176/R-1W	CSO-1	CSO-2	CSO-6	OCSOT Phase III	SW BLOOD/R-1	SW BLOOD/R-2	SW BLOOD/R-3	SW BLOOD/R-4	R-607/R-7W	R-81/R-20	R-81/M-25	R-81/S-6	Table G-2.1	Fault Impact	Isolation/reference
Radar Set AN/SPS-48C:																	NS 0967-LP-014-1140 V3 P2
θs		X	X	X	X	X	X	X	X	X				X			NS 0967-LP-014-1140 V3 P2
Lb		X															NS 0967-LP-014-1140 V3 P2
End of Sweep										X	X						NS 0967-LP-014-1140 V3 P2
SIF Trigger										X	X						NS 0967-LP-014-1140 V3 P2
Height		X	X	X	X	X	X	X	X	X							NS 0967-LP-014-1140 V3 P2
Track Number		X	X	X	X	X	X	X	X	X							NS 0967-LP-014-1140 V3 P2
Cqo	X																NS 0967-LP-014-1140 V3 P2
Eio	X																NS 0967-LP-014-1140 V3 P2
Zdo	X																NS 0967-LP-014-1140 V3 P2
Nb																	NS 0967-LP-014-1140 V3 P2
Nb Enable																	NS 0967-LP-014-1140 V3 P2
Sin By 5												X					NS 0967-LP-014-1140 V3 P2
Cos By 5												X					NS 0967-LP-014-1140 V3 P2
Ro											X						NS 0967-LP-014-1140 V3 P2
Rmax											X						NS 0967-LP-014-1140 V3 P2
Rss		X									X						NS 0967-LP-014-1140 V3 P2
Vo		X	X	X	X	X	X	X	X	X	X						NS 0967-LP-014-1140 V3 P2
V_1											X						NS 0967-LP-014-1140 V3 P2

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 44. Sample test fault directory.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.3.6 for textual requirements.

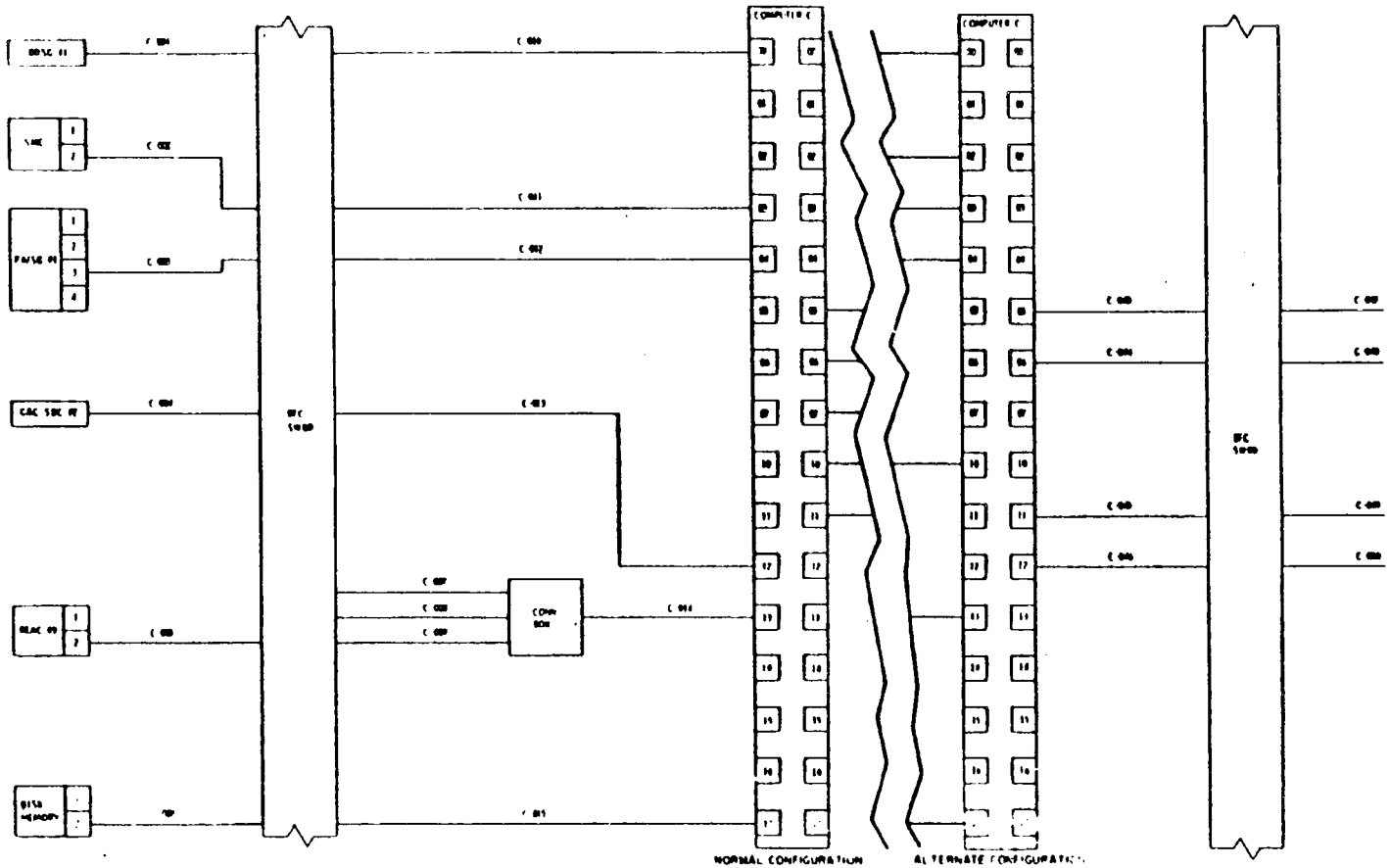
Interface		Unit A		Unit B		Reference document
No.	Cable	Conn	Equip	Conn	Equip	
C-030	G-GLC4	J6	LSMC No. 2	5J3	DFC switch-board	OP 4165 V3 P9
	G-GLC3	5J1	DFC switch-board	J8	LSMC No. 2	
C-031	IR-DD219	J2	DEAC No. 4 (chan 2)	47J3	DFC switch-board	OP 4165 V3 P9
	IR-DD220	47J1	DFC switch-board	J1	DEAC No. 4 (chan 2)	
C-032	IR-DD222	J4	Disk memory (chan 2)	39J1	DFC switch-board	OP 4165 V3 P9
	IR-DD223	39J3	DFC switch-board	J3	Disk memory (chan 2)	
C-033	IR-DD189	34J7	DFC switch-board	TB1-4,9	UFCC computer conn box	OP 4165 V3 P9
	IR-DD190	TB5-9	UFCC computer conn box	34J5	DFC switch-board	
C-034	IR-DD191	35J4	DFC switch-board	TB1-4,9	UFCC computer conn box	OP 4165 V3 P9
	IR-DD192	TB5-9	UFCC computer conn box	35J2	DFC switch-board	
C-035	IR-DD193	47J8	DFC switch-board	TB1-4,9	UFCC computer conn box	OP 4165 V3 P9
	IR-DD194	TB5-9	UFCC computer conn box	47J6	DFC switch-board	
C-036	IR-DD57	7J3	DFC switch-board	J41	UFCC (chan 0)	OP 4165 V3 P9
	IR-DD58	J42	UFCC (chan 0)	7J1	DFC switch-board	
C-037	IR-DD59	14J7	DFC switch-board	J57	UFCC (chan 2)	OP 4165 V3 P9
	IR-DD60	J58	UFCC (chan 2)	14J5	DFC switch-board	
C-038	IR-DD61	46J8	DFC switch-board	J57	UFCC (chan 3)	OP 4165 V3 P9
	IR-DD62	J58	UFCC (chan 3)	46J6	DFC switch-board	

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 45: Sample digital intersubsystem interface cable list.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.3.6 for textual requirements.



Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 46. Sample digital intersubsystem interface diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.3.6 for textual requirements.

INTERSUBSYSTEM INTERCONNECTION TABULATION (ANALOG)

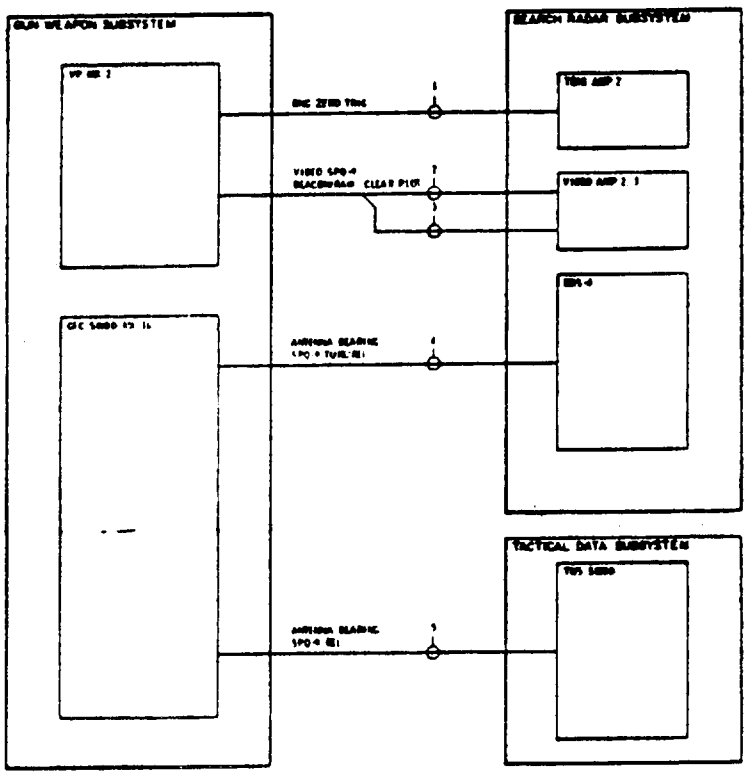
REFERENCE NO. AND FUNCTION	UNIT A AND TERM NO.	CABLE AND WIRE NUMBERS	UNIT B AND TERM NO.
1	VP MK 2		TRIG AMP 2
RANGE ZERO TRIG SPQ-9	J6 - C	2GSB15A111	J25 - BL
SHIELD	- M	111R	- BL
2	VP MK 2		VIDEO AMP 2
VIDEO, SPQ-9 BEACON/RAW SHIELD	J9 - K - T	4GSA15D110 110R	5J42 - A - U
3	VP MK 2		VIDEO AMP 3
VIDEO, SPQ-9 CLEAR PLOT SHIELD	J9 - S - V	13GSA55A111 111R	5J42 - L - O
4	GFC SWBD MK 16		RDS - 4
ANT BRG REL/TRUE S1 SPQ-9 1X S2 S3 115V 60 Hz BA R1 BB R2	JC2AE - L - M - N - K - B	4GSA83A111 112 113 110 110R	20J5 - A - B - E - C - U
5	GFC SWBD MK 16		TDS SWBD
ANT BRG REL S1 SPQ-9 1X S2 S3 SHIELD 115V 60 Hz R1 SYNCHRO REF R2 STUB SHIELD	JC2AA - m - n - w - x - AH - AT - AU - BC	4GSA95E111 112 113 111S 4GSA95E110 110R 110T 110S	JB2 - 3 - 4 - 5 - 1 - 7 - 8 - 9 - 2

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 47. Sample analog interface interconnection tabulation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.6.3.6 for textual requirements.

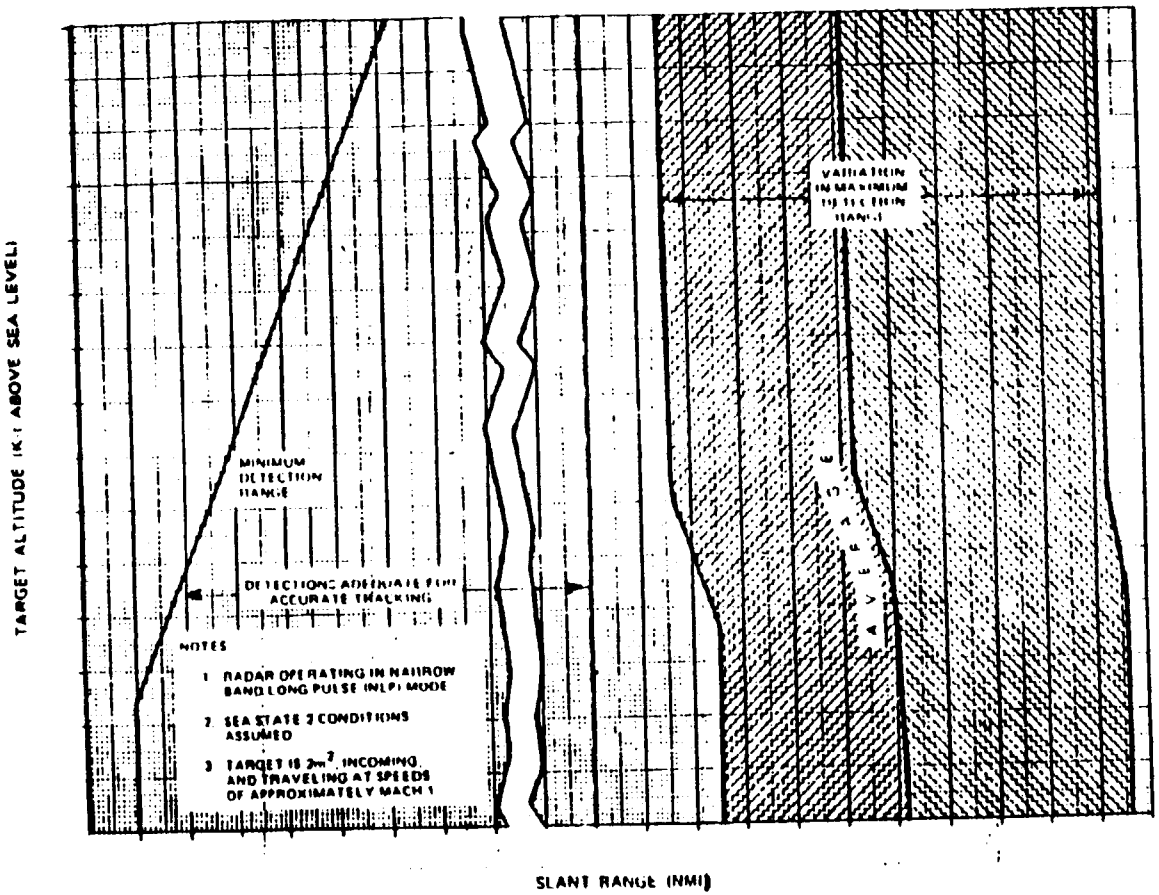


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 48. Sample analog intersubsystem interface diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.7.2.2 for textual requirements.



Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 49. Sample search radar air target detection capabilities in a non-ECM and medium fog environment.

Source: https://assist.dla.mil - Downloaded: 20261019T23:02Z
Check the source to verify that this is the current version before use.

See 3.7.2.2 for textual requirements.

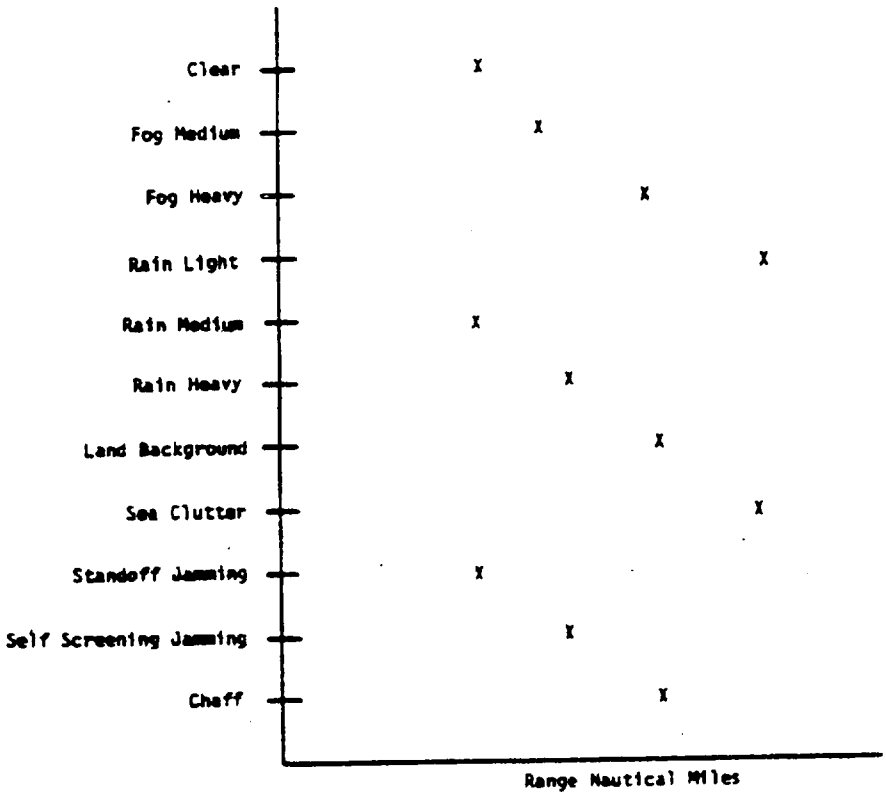
Environment (sea state 2)	Operating mode	Positive detection and tracking range (NMI)	Maximum expected range (NMI)
Clear weather	Narrow-bank, long pulse (NLP)		
Fog - medium	Narrow-bank, long pulse (NLP)		
- heavy	Narrow-band, long pulse (NLP)		
Rain	Wide-band, short pulse (WSP)		
Land	Wide-band, short pulse (WSP)		

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 50. Sample summary of radar set detection capabilities in a non-ECM environment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.7.2.2 for textual requirements.

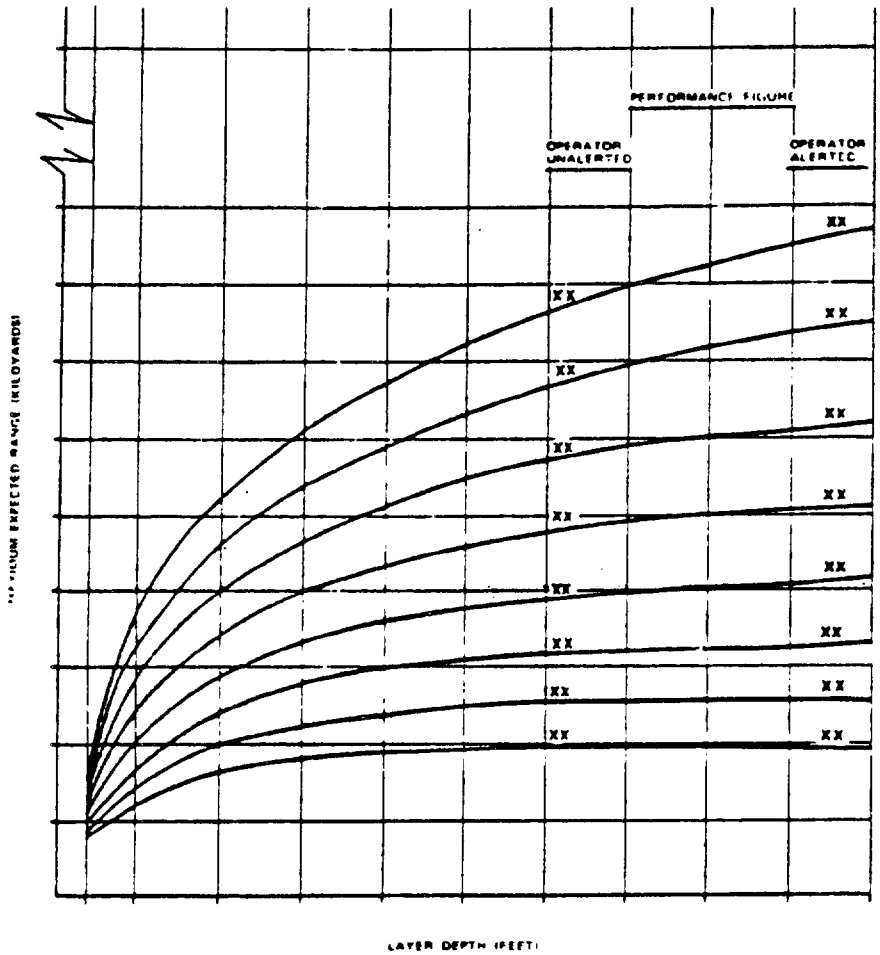


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 51. Sample search radar antiship cruise missile detection capabilities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.7.2.5 for textual requirements.

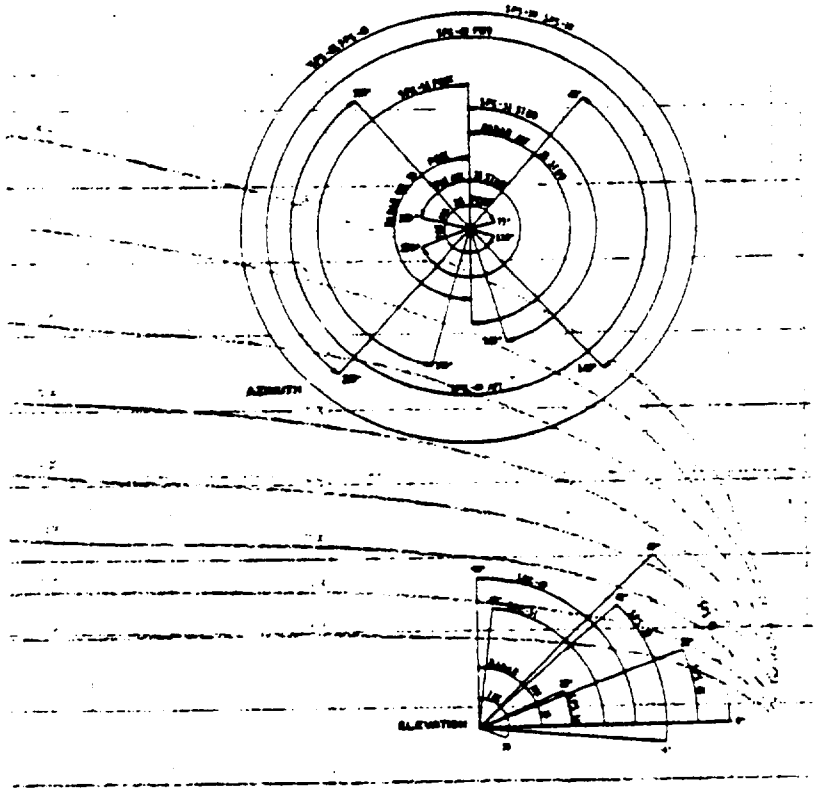


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 52. Sample graph of sonar maximum expected detection ranges versus layer depth for an alerted and unalerted operator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.7.2.9 for textual requirements.



Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 53. Sample figure of air target detection zones for various sensors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.7.3.2 for textual requirements.

Sensor	Bearing (°)	Range (yd)	Elevation (°)	Remarks
Primary air Search radar	± x.xx	± xxx	x.x	Bearing and elevation resolution ± 1/2 beamwidth.
Primary surface search radar	± x.x	± xxxx	NA	Bearing resolution ± 1/2 beamwidth; Range error increases as track range increases.
Secondary air-surface search radar	± x.x	± xx	± x.x	Bearing and elevation resolution ± 1/2 beamwidth.
Target designation xmitter	± x.x (sea state 2)	NA	± x.x (sea state 2)	Bearing and elevation accuracy will degrade as sea state increases. Range data is estimated and is not used in TDS. Range estimates are used as input data for GFCS Mk 68 and GMFCS Mk 76 designation.
IFF	± x.x	± xxx	(± xxx ft altitude)	Altitude data not presently implemented
ESM receiver	± x.x	NA	NA	Bearing accuracy exclusive of antenna system. Antenna error is inversly proportional to frequency and varies from x.x° at high frequencies to xx° at low frequencies.
ESM direction finder	± x.x	NA	NA	ESM bearing input to TDS via ECM universal keyset accuracy ± x.x°

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 54. Sample table of sensor position accuracy and resolution.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.7.3.3 for textual requirements.

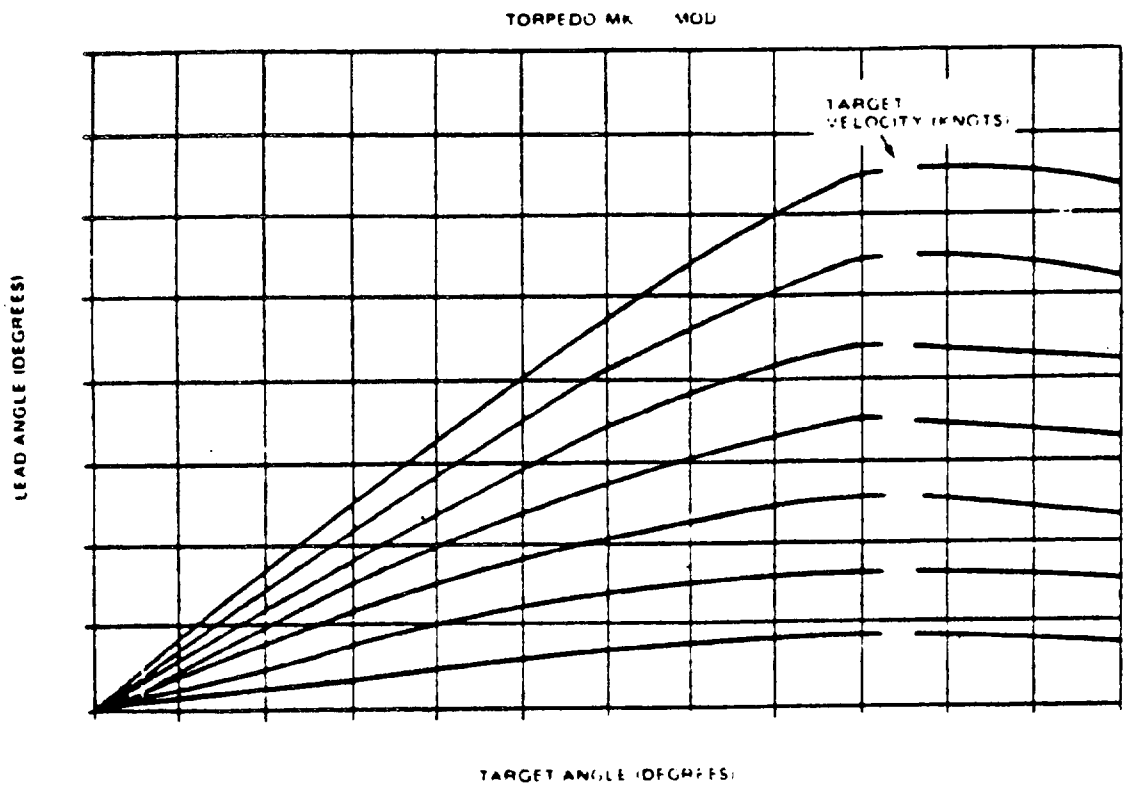
Track data Category	Capabilities (number)
Aircraft, command tracked	
Strike aircraft	
QR tracks	
Bearing lines, local ESM	
Bearing lines	
Bearing lines, total ESM	
Electronic warfare threats	
ESM fixes	
Link-11 PUs	
Local interceptors	
Local strikes	
Simulated target reports	
Special points	
Special tracks	
Tracks, remote TDS	
Tracks, local TDS	
Tracks, system	

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 55. Sample table of TDS program capabilities.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.7.5.3 for textual requirements.

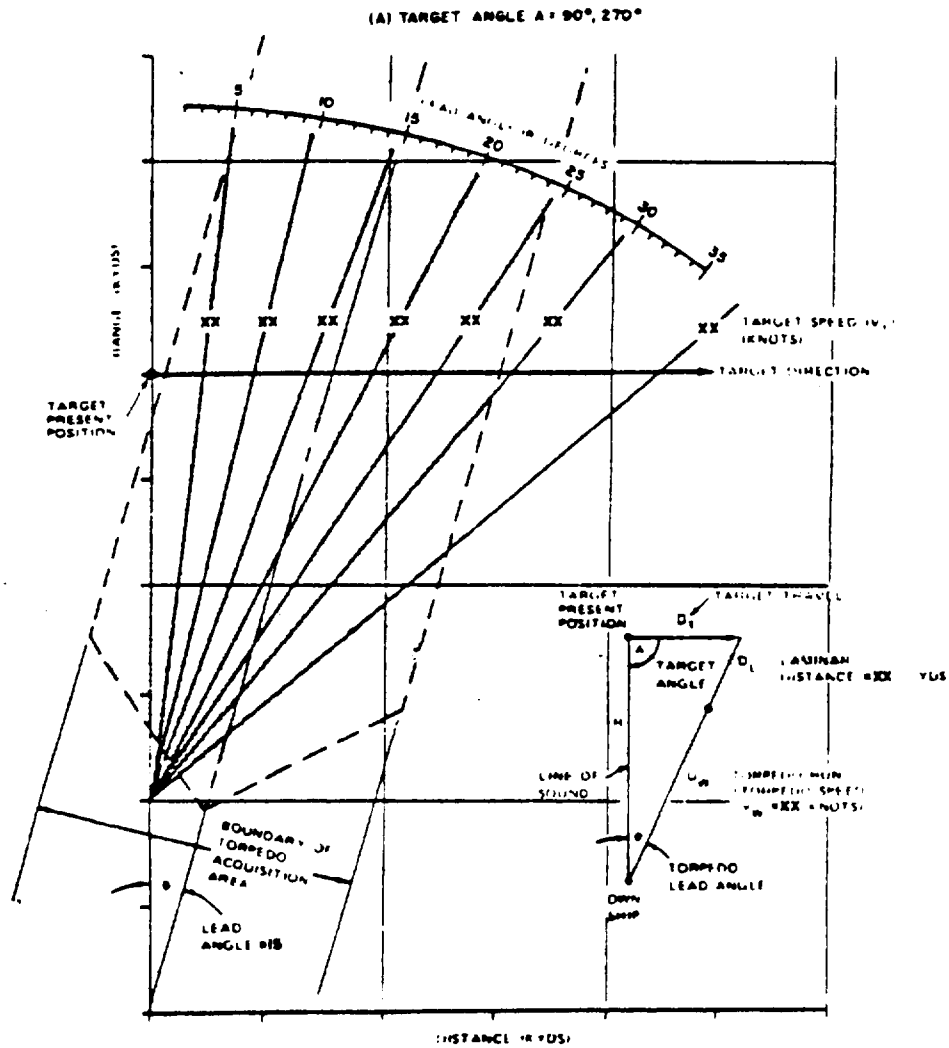


Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 56. Sample OTS torpedo lead angle dependency on target angle/target velocity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.7.5.3 for textual requirements.



Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 57. Sample OTS torpedo attack geometry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.7.5.3 for textual requirements.

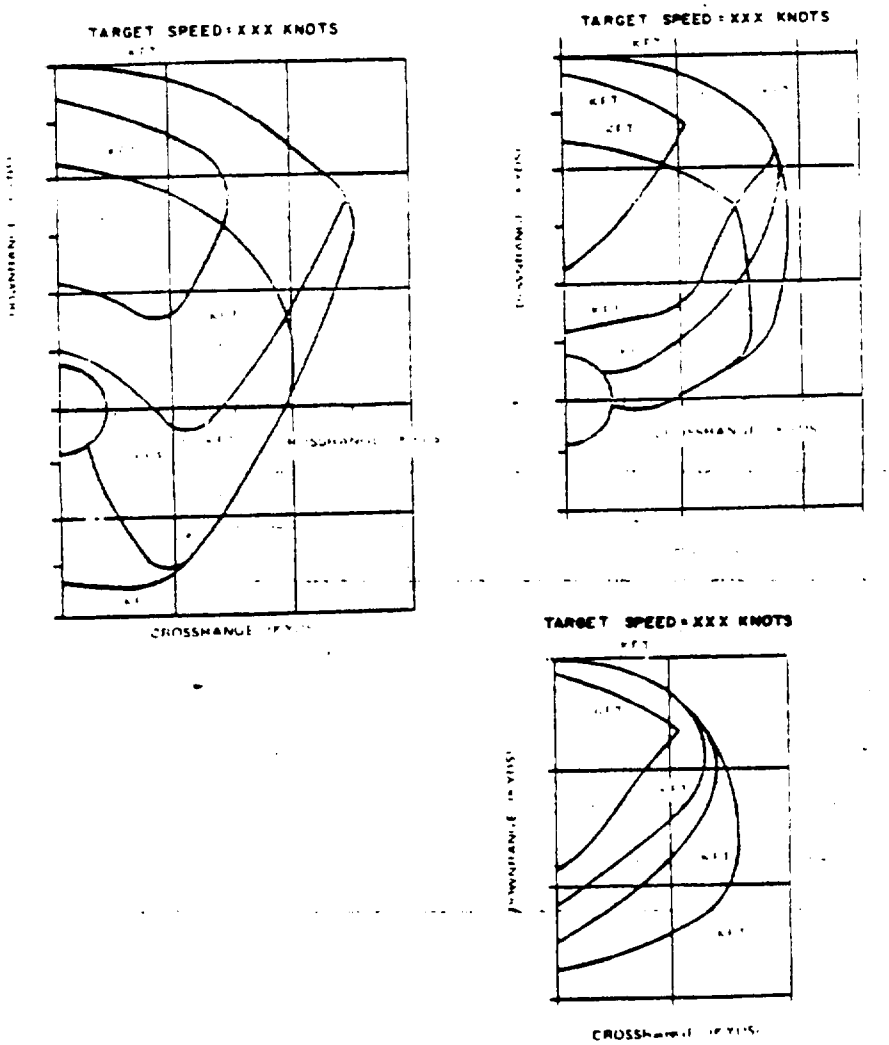
OTS Torpedo attack capability	Target angle (degrees)		
	30/330	60/300	90/270
Selected deflection lead angle	5°	10.0°	15°
OTS Torpedo Mk xx Mod x:			
Target present range			
Intercept range coverage			
Target speed coverage			

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 58. Sample summary of OTS torpedo attack capability for selected deflection lead angles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.7.5.3 for textual requirements.



Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 59. Sample target crossrange engageability envelope variation with target altitude.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

See 3.7.6.4 for textual requirements.

Type	Equipment	Frequency	Power (peak)	Status	Control station	EMCON responsi-bility	Index
Communi-cations	WRT-1A WRC-1B SRC-23A URC-32/32B WRT-2 SRC-20-20A SRC-21 SRC-31A URC-9 URC-80 SAT-2			Off rad Off rad Off rad Off dummy rad Off dummy rad Off rad Off rad Off rad Off rad Off rad Off rad			
Search	SPS-48C SPS-40B SPS-55 Seafarer (LN 66)			Off stby dummy rad Off stby dummy rad Off stby dummy rad Off on/rad			
FC/GM/GL	SPG-51D Track CWI SPG-60 Track SPQ-9A			 Off dummy rad Off dummy rad Off dummy rad Off stby rad			
NAV/ID	UPX-23 APX-72 SRN-9			Off rad Off rad Off rad			
DECM							
Acous-tics	SQS-53A UNQ-1D WQC-2 SLQ-25 (nixie)			Off stby rad Off stby rad Off rad Off rad			

Sample arrangement only. Size, legibility, and microfiche requirements do not necessarily conform to minimum specification requirements.

FIGURE 60. Sample table of radiating equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

INDEX

			Page
Paragraph	3.7.1.6	AAW mission	16
	3.7.3.2	Accuracy of entry data	17
	3.6.3.3 and 3.7.5.3	Additional sections	14 and 19
	3.6.3.7	Alinement, combat system	15
	3.8	Appendix A - supplementary data	21
	2	Applicable documents	1
	3.7.1.7	ASW mission	16
	3.2.3	Backbone	4
	3.1.2.6.1	Backbone stock	3
	3.2.2	Back cover	4
	3.6.3.8, 3.7.1.13, 3.7.2.10, 3.7.3.7, 3.7.4.8, 3.7.5.5, 3.7.6.7, 3.7.7.4, and 3.7.8.6	Bibliography	15, 16, 17, 18, 19, 20, and 21
	3.1.2.6	Binder	3
	6.3.1	Bridge-the-gap data	24
	3.1.2.5(c)	Callouts	3
	3.7 and 3.7.4	Capabilities and limitations	15 and 18
	3.1.3.6	Changes	3
	3.4.1	Chapter 1 - introduction	8
	3.4.2	Chapter 2 - combat subsystem descriptions	9
	3.4.3	Chapter 3 - combat system operational description	10
	3.5.1	Chapter 4 - operational sequence	10
	3.6.1	Chapter 5 - readiness assessment	12
	3.6.2	Chapter 6 - fault detection and impact evaluation	13
	3.6.3	Chapter 7 - fault isolation	13
	3.7.1	Chapter 8 - ship mission capabilities and limitations	15
	3.7.2	Chapter 9 - detection capabilities and limitations	16
	3.7.3	Chapter 10 - threat processing capabilities and limitations	17
	3.7.4	Chapter 11 - threat evaluation and threat-to-weapon pairing capabilities and limitations	18
	3.7.5	Chapter 12 - engagement and engagement assessment	19
	3.7.6	Chapter 13 - emission control	20
	3.7.7	Chapter 14 - communications capabilities and limitations	20
	3.7.8	Chapter 15 - navigation capabilities and limitations	21
	6.3.2	Combat subsystem	24
	6.3.3	Combat system	24
	3.6.3.7	Combat system alinement	15
	6.3.4	Combat system baseline	24
	3.4.1.2	Combat system description	8
	6.3.5	Combat system functions	24
	6.3.6	Combat system maintenance	24
	3.4.3	Combat system operational description	10
	3.7.1.4	Combat system operational modes	16
	3.6	Combat system readiness	12
	3.4.2.9	Communications	9
	3.7.7	Communications capabilities and limitations	20
	3.1.2.4.2	Computer output typesetting	2
	3.5.1.12	Condition III considerations	12
	6.3.7	Condition watch I	24
	6.3.8	Condition watch III	25
	3.5.1.9	Controlled aircraft engagement and engagement assessment	11
	3.7.6.2	Cover and deception	20
	3.1.2.6.1	Cover stock	3

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

INDEX (Continued)

			Page
Paragraph	3.3.1	Data level descriptions	5
	6.2.2	Data requirements	23
	3.1.3.1	Data research and analysis source data	3
	3.1.3	Data, technical	3
	6.3	Definitions	24
	3.4.3.2 and 3.5.1.2	Detection and entry	10 and 11
	3.7.2	Detection capabilities and limitations	16
	3.7.5.2	Electronic warfare ECM engagement and engagement assessment	19
	3.5.1.5	Electronic warfare engagement and engagement assessment	11
	3.4.2.4	Electronic warfare subsystem	9
	3.7.2.3	Electronic warfare subsystem support measure detection	17
	3.7.4.4	Electronics warfare threat evaluation	19
	3.7.6.5	EMCON monitoring	20
	3.7.6.3	EMCON plans	20
	3.7.6	Emission control	20
	3.4.3.5 and 3.7.5	Engagement and engagement assessment	10 and 19
	3.7.7.3	External communications	21
	3.4.2.9	External communication subsystem	9
	3.4.2.10	External navigation subsystem	9
	3.6.1.2	Factors, contributing to readiness assessment	12
	3.6.2	Fault detection	13
	3.6.3	Fault isolation	13
	3.1.2.5(a)	Foldouts	2
	3.2.5	Foreword	5
	3.1.1, 3.1.2, and 3.1.2.5(d)	Format	2 and 3
	3.2.1	Front cover	4
	3.2	Front matter	4
	3.1	General requirements	2
	3.9	Glossary	21
	3.5.1.6	Gun engagement and engagement assessment	11
	3.4.2.5	Gun weapon subsystem	9
	3.7.2.4	Gun weapon subsystem radar detection	17
	3.1.2.5(b)	Halftone	3
	3.7.3.5	Identification and classification	18
	3.7.2.2.1	IFF detection	17
	3.1.2.2	Illustrations	2
	3.1.2.1	Illustration numbering	2
	3.6.2.3	Impact evaluation	13
	4.5	Inspection of preparation for delivery	23
	6.1	Intended use	23
	3.7.7.2	Internal communications	20
	3.6.3.6	Intersubsystem interface data	14
	3.7.2.1	Introduction to combat system detection capabilities and limitations	16
	3.4.2.1	Introduction to combat system	9
	3.4.3.1	Introduction to combat system functions	10
	3.7.4.1	Introduction to combat system threat evaluation and threat-to-weapon pairing capabilities and limitations	18
	3.7.3.1	Introduction to combat system threat processing capabilities and limitations	17
	3.7.7.1	Introduction to communications capabilities and limitations	20
	3.4.1.1	Introduction to CSTOM	8
	3.7.6.1	Introduction to EMCON	20
	3.7.5.1	Introduction to engagement and engagement assessment	19

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:02Z
Check the source to verify that this is the current version before use.

INDEX (Continued)

			Page
Paragraph	3.6.2.1	Introduction to fault detection and impact evaluation	13
	3.6.3.1	Introduction to fault isolation	13
	3.7.8.1	Introduction to navigation capabilities and limitations	21
	3.6.1.1	Introduction to readiness assessment	12
	3.4.2.1	Introduction to the combat system	9
	3.7.1.1	Introduction to the combat system capabilities and limitations	15
	2.1	Issues of documents	1
	3.4.2.8	LAMPS	9
	3.7.2.7	LAMPS detection	17
	3.5.1.10	LAMPS engagement and engagement assessment	11
	6.3.9	Level of coverage	25
	3.4.2.8	Light-airborne multi-purpose system (LAMPS) subsystem	9
	3.4.1.3	Location of combat system components	8
	3.6.1.5	Maintenance support elements	12
	3.1.4	Manual arrangement	4
	3.1.3.3	Manuscript copy	3
	3.1.2.4.1	Mechanically reproduced copy preparation methods	2
	3.1.2.5	Microform compatibility	2
	3.3.2	Minimum data requirements	5
	3.5.1.8	Missile engagement and engagement assessment	11
	3.4.2.7	Missile weapon subsystem	9
	3.7.2.6	Missile weapon subsystem radar detection	17
	3.4.2.10	Navigation	9
	3.7.8	Navigation capabilities and limitations	21
	6	Notes	23
	3.7.8.3	OMEGA navigation system	21
	3.7.1.3	Operating environments	15
	3.6.2.2	Operational fault directories	13
	6.3.10	Operational mode network	25
	3.5 and 3.5.1	Operational sequences	10
	3.5.1.1 and 6.3.11	Operational sequence networks	10 and 25
	3.7.2.4.1	Optical detection	17
	6.2	Ordering data	23
	2.2	Other publications	1
	5	Packaging	23
	5.1	Packaging and packing	23
	3.1.2.1	Page numbering	2
	3.1.2.6 and 3.1.2.6.2	Paper stock	3
	3.1.2.1	Paragraph numbering	2
	3.6.1.4	Planned maintenance system concept	12
	3.1.2.3	Planned maintenance system (PMS) test references	2
	3.1.3.4	Preliminary manual	3
	3.1.2.6 and 3.1.2.6.3	Printing	3
	3.1.2.5(c)	Print size	3
	6.2.1	Procurement requirements	23
	3.5.1.1	Purpose and use of the operational mode network	10
	4.1.1	Quality assurance	22
	4	Quality assurance provisions	21

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

INDEX (Continued)

			Page
Paragraph	3.7.6.4	Radiation characteristics	20
	3.7.8.4	Radio direction finders	21
	3.6.1 and 6.3.12	Readiness assessment	12 and 25
	3.6.1.6	Readiness assessment diagram description	13
	3.5.1.11	Reduced capability conditions	12
	3.7.1.11	Reduced capability configurations	16
	4.4	Rejection criteria	22
	3.1.2.4	Reproducible copy preparation methods	2
	3	Requirements	2
	4.1	Responsibility for inspection	21
	3.1.3.7	Revisions	3
	3.7.8.2	Satellite navigation system	21
	1 and 1.1	Scope	1
	3.4.2.2	Search radar subsystem	9
	3.7.2.2	Search radar subsystem detection	16
	3.6.3.2	Search radar subsystem fault isolation	14
	3.4.1.2	Ship and combat system description	8
	3.7.1.5	Ship characteristics	16
	3.6.1.3	Ship electronic readiness team	12
	3.7.1	Ship mission capabilities and limitations	15
	6.3.13	Ships electronic readiness team (SERT)	25
	3.7.1.9	Shore mission	16
	3.1.3.5	Standard (final) manual	3
	3.1.3.9	Status reports	3
	3.3	Structure of CSTOM	5
	3.1.1 and 3.1.2.5(d)	Style	2 and 3
	3.7.1.12	Summary of combat system capabilities and limitations	16
	3.7.2.9	Summary of combat system detection capabilities and limitations	17
	3.7.6.6	Summary of combat system EMCON capabilities and limitations	20
	3.7.5.4	Summary of combat system engagement and engagement assessment capabilities and limitations	20
	3.7.4.7	Summary of combat system threat evaluation and threat-to-weapon pairing capabilities and limitations	19
	3.7.3.6	Summary of combat system threat processing capabilities and limitations	18
	6.4	Supersession data	25
	3.1.3.8	Supplements	3
	3.4.2.11	Support subsystems	9
	3.6.3.4	Support subsystem fault isolation	14
	3.7.1.8	SUW mission	16
	3.1.2.5(c)	Symbols	3
	3.6.1.6	Synoptic test descriptions	13
	3.1.2.1	Table numbering	2
	3.1.2.5(c)	Tabular material	3
	3.7.8.5	TACAN	21
	3.4.2.3, 3.7.1.10, and 6.3.14	Tactical data subsystem	9, 16, and 25
	3.7.2.8	Tactical data subsystem remote detection	17
	3.7.4.2	Tactical data subsystem threat evaluation	18
	3.7.3.3	Tactical data subsystem tracking and correlation	18
	3.1.3.2	Technical manual outline	3
	3.6.3.5	Test fault directories	14
	3.7.1.2	Threat categories	15
	3.4.3.4 and 3.5.1.4	Threat evaluation and threat-to-weapon pairing	10 and 11
	3.7.3	Threat processing capabilities and limitations	17
	3.7.4.6	Threat-to-weapon pairing	19
	3.2.4	Title page	4
	3.4.3.3 and 3.5.1.3	Tracking and identification	10 and 11
	3.6.1.5	Training and maintenance support elements	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

INDEX (Continued)

			Page
Paragraph	3.5.1.7	Underwater engagement and engagement assessment	11
	3.4.2.6	Underwater weapon subsystem	9
	3.7.2.5	Underwater weapon subsystem sonar detection	17
	4.3 and 6.3.15	Validation	22 and 25
	4.2	Validation plan	22
	4.2.1	Validation plan (general)	22
	4.2.2	Validation plan (specific)	22
	6.3.16	Verification	25
	3.3.3	Volume arrangement	5
	3.4	Volume 1 - Combat system description	8
	3.5	Volume 2 - Operational sequences	10
	3.6	Volume 3 - Combat system readiness	12
	3.7	Volume 4 - Capabilities and limitations	15
	3.4.2.4	Warfare subsystem, electronic	9
	3.7.3.4	Weapon direction system entry and tracking	18
	3.7.4.3	Weapon direction system threat evaluation	18
	3.7.4.5	Weapon subsystem threat evaluation	19

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

IONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in ting comments and suggestions for improvements. All users of military standardization documents are invited to provide ions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and . In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was id, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the ns. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an vledgement will be mailed to you within 30 days to let you know that your comments were received and are being ered.

: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of cation requirements on current contracts. Comments submitted on this form do not constitute or imply authorization ve any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

TMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

AL BUSINESS
TY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
Naval Ship Engineering Center (SEC 6124)
Department of the Navy
Washington DC 20362

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:02Z
Check the source to verify that this is the current version before use.